Quantum Information Theory

Lecture notes

(based on the courses of Prof. Michael Walter)

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1 The formalism of quantum information theory

1.1 Hilbert spaces and Dirac notation

Definition 1.1.1: To every quantum system we associate a *Hilbert space* $\mathcal{H}$, i.e., a complex vector space with an inner product $\langle\cdot|\cdot\rangle$ which is complete with respect to the norm $\|\psi\| := \sqrt{\langle\psi|\psi\rangle}$.

Remark 1.1.2: In this course, all Hilbert spaces are finite-dimensional, $\dim\mathcal{H} < \infty$. Hence completeness is automatic and $\mathcal{H} \cong \mathbb{C}^d$ with $d = \dim\mathcal{H}$.

We use *Dirac notation*: vectors are written as *kets* $|\psi\rangle \in \mathcal{H}$, and $\langle\psi| := \langle\psi|\cdot\rangle$ denotes the corresponding linear functional (*bra*).

Example 1.1.3: A *qubit* is described by $\mathcal{H} = \mathbb{C}^2$ with standard basis

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix},$$

labeled by the values of a bit, i.e., by $\Sigma = \{0,1\}$.

Example 1.1.4: More generally, for a finite set Σ we consider $\mathcal{H} = \mathbb{C}^\Sigma$ with orthonormal basis $\{|x\rangle\}_{x\in\Sigma}$, called the *standard* or *computational basis*.

Remark 1.1.5: If $\|\psi\| = 1$, then $|\psi\rangle\langle\psi|$ is the orthogonal projection onto $\mathbb{C}|\psi\rangle$. Indeed, $|\psi\rangle\langle\psi|\,|\psi\rangle = |\psi\rangle\langle\psi|\psi\rangle = |\psi\rangle$, while $|\psi\rangle\langle\psi|\,|\phi\rangle = 0$ if $\langle\psi|\phi\rangle = 0$. For example, $|0\rangle\langle 0| = \begin{pmatrix}1&0\\0&0\end{pmatrix}$.

1.2 Operators on Hilbert spaces

Definition 1.2.1: For Hilbert spaces $\mathcal{H},\mathcal{K}$ we write $\mathrm{L}(\mathcal{H},\mathcal{K}) := \{A : \mathcal{H} \to \mathcal{K} \text{ linear}\}$ and $\mathrm{L}(\mathcal{H}) := \mathrm{L}(\mathcal{H},\mathcal{H})$. The *adjoint* of $A \in \mathrm{L}(\mathcal{H},\mathcal{K})$ is the unique operator $A^\dagger \in \mathrm{L}(\mathcal{K},\mathcal{H})$ such that

$$\langle\psi|A^\dagger|\phi\rangle = \overline{\langle\phi|A|\psi\rangle} \qquad \text{for all } |\psi\rangle\in\mathcal{H},\ |\phi\rangle\in\mathcal{K}.$$

In coordinates, $A^\dagger = \overline{A}^T$ is the conjugate transpose.

Definition 1.2.2: An operator $A \in \mathrm{L}(\mathcal{H})$ is called

(i) *normal* if $AA^\dagger = A^\dagger A$,

(ii) *Hermitian* if $A = A^\dagger$,

(iii) *unitary* if $UU^\dagger = U^\dagger U = I$ (the identity operator),

(iv) *positive semidefinite* (PSD) if A is Hermitian and all its eigenvalues are nonnegative. It is called *positive definite* if it is moreover invertible.

We write $\mathrm{PSD}(\mathcal{H}) := \{A \in \mathrm{L}(\mathcal{H}) \mid A \text{ PSD}\}$, and $A \geq B$ if and only if $A - B$ is PSD.

Theorem 1.2.1: Spectral theorem

Let $A \in \mathrm{L}(\mathcal{H})$ be normal. Then there exist an orthonormal basis $\{|\psi_i\rangle\}$ of eigenvectors and eigenvalues $a_i \in \mathbb{C}$ such that

$$A = \sum_i a_i|\psi_i\rangle\langle\psi_i| = \sum_{a\in\sigma(A)} a\,P_a,$$

where $\sigma(A)$ denotes the spectrum and P_a the orthogonal projection onto the eigenspace of a.

Remark 1.2.3: Hermitian and unitary operators are normal. For Hermitian A the eigenvalues are real, $a_i \in \mathbb{R}$, while for unitary A they satisfy $|a_i| = 1$. For a Hermitian operator A, the following are equivalent: A is PSD; $A = B^\dagger B$ for some operator B; $\langle\psi|A|\psi\rangle \geq 0$ for all $|\psi\rangle \in \mathcal{H}$. For a function f and normal A as above we define $f(A) := \sum_i f(a_i)|\psi_i\rangle\langle\psi_i|$, e.g., $\sqrt{A}$ or $\log A$ for PSD A.

1.3 Quantum states

Definition 1.3.1: A *(quantum) state* on $\mathcal{H}$ is an element of

$$\mathcal{S}(\mathcal{H}) := \{\rho \in \mathrm{PSD}(\mathcal{H}) \mid \operatorname{tr}\rho = 1\}.$$

Elements of $\mathcal{S}(\mathcal{H})$ are also called *density operators*. A state is called

(i) *pure* if $\rho = |\psi\rangle\langle\psi|$ for some unit vector $|\psi\rangle \in \mathcal{H}$,

(ii) *mixed* otherwise.

Example 1.3.2: (i) *Classical states*: $\rho = \sum_{x\in\Sigma} p_x|x\rangle\langle x|$ on $\mathbb{C}^\Sigma$, where $\{p_x\}$ is a probability distribution.

(ii) The *maximally mixed state* $\tau = I/d$, where $d = \dim\mathcal{H}$.

Remark 1.3.3: By the spectral theorem, for every state ρ there exist an orthonormal basis $\{|\psi_i\rangle\}$ and a probability distribution $\{p_i\}$ such that $\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i|$. In particular, the *purity* satisfies

$$\operatorname{tr}\rho^2 = \sum_i p_i^2 \leq 1, \qquad \text{with equality if and only if } \rho \text{ is pure.}$$

Lemma 1.3.4: The set $\mathcal{S}(\mathcal{H})$ is convex: if $\{p_j, \rho_j\}_{j\in J}$ is an *ensemble of states* (i.e., $\{p_j\}$ a probability distribution and $\rho_j \in \mathcal{S}(\mathcal{H})$, with J an arbitrary index set), then $\rho = \sum_j p_j\rho_j$ is a state. The extreme points of $\mathcal{S}(\mathcal{H})$ are precisely the pure states.

Proof. Exercise. □

Remark 1.3.5: Ensembles model a *source* which emits the state ρ_j with probability p_j (or ensembles in statistical physics).

1.4 Qubits and the Bloch ball

Definition 1.4.1: The *Pauli matrices* are

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They form a basis of the real vector space of Hermitian 2×2 matrices. Moreover, X, Y, Z are traceless, square to the identity, and satisfy $XY = iZ$ (and cyclic permutations).

Lemma 1.4.2 (Bloch ball): Every Hermitian 2×2 matrix ρ of trace one can be written uniquely as

$$\rho = \frac{1}{2}\big(I + xX + yY + zZ\big) = \frac{1}{2}\begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix}, \qquad r = (x,y,z) \in \mathbb{R}^3,$$

where r is called the *Bloch vector*; we have $x = \operatorname{tr}[X\rho]$, $y = \operatorname{tr}[Y\rho]$, $z = \operatorname{tr}[Z\rho]$. Then:

(i) ρ is a state if and only if $\|r\| \leq 1$ (*Bloch ball*),

(ii) ρ is a pure state if and only if $\|r\| = 1$ (*Bloch sphere*),

(iii) ρ is classical, i.e., $\rho = \mathrm{diag}(p, 1-p)$, if and only if $r = (0, 0, z)$.

The maximally mixed state τ corresponds to $r = 0$, and $|0\rangle\langle 0|$, $|1\rangle\langle 1|$ to the north and south pole.

Proof. The eigenvalues of ρ are of the form $\{\lambda, 1-\lambda\}$ with $\lambda \in \mathbb{R}$, and

$$\lambda(1-\lambda) = \det \rho = \tfrac{1}{4}\big(1 - x^2 - y^2 - z^2\big) = \tfrac{1}{4}\big(1 - \|r\|^2\big).$$

Thus $\rho \geq 0$ iff $\lambda(1-\lambda) \geq 0$ iff $\|r\| \leq 1$, and ρ has rank one iff $\lambda(1-\lambda) = 0$ iff $\|r\| = 1$. □

1.5 Measurements

Definition 1.5.1: A *measurement* (or POVM) on $\mathcal{H}$ with outcomes in a finite set Ω is a map

$$\mu : \Omega \to \mathrm{PSD}(\mathcal{H}) \qquad \text{such that} \qquad \sum_{x \in \Omega} \mu(x) = I.$$

The operators $\mu(x)$ are called *POVM elements*. The measurement is called *projective* if all $\mu(x)$ are orthogonal projections. We write $\mathrm{Meas}(\mathcal{H}, \Omega)$ for the set of such measurements, and often write $\{Q_x\}$, or $\{P_x\}$ in the projective case, instead of $\mu(x)$.

Remark 1.5.2: If μ is projective, then necessarily the projections are pairwise orthogonal, $\mu(x)\mu(y) = 0$ for $x \neq y$ (exercise).

Definition 1.5.3 (Born rule): If we measure $\mu \in \mathrm{Meas}(\mathcal{H}, \Omega)$ in the state $\rho \in \mathcal{S}(\mathcal{H})$, we obtain the outcome $x \in \Omega$ with probability

$$p(x) = \mathrm{tr}[\mu(x)\rho].$$

Remark 1.5.4: (i) By the definitions, p is indeed a probability distribution.

(ii) The Born rule does *not* specify what happens to the quantum state after the measurement.

(iii) The name *positive-operator valued measure* stems from the fact that μ and ρ together determine a probability measure. More generally, Ω can be a measurable space and μ a σ-additive map from measurable subsets to $\mathrm{PSD}(\mathcal{H})$ with $\mu(\Omega) = I$ (cf. Section 12).

Example 1.5.5 (Basis measurements): For an orthonormal basis $\{|\psi_x\rangle\}_{x \in \Sigma}$ of $\mathcal{H}$, $\mu(x) = |\psi_x\rangle\langle\psi_x|$ defines a projective measurement with $p(x) = \langle\psi_x|\rho|\psi_x\rangle$. For the standard basis measurement $\mu(x) = |x\rangle\langle x|$ and a pure state $\rho = |\psi\rangle\langle\psi|$ we get $p(x) = |\langle x|\psi\rangle|^2$.

Example 1.5.6 (Qubit measurements): For a qubit we can measure in the standard basis $\{|0\rangle, |1\rangle\}$, but also in the *Hadamard basis* $\{|+\rangle, |-\rangle\}$, where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. More generally, every unit vector $s \in \mathbb{R}^3$ determines a two-outcome projective measurement

$$\mu(0) = \tfrac{1}{2}\big(I + s_x X + s_y Y + s_z Z\big), \qquad \mu(1) = \tfrac{1}{2}\big(I - s_x X - s_y Y - s_z Z\big),$$

i.e., the projections onto the orthogonal pure states with Bloch vectors $\pm s$. If we measure a state with Bloch vector r, then

$$\Pr(\text{outcome } 0/1) = \mathrm{tr}[\mu(0/1)\rho] = \tfrac{1}{2}\big(1 \pm r \cdot s\big).$$

Remark 1.5.7 (Uncertainty): For $\rho = |0\rangle\langle 0|$, the standard basis measurement yields a deterministic outcome, while the Hadamard basis measurement yields a uniformly random outcome. No state yields deterministic outcomes for both measurements; quantitative versions of this are *uncertainty relations* (exercise).

2 Composite systems and entanglement

2.1 Composite systems

Definition 2.1.1: For a quantum system composed of subsystems ("registers") $1, \ldots, n$, the Hilbert space is the tensor product $\mathcal{H} = \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$.

Example 2.1.2: (i) n qubits: $\mathcal{H} = \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 = (\mathbb{C}^2)^{\otimes n}$.

(ii) $\mathbb{C}^{\Sigma_1} \otimes \cdots \otimes \mathbb{C}^{\Sigma_n} \cong \mathbb{C}^{\Sigma}$ for $\Sigma = \Sigma_1 \times \cdots \times \Sigma_n$, with *product basis* $|x_1, \ldots, x_n\rangle := |x_1\rangle \otimes \cdots \otimes |x_n\rangle$ for $x_i \in \Sigma_i$. We often leave out the commas, e.g., $|00\rangle = |0\rangle \otimes |0\rangle$.

Remark 2.1.3 (Notation): From now on we use subscripts to indicate (sub)systems. E.g., $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$, ρ_{AB} is a state on $\mathcal{H}_{AB}$, X_B an operator on $\mathcal{H}_B$, $\Sigma_{AB} = \Sigma_A \times \Sigma_B$, $\mathcal{S}(AB) = \mathcal{S}(\mathcal{H}_A \otimes \mathcal{H}_B)$, and $|A| := \dim \mathcal{H}_A$.

2.2 Separable and entangled states

Definition 2.2.1: Let $\rho_{AB} \in \mathcal{S}(AB)$.

(i) ρ_{AB} is a *product state* if $\rho_{AB} = \rho_A \otimes \rho_B$.

(ii) ρ_{XY} is *classical* if $\rho_{XY} = \sum_{x,y} p(x,y) |xy\rangle\langle xy| = \sum_{x,y} p(x,y) |x\rangle\langle x| \otimes |y\rangle\langle y|$ for a joint probability distribution p.

(iii) ρ_{AB} is *separable* if

$$\rho_{AB} = \sum_{j \in J} p_j \, \rho_A^{(j)} \otimes \rho_B^{(j)}$$

for a probability distribution $\{p_j\}$ and states $\rho_A^{(j)}, \rho_B^{(j)}$ (which need *not* be orthogonal).

(iv) ρ_{AB} is *entangled* if it is not separable.

Remark 2.2.2: Product states and classical states are separable. Separable states are exactly the (average) states that Alice and Bob can prepare by sampling j with probability p_j, communicating j classically, and preparing $\rho_A^{(j)}$ and $\rho_B^{(j)}$ locally ("LOCC": local operations and classical communication).

Example 2.2.3: The *maximally entangled state* or *EPR pair*

$$|\Phi_{AB}^+\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big), \qquad \rho_{AB} = \tfrac{1}{2}\big(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|\big)$$

is entangled.

Remark 2.2.4: (i) To create entangled states, quantum communication (interaction) is needed.

(ii) Entanglement is necessary for quantum advantages in computation, communication and cryptography.

(iii) A *pure* state $|\psi_{AB}\rangle$ is entangled if and only if it is not a product $|\alpha\rangle \otimes |\beta\rangle$ (since convex decompositions of pure states are trivial).

2.3 Partial trace and reduced states

Suppose Alice wants to perform a measurement $\mu_A \in \mathrm{Meas}(\mathcal{H}_A, \Omega)$ on subsystem A of a system in state ρ_{AB}. This is described by the measurement

$$\mu_A \otimes I_B : \Omega \to \mathrm{PSD}(\mathcal{H}_A \otimes \mathcal{H}_B), \qquad \omega \mapsto \mu_A(\omega) \otimes I_B.$$

Evaluating the trace in a product basis,

$$\Pr(\text{outcome } x) = \mathrm{tr}\big[\rho_{AB}(\mu_A(x) \otimes I_B)\big] = \mathrm{tr}\Big[\Big(\sum_b (I_A \otimes \langle b|)\, \rho_{AB}\, (I_A \otimes |b\rangle)\Big)\mu_A(x)\Big],$$

where the operator in parentheses acts on $\mathcal{H}_A$. This motivates:

Definition 2.3.1: The *partial trace* $\mathrm{tr}_B : \mathrm{L}(\mathcal{H}_A \otimes \mathcal{H}_B) \to \mathrm{L}(\mathcal{H}_A)$ is defined by

$$\mathrm{tr}_B[M_{AB}] := \sum_b (I_A \otimes \langle b|)\, M_{AB}\, (I_A \otimes |b\rangle),$$

for any orthonormal basis $\{|b\rangle\}$ of $\mathcal{H}_B$. Concretely, $\langle a|\,\mathrm{tr}_B[M_{AB}]|\tilde{a}\rangle = \sum_b \langle a, b|M_{AB}|\tilde{a}, b\rangle$. For $\rho_{AB} \in \mathcal{S}(AB)$, the *reduced state* is $\rho_A := \mathrm{tr}_B[\rho_{AB}]$.

Thus, if we measure μ_A in the state ρ_{AB}, then $\Pr(\text{outcome } x) = \mathrm{tr}[\rho_A\, \mu_A(x)]$, i.e., ρ_A is a good description of the state of subsystem A.

Lemma 2.3.2 (Rules for the partial trace): (i) $\mathrm{tr}_B[M_A \otimes N_B] = M_A\, \mathrm{tr}[N_B]$ (this fully determines tr_B by linearity),

(ii) $\mathrm{tr}_B[(M_A \otimes I_B) N_{AB} (\tilde{M}_A \otimes I_B)] = M_A\, \mathrm{tr}_B[N_{AB}] \tilde{M}_A$,

(iii) $\mathrm{tr}_B[(I_A \otimes M_B) N_{AB}] = \mathrm{tr}_B[N_{AB}(I_A \otimes M_B)]$,

(iv) $\mathrm{tr}[M_{AB}(N_A \otimes I_B)] = \mathrm{tr}\big[\mathrm{tr}_B[M_{AB}] N_A\big]$,

(v) if ρ_{AB} is a state, then $\rho_A = \mathrm{tr}_B[\rho_{AB}]$ is a state.

Proof. Exercise. For (v), use (iv) with $N_A = I_A$ to see $\mathrm{tr}\,\rho_A = 1$, and with $N_A = |\psi\rangle\langle\psi| \geq 0$ to see $\rho_A \geq 0$. □

Example 2.3.3: For the EPR pair, $\rho_A = \mathrm{tr}_B[|\Phi^+_{AB}\rangle\langle\Phi^+_{AB}|] = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = I/2$. Thus even if ρ_{AB} is pure, ρ_A can be mixed!

2.4 Purifications

Conversely, every mixed state arises as the reduced state of a pure state:

Theorem 2.4.1: Purification

For every $\sigma_A \in \mathcal{S}(\mathcal{H}_A)$ there exist a Hilbert space $\mathcal{H}_R$ and a unit vector $|\psi_{AR}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_R$ such that

$$\mathrm{tr}_R\big[|\psi_{AR}\rangle\langle\psi_{AR}|\big] = \sigma_A.$$

Such $|\psi_{AR}\rangle$ is called a *purification* of σ_A. The condition $\dim \mathcal{H}_R \geq \mathrm{rk}\,\sigma_A$ is necessary and sufficient for the existence of a purification on $\mathcal{H}_A \otimes \mathcal{H}_R$.

Proof. If $\sigma_A = \sum_{i=1}^r p_i |\psi_i\rangle\langle\psi_i|$ is a spectral decomposition with $r = \mathrm{rk}\,\sigma_A$, then $|\psi_{AR}\rangle := \sum_i \sqrt{p_i}|\psi_i\rangle \otimes |i\rangle$ with $\mathcal{H}_R = \mathbb{C}^r$ (or any space containing an orthonormal family $\{|i\rangle\}$) is a purification. Necessity of $\dim \mathcal{H}_R \geq \mathrm{rk}\,\sigma_A$ follows from the Schmidt decomposition below. □

Example 2.4.1 (Standard purification): If $\mathcal{H}_R = \mathcal{H}_A$, then

$$|\psi_{AR}^{\mathrm{std}}\rangle := \left(\sqrt{\sigma_A} \otimes I_R\right) \sum_x |x\rangle \otimes |x\rangle$$

is a purification for any basis $\{|x\rangle\}$. Indeed, $\mathrm{tr}_R[|\psi_{AR}^{\mathrm{std}}\rangle\langle\psi_{AR}^{\mathrm{std}}|] = \sum_{x,y} \sqrt{\sigma_A}|x\rangle\langle y|\sqrt{\sigma_A}\ \mathrm{tr}[|x\rangle\langle y|] = \sqrt{\sigma_A}\, I\, \sqrt{\sigma_A} = \sigma_A$. If we choose $\mathcal{H}_R = \mathcal{H}_A^*$, this is truly canonical: $|\psi^{\mathrm{std}}\rangle = \sqrt{\sigma_A} \in \mathrm{L}(\mathcal{H}_A) \cong \mathcal{H}_A \otimes \mathcal{H}_A^*$.

Remark 2.4.2 (Church of the larger Hilbert space): Mixed states are not strictly needed: every state can be regarded as the reduced state of a pure state on a larger system. This viewpoint is conceptually meaningful, as we will see repeatedly.

2.5 Singular value and Schmidt decomposition

Theorem 2.5.1: Singular value decomposition

For every $M \in \mathrm{L}(\mathcal{H}, \mathcal{K})$ there exist $r \geq 0$, numbers $s_1, \ldots, s_r > 0$ (the *singular values*) and orthonormal families $\{|e_j\rangle\} \subset \mathcal{K}$, $\{|g_j\rangle\} \subset \mathcal{H}$ such that

$$M = \sum_{j=1}^{r} s_j |e_j\rangle\langle g_j|.$$

Proof. Write $M^\dagger M = \sum_{j=1}^r s_j^2 |g_j\rangle\langle g_j|$ (spectral decomposition, keeping only the nonzero eigenvalues), and set $|e_j\rangle := M|g_j\rangle/s_j$. Then $\langle e_j|e_k\rangle = \langle g_j|M^\dagger M|g_k\rangle/(s_j s_k) = \delta_{jk}$ and $M = \sum_j M|g_j\rangle\langle g_j| = \sum_j s_j|e_j\rangle\langle g_j|$. □

Remark 2.5.1: If $M = M^\dagger$, then the s_j are the absolute values of the nonzero eigenvalues of M, and $|e_j\rangle = \pm|g_j\rangle$.

Theorem 2.5.2: Schmidt decomposition

Every $|\psi_{AB}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ can be written as

$$|\psi_{AB}\rangle = \sum_{j=1}^{r} s_j\, |e_j\rangle \otimes |f_j\rangle,$$

with *Schmidt coefficients* $s_j > 0$, orthonormal $\{|e_j\rangle\} \subset \mathcal{H}_A$ and orthonormal $\{|f_j\rangle\} \subset \mathcal{H}_B$. The number r is called the *Schmidt rank*. In particular, the reduced states

$$\rho_A = \sum_j s_j^2 |e_j\rangle\langle e_j|, \qquad \rho_B = \sum_j s_j^2 |f_j\rangle\langle f_j|$$

have the same nonzero eigenvalues.

Proof. Fix orthonormal bases $\{|a\rangle\}$, $\{|b\rangle\}$ and consider the linear "vectorization" isomorphism $\mathcal{H}_A \otimes \mathcal{H}_B \to \mathrm{L}(\mathcal{H}_B, \mathcal{H}_A)$, $|a\rangle \otimes |b\rangle \mapsto |a\rangle\langle b|$. It maps $|e\rangle \otimes |f\rangle \mapsto |e\rangle\langle \bar{f}|$, where $|\bar{f}\rangle$ is the entrywise complex conjugate of $|f\rangle$. Apply the singular value decomposition to the image $M = \sum_j s_j |e_j\rangle\langle g_j|$ of $|\psi_{AB}\rangle$ and set $|f_j\rangle := |\bar{g}_j\rangle$, which is again orthonormal. The formulas for the reduced states follow by direct computation. □

The Schmidt decomposition implies that purifications are essentially unique:

Corollary 2.5.2 (Uniqueness of purifications): Let $|\psi_{AR}\rangle$ and $|\tilde{\psi}_{AS}\rangle$ be purifications of the same ρ_A with $\dim \mathcal{H}_R \leq \dim \mathcal{H}_S$. Then there is an isometry $V : \mathcal{H}_R \to \mathcal{H}_S$ (i.e., $V^\dagger V = I_R$) such that

$$|\tilde{\psi}_{AS}\rangle = (I_A \otimes V)|\psi_{AR}\rangle.$$

In particular, if $\mathcal{H}_R = \mathcal{H}_S$, then V can be chosen unitary.

Proof. Let $\rho_A = \sum_j s_j^2 |e_j\rangle\langle e_j|$ with $s_j > 0$. Expanding in the basis $\{|e_j\rangle\}$, write $|\psi_{AR}\rangle = \sum_j |e_j\rangle \otimes |\phi_j\rangle$. Then $\rho_A = \sum_{j,k} \langle\phi_k|\phi_j\rangle |e_j\rangle\langle e_k|$, hence $\langle\phi_k|\phi_j\rangle = s_j^2 \delta_{jk}$, i.e., $|\phi_j\rangle = s_j |f_j\rangle$ with orthonormal $\{|f_j\rangle\}$. Similarly $|\tilde{\psi}_{AS}\rangle = \sum_j s_j |e_j\rangle \otimes |\tilde{f}_j\rangle$. Any isometry V with $V|f_j\rangle = |\tilde{f}_j\rangle$ does the job. □

2.6 Entanglement of pure states

Lemma 2.6.1: For $|\psi_{AB}\rangle$ and $\rho_{AB} = |\psi_{AB}\rangle\langle\psi_{AB}|$, the following are equivalent:

(i) $|\psi_{AB}\rangle$ is unentangled (separable), i.e., $|\psi_{AB}\rangle = |\alpha\rangle \otimes |\beta\rangle$,

(ii) the Schmidt rank is one,

(iii) ρ_A (or ρ_B) is pure,

(iv) $\rho_{AB} = \rho_A \otimes \rho_B$.

Proof. Immediate from the Schmidt decomposition. □

Definition 2.6.2: A pure state $|\psi_{AB}\rangle$ with $d := |A| = |B|$ is *maximally entangled* if

$$|\psi_{AB}\rangle = \frac{1}{\sqrt{d}} \sum_{j=1}^{d} |e_j\rangle \otimes |f_j\rangle$$

for orthonormal bases $\{|e_j\rangle\}$, $\{|f_j\rangle\}$; equivalently, $\rho_A = I/|A|$ and $\rho_B = I/|B|$ (this requires $|A| = |B|$). The *standard maximally entangled state* is

$$|\Phi^+_{AB}\rangle = \frac{1}{\sqrt{d}} \sum_x |x, x\rangle \in \mathbb{C}^d \otimes \mathbb{C}^d.$$

Lemma 2.6.3 (Transpose trick): For all $M, N \in \mathrm{L}(\mathbb{C}^d)$ (transpose with respect to the standard basis),

$$(M \otimes I)|\Phi^+\rangle = (I \otimes M^T)|\Phi^+\rangle, \qquad \langle\Phi^+|M \otimes N|\Phi^+\rangle = \frac{1}{d} \operatorname{tr}\big[MN^T\big].$$

Proof. Exercise. □

Remark 2.6.4: In general, the *entanglement spectrum* $\{s_j^2\}$ (the nonzero eigenvalues of ρ_A and ρ_B) characterizes the entanglement in $|\psi_{AB}\rangle$, e.g., via the entanglement entropy or the purity $\operatorname{tr} \rho_A^2 = \operatorname{tr} \rho_B^2$ (exercise).

Lemma 2.6.5: Let ρ_{AB} be a (possibly mixed) state such that ρ_A is pure. Then $\rho_{AB} = \rho_A \otimes \rho_B$.

Proof. Exercise. (Purify ρ_{AB} and apply the previous lemma to the bipartition $A|BR$.) □

Corollary 2.6.6 (Monogamy of entanglement): If ρ_{ABC} is such that ρ_{AB} is pure, then $\rho_{ABC} = \rho_{AB} \otimes \rho_C$ and in particular $\rho_{AC} = \rho_A \otimes \rho_C$. If both ρ_{AB} and ρ_{AC} are pure, then $\rho_{AB} = \rho_A \otimes \rho_B$ and $\rho_{AC} = \rho_A \otimes \rho_C$ (in fact $\rho_{ABC} = \rho_A \otimes \rho_B \otimes \rho_C$). In words: A cannot be purely entangled with both B and C.

3 Distance measures

3.1 Operator norms

Definition 3.1.1: Let $M \in \mathrm{L}(\mathcal{H}, \mathcal{K})$ with singular values $\{s_i\}$. We define

$$\begin{aligned} \text{the } \textit{trace norm} \quad & \|M\|_1 := \sum_i s_i = \operatorname{tr} \sqrt{M^\dagger M}, \\ \text{the } \textit{Frobenius norm} \quad & \|M\|_2 := \Big(\sum_i s_i^2\Big)^{1/2} = \sqrt{\operatorname{tr}[M^\dagger M]}, \\ \text{the } \textit{operator norm} \quad & \|M\|_\infty := \max_i s_i = \max_{\|\phi\|=1} \|M|\phi\rangle\|. \end{aligned}$$

We often simply write $\|M\| = \|M\|_\infty$. The Frobenius norm is induced by the *Hilbert–Schmidt inner product* $(M, N) := \operatorname{tr}[M^\dagger N]$.

Lemma 3.1.2: (i) All three norms are invariant under $M \mapsto VMW^\dagger$ for isometries V, W, and they are submultiplicative.

(ii) $|\operatorname{tr}[AB]| \le \|A\|_2 \|B\|_2$ (Cauchy–Schwarz) and $|\operatorname{tr}[AB]| \le \|A\|_1 \|B\|_\infty$ (Hölder).

(iii) For $A \in \mathrm{L}(\mathcal{H})$,

$$\|A\|_1 = \max_{\|B\|_\infty \le 1} |\operatorname{tr}[AB]| = \max_{U \text{ unitary}} |\operatorname{tr}[AU]|. \tag{1}$$

Proof. For (iii): "$\ge$" follows from Hölder's inequality, and the second maximum is clearly at most the first. Conversely, if $A = \sum_i s_i |e_i\rangle\langle g_i|$ is a singular value decomposition, choose a unitary U with $U|e_i\rangle = |g_i\rangle$; then $\operatorname{tr}[AU] = \sum_i s_i \langle g_i|U|e_i\rangle = \|A\|_1$. □

3.2 Trace distance

Definition 3.2.1: The *(normalized) trace distance* between $\rho, \sigma \in \mathcal{S}(\mathcal{H})$ is

$$T(\rho, \sigma) := \tfrac{1}{2}\|\rho - \sigma\|_1 \in [0, 1].$$

Lemma 3.2.2: (i) For pure states $\rho = |\psi\rangle\langle\psi|$, $\sigma = |\phi\rangle\langle\phi|$: $T(\rho, \sigma) = \sqrt{1 - |\langle\phi|\psi\rangle|^2}$.

(ii) $T(\rho, \sigma) = \max_{0 \le Q \le I} \operatorname{tr}[Q(\rho - \sigma)]$.

Proof. (i) is an exercise; (ii) is a homework problem. Statement (ii) has a nice operational interpretation as the optimal bias for distinguishing ρ and σ (Helstrom's theorem). □

3.3 Fidelity and purified distance

Definition 3.3.1: The *fidelity* and the *purified distance* between $\rho, \sigma \in \mathcal{S}(\mathcal{H})$ are

$$F(\rho, \sigma) := \|\sqrt{\rho}\sqrt{\sigma}\|_1 = \operatorname{tr} \sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}, \qquad P(\rho, \sigma) := \sqrt{1 - F(\rho, \sigma)^2}.$$

Remark 3.3.2: Note that $\sqrt{\rho}\sqrt{\sigma}$ is not Hermitian in general.

(i) If $\rho = |\psi\rangle\langle\psi|$ is pure, then $F(\rho, \sigma) = \sqrt{\langle\psi|\sigma|\psi\rangle}$ (using $\sqrt{\rho} = \rho$).

(ii) If also $\sigma = |\phi\rangle\langle\phi|$ is pure, then $F(\rho, \sigma) = |\langle\psi|\phi\rangle|$ and $P(\rho, \sigma) = T(\rho, \sigma)$.

(iii) F and P are symmetric, $0 \le F, T \le 1$, and $F(\rho, \sigma) = 1$ iff $\rho = \sigma$, while $F(\rho, \sigma) = 0$ iff $\rho\sigma = 0$. Thus F is a similarity measure, while P is a distance measure (indeed a metric).

Theorem 3.3.1: Uhlmann's theorem

Let $\rho_A, \sigma_A \in \mathcal{S}(\mathcal{H}_A)$ have purifications on $\mathcal{H}_A \otimes \mathcal{H}_R$. Then

$$F(\rho_A, \sigma_A) = \max\Big\{ |\langle \psi_{AR} | \phi_{AR} \rangle| \;\Big|\; |\psi_{AR}\rangle, |\phi_{AR}\rangle \text{ purifications of } \rho_A, \sigma_A \Big\}.$$

By uniqueness of purifications, the right-hand side equals $\max_{U_R} |\langle \psi_{AR} | I_A \otimes U_R | \phi_{AR} \rangle|$ for arbitrary fixed purifications, where the maximum is over unitaries on $\mathcal{H}_R$.

Proof. For simplicity, assume $\mathcal{H}_R = \mathcal{H}_A$ and take the standard purifications $|\psi^{\text{std}}\rangle = (\sqrt{\rho_A} \otimes I) \sum_x |xx\rangle$, $|\phi^{\text{std}}\rangle = (\sqrt{\sigma_A} \otimes I) \sum_x |xx\rangle$. Then

$$|\langle \psi^{\text{std}} | I_A \otimes U | \phi^{\text{std}} \rangle| = \Big| \sum_{x,y} \langle x | \sqrt{\rho_A} \sqrt{\sigma_A} | y \rangle \langle x | U | y \rangle \Big| = |\operatorname{tr}\big[\sqrt{\rho_A} \sqrt{\sigma_A}\, U^T \big]|,$$

and U^T is an arbitrary unitary. Maximizing over U gives $\| \sqrt{\rho_A} \sqrt{\sigma_A} \|_1 = F(\rho_A, \sigma_A)$ by (1). (For $\mathcal{H}_R \neq \mathcal{H}_A$, see the lecture notes.) □

Lemma 3.3.3 (Monotonicity): For all $\rho_{AB}, \sigma_{AB} \in \mathcal{S}(AB)$,

$$T(\rho_{AB}, \sigma_{AB}) \geq T(\rho_A, \sigma_A), \qquad P(\rho_{AB}, \sigma_{AB}) \geq P(\rho_A, \sigma_A), \qquad F(\rho_{AB}, \sigma_{AB}) \leq F(\rho_A, \sigma_A).$$

Proof. Exercise. (For F, use Uhlmann's theorem.) □

Theorem 3.3.2: Fuchs–van de Graaf inequalities

For all $\rho, \sigma \in \mathcal{S}(\mathcal{H})$,

$$1 - F(\rho, \sigma) \leq T(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)^2}, \qquad \text{hence} \qquad \tfrac{1}{2} P(\rho, \sigma)^2 \leq T(\rho, \sigma) \leq P(\rho, \sigma).$$

The upper bound is an equality for pure states.

Proof. The lower bound is an exercise. For the upper bound, by Uhlmann's theorem choose purifications $|\psi_{AR}\rangle, |\phi_{AR}\rangle$ of ρ_A, σ_A with $F(\rho_A, \sigma_A) = F(\psi_{AR}, \phi_{AR})$. By monotonicity and the formula for pure states,

$$T(\rho_A, \sigma_A) \leq T(\psi_{AR}, \phi_{AR}) = \sqrt{1 - F(\psi_{AR}, \phi_{AR})^2} = \sqrt{1 - F(\rho_A, \sigma_A)^2}.$$

Finally, $P^2 = (1 - F)(1 + F) \leq 2(1 - F) \leq 2T$. □

4 Nonlocal games

4.1 Nonlocal games and classical strategies

In a *nonlocal game* (or *Bell game*), a referee sends questions x and y to two players, Alice and Bob, who reply with answers a and b. The referee then checks a winning condition. Alice and Bob play together (cooperate); they *cannot communicate during the game*, but they can meet before the game starts to agree on a strategy (e.g., share randomness or a quantum state).

Definition 4.1.1: A *nonlocal game* G is specified by

- finite question sets $\mathcal{X}$ and $\mathcal{Y}$ and answer sets $\mathcal{A}$ and $\mathcal{B}$,
- a probability distribution $p(x, y)$ on $\mathcal{X} \times \mathcal{Y}$ (typically uniform),
- a winning condition $W(a, b | x, y) \in \{0, 1\}$.

Any strategy determines a *correlation* $p(a,b|x,y)$, the probability that the players answer a,b given questions x,y. The goal is to maximize

$$p_{\text{win}} = \sum_{x,y,a,b} p(x,y)\, p(a,b|x,y)\, W(a,b|x,y).$$

Example 4.1.2 (CHSH game): $\mathcal{X} = \mathcal{Y} = \mathcal{A} = \mathcal{B} = \{0,1\}$, $p(x,y) = 1/4$, and the players win iff $a \oplus b = x \cdot y$, where $\oplus$ denotes addition mod 2 (XOR). That is, they need to give equal answers unless $x = y = 1$, in which case they need to give different answers.

How well Alice and Bob can do depends on the "laws of nature" we allow them to use.

Definition 4.1.3: (i) A *classical deterministic strategy* consists of functions $f : \mathcal{X} \to \mathcal{A}$, $g : \mathcal{Y} \to \mathcal{B}$, i.e., $p(a,b|x,y) = \delta_{a,f(x)}\delta_{b,g(y)}$.

(ii) A *classical randomized strategy* (or *local hidden variable model*) is given by a distribution $p(\lambda)$ of shared randomness ("hidden variable") and conditional distributions $p(a|x,\lambda)$, $p(b|y,\lambda)$, leading to the *classical correlation*

$$p(a,b|x,y) = \sum_{\lambda} p(\lambda)\, p(a|x,\lambda)\, p(b|y,\lambda).$$

The *classical value* $\omega(G) = \omega_{\text{cl}}(G)$ is the maximal winning probability over all classical strategies.

Remark 4.1.4: Randomness does not help: WLOG $p(\cdot|x,\lambda)$ and $p(\cdot|y,\lambda)$ are deterministic (redefine λ to include the local randomness), and then $p_{\text{win}}^{\text{rand}} = \mathbb{E}_\lambda\, p_{\text{win}}^{\text{det}} \le \max_\lambda p_{\text{win}}^{\text{det}}$. Hence the classical value is attained by a deterministic strategy.

Lemma 4.1.5: $\omega_{\text{cl}}(\text{CHSH}) = 3/4$.

Proof. Answering $a = b = 0$ always wins with probability $3/4$. Suppose f, g were a perfect deterministic strategy, i.e., $f(x) \oplus g(y) = xy$ for all x, y. Summing over all $x, y \in \{0,1\}$ gives

$$1 = \sum_{x,y} xy \equiv \sum_{x,y} \big(f(x) + g(y)\big) = 2\sum_x f(x) + 2\sum_y g(y) \equiv 0 \pmod 2,$$

a contradiction. Thus at least one of the four question pairs is answered wrongly. □

Remark 4.1.6: The statement "$p_{\text{win}} \le 3/4$ for every classical strategy" is an example of a *Bell inequality*.

4.2 Quantum strategies

Definition 4.2.1: A *quantum strategy* consists of a state $\rho_{AB} \in \mathcal{S}(AB)$ (for some $\mathcal{H}_A, \mathcal{H}_B$) and measurements $\mu_A(\cdot|x) \in \text{Meas}(\mathcal{H}_A, \mathcal{A})$ for each $x \in \mathcal{X}$ and $\mu_B(\cdot|y) \in \text{Meas}(\mathcal{H}_B, \mathcal{B})$ for each $y \in \mathcal{Y}$. It gives rise to the *quantum correlation*

$$p(a,b|x,y) = \text{tr}\big[\rho_{AB}\big(\mu_A(a|x) \otimes \mu_B(b|y)\big)\big].$$

The *quantum value* $\omega^*(G) = \omega_q(G)$ is the supremum of p_{win} over all quantum strategies.

Remark 4.2.2: (i) WLOG ρ_{AB} is pure (use a purification, or the same convexity argument as above), and WLOG the measurements are projective (every POVM can be implemented by an isometry followed by a projective measurement; exercise).

(ii) Clearly $\omega_q \ge \omega_{\text{cl}}$.

(iii) Entanglement is necessary for $\omega_q > \omega_{\text{cl}}$: if $\rho_{AB} = \sum_\lambda p(\lambda)\rho_A^\lambda \otimes \rho_B^\lambda$ is separable, then $p(a,b|x,y) = \sum_\lambda p(\lambda)\ \text{tr}[\rho_A^\lambda \mu_A(a|x)]\ \text{tr}[\rho_B^\lambda \mu_B(b|y)]$ is a classical correlation.

4.3 A quantum strategy for CHSH

We use the EPR pair $|\Phi^+_{AB}\rangle$. Recall that $\rho_r = \frac{1}{2}(I + r_x X + r_y Y + r_z Z)$ denotes the state with Bloch vector r; then $\mathrm{tr}[\rho_r \rho_s] = \frac{1}{2}(1 + r \cdot s)$ and, by the transpose trick (using $Y^T = -Y$),

$$\langle \Phi^+ | \rho_r \otimes \rho_s | \Phi^+ \rangle = \tfrac{1}{4}\big(1 + r \cdot \tilde{s}\big), \qquad \tilde{s} = (s_x, -s_y, s_z).$$

For simplicity we work in the X-Z-plane: let $r(\theta) = (\sin\theta, 0, \cos\theta)$ and $\rho_\theta := \rho_{r(\theta)}$, so that $\rho_{\theta+\pi} = \rho_{-r(\theta)}$. Alice measures $\mu_A(0|x) = \rho_{\alpha_x}$, $\mu_A(1|x) = \rho_{\alpha_x+\pi}$, and Bob similarly with angles β_y. Then

$$p(0,0|x,y) = p(1,1|x,y) = \tfrac{1}{4}\big(1 + \cos(\alpha_x - \beta_y)\big),$$

so that

$$p(\text{same}|x,y) = \tfrac{1}{2}\big(1 + \cos(\alpha_x - \beta_y)\big), \qquad p(\text{different}|x,y) = \tfrac{1}{2}\big(1 - \cos(\alpha_x - \beta_y)\big).$$

The angles should be close unless $x = y = 1$. Choosing $\alpha_0 = 0$, $\alpha_1 = \pi/2$, $\beta_0 = \pi/4$, $\beta_1 = -\pi/4$ gives $\cos(\alpha_x - \beta_y) = 1/\sqrt{2}$ if $xy = 0$ and $-1/\sqrt{2}$ if $xy = 1$. Hence

$$p_{\text{win}} = \frac{1}{2}\Big(1 + \frac{1}{\sqrt{2}}\Big) = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 85\,\%.$$

Theorem 4.3.1: Quantum advantage for CHSH

$\omega_q(\text{CHSH}) \geq \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 85\% > 75\% = \omega_{\text{cl}}(\text{CHSH})$.

Remark 4.3.1: Natural questions: Can we do even better? (Not for CHSH, see below.) Why this strange value? Are optimal strategies special? (Yes, see self-testing.) How about infinite-dimensional strategies? (They can sometimes do better.) How to compute ω_q? (It is not computable in general, but…)

4.4 XOR games and Tsirelson's theorem

Definition 4.4.1: An *XOR game* is a nonlocal game with answers $a, b \in \{0,1\}$ whose winning condition only depends on $a \oplus b$. We can write

$$W(a,b|x,y) = \tfrac{1}{2}\big(1 + (-1)^{a+b} S_{xy}\big), \qquad S_{xy} = \begin{cases} +1 & \text{if } a = b \text{ wins}, \\ -1 & \text{if } a \neq b \text{ wins}, \end{cases}$$

with *sign matrix* S. The *game matrix* is $G_{xy} := p(x,y) S_{xy}$; it fully specifies the game. Instead of p_{win} it is natural to study the *bias*

$$\beta := p_{\text{win}} - p_{\text{lose}} = 2p_{\text{win}} - 1 = \sum_{x,y,a,b} G_{xy}\, p(a,b|x,y)\, (-1)^{a+b}.$$

Example 4.4.2: For CHSH, $S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $G = \frac{1}{4}S$.

It is convenient to label answers by ± 1, i.e., $A = (-1)^a$, $B = (-1)^b$. For classical deterministic strategies $A_x, B_y \in \{\pm 1\}$,

$$\beta_{\text{cl}}(G) = \max_{A_x, B_y \in \{\pm 1\}} \sum_{x,y} G_{xy} A_x B_y.$$

In general, approximating the classical value is NP-hard. We will see that the quantum value of XOR games is easy to compute.

In the quantum case, a two-outcome projective measurement $\{\mu(0),\mu(1)\}$ is equivalent to a *binary observable* $O=\mu(0)-\mu(1)$, i.e., $O=O^\dagger$, $O^2=I$ (eigenvalues ±1). For a quantum strategy with pure state $|\psi_{AB}\rangle$ and binary observables A_x, B_y,

$$\beta_q(G)=\sup\sum_{x,y}G_{xy}\langle\psi_{AB}|A_x\otimes B_y|\psi_{AB}\rangle,$$

since $\langle\psi|A_x\otimes B_y|\psi\rangle=\sum_{a,b}(-1)^{a+b}p(a,b|x,y)$.

Example 4.4.3: The CHSH strategy above corresponds to $A_0=Z$, $A_1=X$, $B_0=(Z+X)/\sqrt{2}$, $B_1=(Z-X)/\sqrt{2}$.

Key idea: vector strategies. Define unit vectors

$$|v_x\rangle:=(A_x\otimes I)|\psi_{AB}\rangle,\qquad |w_y\rangle:=(I\otimes B_y)|\psi_{AB}\rangle,$$

so that $\beta=\sum_{x,y}G_{xy}\langle v_x|w_y\rangle$. This reduces the problem from operators to vectors.

Theorem 4.4.1: Tsirelson bound

$\omega_q(\mathrm{CHSH})=\frac{1}{2}\big(1+\frac{1}{\sqrt{2}}\big)$, i.e., $\beta_q(\mathrm{CHSH})=1/\sqrt{2}$.

Proof. For any quantum strategy, using Cauchy–Schwarz,

$$\begin{aligned}\beta&=\tfrac{1}{4}\big(\langle v_0|w_0\rangle+\langle v_0|w_1\rangle+\langle v_1|w_0\rangle-\langle v_1|w_1\rangle\big)=\tfrac{1}{4}\left\langle\begin{pmatrix}v_0\\v_1\end{pmatrix},\begin{pmatrix}w_0+w_1\\w_0-w_1\end{pmatrix}\right\rangle\\&\le\tfrac{1}{4}\sqrt{1+1}\,\sqrt{\|w_0+w_1\|^2+\|w_0-w_1\|^2}=\tfrac{1}{4}\sqrt{2}\cdot 2=\tfrac{1}{\sqrt{2}},\end{aligned}$$

since $\|w_0+w_1\|^2+\|w_0-w_1\|^2=2\|w_0\|^2+2\|w_1\|^2=4$. The strategy above attains this bound. □

Theorem 4.4.2: Tsirelson's theorem

For every XOR game with game matrix G,

$$\beta_q(G)=\sup_{\substack{d\in\mathbb{N},\ v_x,w_y\in\mathbb{R}^d\\ \|v_x\|=\|w_y\|=1}}\sum_{x,y}G_{xy}\langle v_x,w_y\rangle=\max_{\substack{v_x,w_y\in\mathbb{R}^{|\mathcal{X}|+|\mathcal{Y}|}\\ \|v_x\|=\|w_y\|=1}}\sum_{x,y}G_{xy}\langle v_x,w_y\rangle.$$

That is, the quantum value is obtained by replacing the signs $A_x,B_y\in\{\pm1\}$ in the classical problem by unit vectors. Moreover, $\beta_q(G)$ can be computed efficiently.

Proof. Denote the three quantities by (1), (2), (3).

(1)≤(2): Use the vector strategy and identify $\mathbb{C}^d\cong\mathbb{R}^{2d}$; since β is real, $\langle v_x|w_y\rangle$ may be replaced by its real part, which is the real inner product.

(2)=(3): Both only depend on the Gram matrix of the vectors, and n vectors always fit into $\mathbb{R}^n$ (any $n\times n$ PSD matrix is a Gram matrix of vectors in $\mathbb{R}^n$).

(3)≤(1): On $(\mathbb{C}^2)^{\otimes n}$ there are binary observables $C_1,\dots,C_n$ with $C_j^2=I$ and $C_jC_k=-C_kC_j$ for $j\neq k$, e.g., $C_j=X^{\otimes(j-1)}\otimes Z\otimes I^{\otimes(n-j)}$. For a unit vector $v\in\mathbb{R}^n$, $C(v):=\sum_j v_jC_j$ is again a binary observable, since $C(v)^2=\sum_j v_j^2I=I$. Let Alice use $A_x=C(v_x)$, Bob $B_y=C(w_y)^T$, and let them share the maximally entangled state $|\Phi^+\rangle$ on $(\mathbb{C}^2)^{\otimes n}\otimes(\mathbb{C}^2)^{\otimes n}$, $D=2^n$. By the transpose trick,

$$\langle\Phi^+|A_x\otimes B_y|\Phi^+\rangle=\frac{1}{D}\sum_{j,k}(v_x)_j(w_y)_k\operatorname{tr}[C_jC_k]=\langle v_x,w_y\rangle,$$

since $\operatorname{tr}[C_jC_k]=D\delta_{jk}$ (for $j\neq k$, $\operatorname{tr}[C_jC_k]=\operatorname{tr}[C_kC_j]=-\operatorname{tr}[C_jC_k]$). □

4.5 Self-testing

The optimal CHSH strategy is essentially unique: up to local isometries, the state is $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\otimes|\text{junk}\rangle$ and A_0, A_1 act as Z, X, etc. We say that CHSH is a (robust) *self-test* for the optimal quantum strategy, or that the optimal strategy is *rigid*. This is remarkable, fundamental, and useful, e.g., for device-independent quantum cryptography. The plan is:

(i) Analyze the (near) equality conditions in the proof of the Tsirelson bound: $A_0A_1 \approx -A_1A_0$ on $|\psi_{AB}\rangle$, etc.

(ii) Show that these imply that Alice and Bob measure (nearly) the same observables as in the optimal strategy.

(iii) Conclude that they also share (nearly) the same state.

Warm-up: the exact case. Suppose $\beta = 1/\sqrt{2}$ exactly. Then we have equality in the Cauchy–Schwarz inequality, so

$$(A_0 \otimes I)|\psi\rangle = \Big(I \otimes \frac{B_0 + B_1}{\sqrt{2}}\Big)|\psi\rangle, \qquad (A_1 \otimes I)|\psi\rangle = \Big(I \otimes \frac{B_0 - B_1}{\sqrt{2}}\Big)|\psi\rangle.$$

Using these relations twice,

$$(\{A_0, A_1\}\otimes I)|\psi\rangle = \Big(I\otimes\tfrac{1}{2}\big[(B_0-B_1)(B_0+B_1)+(B_0+B_1)(B_0-B_1)\big]\Big)|\psi\rangle = \big(I\otimes(B_0^2-B_1^2)\big)|\psi\rangle = 0,$$

where $\{A_0, A_1\} = A_0A_1 + A_1A_0$ is the anti-commutator. Equivalently, $\mathrm{tr}[\{A_0, A_1\}^2\rho_A] = 0$.

Assume for simplicity that ρ_A, ρ_B have full rank. Then $\{A_0, A_1\} = 0$, i.e., $A_0A_1 = -A_1A_0$: A_0 has eigenvalues ±1 and A_1 exchanges the two eigenspaces, which must therefore have the same dimension. Hence (exercise) there is a unitary $U_A : \mathcal{H}_A \to \mathbb{C}^2 \otimes \mathbb{C}^{m_A}$ such that

$$U_A A_0 U_A^\dagger = Z \otimes I, \qquad U_A A_1 U_A^\dagger = X \otimes I,$$

and similarly a unitary U_B for $(B_0 \pm B_1)/\sqrt{2}$. Since $|\psi\rangle = (A_0 \otimes \frac{B_0+B_1}{\sqrt{2}})|\psi\rangle$, the state $|\tilde{\psi}\rangle := (U_A \otimes U_B)|\psi\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^{m_A} \otimes \mathbb{C}^2 \otimes \mathbb{C}^{m_B}$ satisfies (after reordering the tensor factors)

$$|\tilde{\psi}\rangle = (Z \otimes Z \otimes I_{m_A} \otimes I_{m_B})|\tilde{\psi}\rangle, \qquad |\tilde{\psi}\rangle = (X \otimes X \otimes I_{m_A} \otimes I_{m_B})|\tilde{\psi}\rangle.$$

Since $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is the unique vector in $\mathbb{C}^2 \otimes \mathbb{C}^2$ stabilized by $X \otimes X$ and $Z \otimes Z$ (the Bell states form a joint eigenbasis), we conclude $|\tilde{\psi}\rangle = |\Phi^+\rangle \otimes |\text{junk}\rangle$.

Theorem 4.5.1: Self-testing for CHSH

Consider any quantum strategy for CHSH given by a pure state $|\psi_{AB}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ and binary observables $\{A_x\}$, $\{B_y\}$ such that $p_{\text{win}} \geq \frac{1}{2}\big(1 + \frac{1}{\sqrt{2}} - \varepsilon\big)$. Then there are isometries $V_A : \mathcal{H}_A \to \mathbb{C}^2 \otimes \mathcal{H}_{\hat{A}}$, $V_B : \mathcal{H}_B \to \mathbb{C}^2 \otimes \mathcal{H}_{\hat{B}}$ and a state $|\tilde{\psi}\rangle \in \mathcal{H}_{\hat{A}} \otimes \mathcal{H}_{\hat{B}}$ such that

$$\big\|(V_A \otimes V_B)|\psi_{AB}\rangle - |\Phi^+\rangle \otimes |\tilde{\psi}\rangle\big\| = O(\sqrt{\varepsilon}),$$
$$\big\|(V_A \otimes I_B)(A_0 \otimes I_B)|\psi_{AB}\rangle - \big((Z \otimes I)V_A \otimes I_B\big)|\psi_{AB}\rangle\big\| = O(\sqrt{\varepsilon}),$$

and similarly for A_1 (with X) and for Bob's observables. In words: the optimal strategy is robustly unique up to local isometries.

Proof sketch. (i) *Approximate Cauchy–Schwarz.* If $\langle\alpha,\beta\rangle = |\langle\alpha,\beta\rangle| \geq \|\alpha\|\|\beta\|(1-\varepsilon)$, then

$$\Big\| |\alpha\rangle - \tfrac{\|\alpha\|}{\|\beta\|}|\beta\rangle \Big\|^2 = 2\|\alpha\|^2 - 2\tfrac{\|\alpha\|}{\|\beta\|}|\langle\alpha,\beta\rangle| \leq 2\|\alpha\|^2\varepsilon.$$

In our case $|\alpha\rangle = ((A_0\otimes I)|\psi\rangle, (A_1\otimes I)|\psi\rangle)$ and $\|\alpha\|/\|\beta\| = \sqrt{2}/2 = 1/\sqrt{2}$, hence

$$\Big\|(A_0\otimes I)|\psi\rangle - \Big(I\otimes\frac{B_0+B_1}{\sqrt{2}}\Big)|\psi\rangle\Big\| = O(\sqrt{\varepsilon}), \quad \Big\|(A_1\otimes I)|\psi\rangle - \Big(I\otimes\frac{B_0-B_1}{\sqrt{2}}\Big)|\psi\rangle\Big\| = O(\sqrt{\varepsilon}),$$

and as in the warm-up $\|(\{A_0,A_1\}\otimes I)|\psi\rangle\| = O(\sqrt{\varepsilon})$.

(ii) *Construction of the isometries.* Let

$$V_A := \Big(I\otimes\tfrac{1}{2}\big(I\otimes I + Z\otimes A_0 + X\otimes A_1 + ZX\otimes A_0A_1\big)\Big)\big(|\Phi^+\rangle\otimes I_{\mathcal{H}_A}\big) : \mathcal{H}_A \to \mathbb{C}^2\otimes(\mathbb{C}^2\otimes\mathcal{H}_A),$$

i.e., $\mathcal{H}_{\hat{A}} = \mathbb{C}^2\otimes\mathcal{H}_A$. By the transpose trick, equivalently $V_A = \frac{1}{2}(I\otimes I\otimes I + Z\otimes I\otimes A_0 + X\otimes I\otimes A_1 + XZ\otimes I\otimes A_0A_1)(|\Phi^+\rangle\otimes I)$. (If A_0, A_1 were exactly Z, X, this would be a SWAP.) It is an isometry:

$$V_A^\dagger V_A = \tfrac{1}{2}\operatorname{tr}_{\mathbb{C}^2}\Big[\tfrac{1}{4}(\cdots)^\dagger(\cdots)\Big] = \tfrac{1}{8}\cdot 2\cdot\big(I + A_0^2 + A_1^2 + A_1A_0A_0A_1\big) = I,$$

using that I, Z, X, ZX are orthogonal with respect to the Hilbert–Schmidt inner product. Using $A_0^2 = I$ and $A_1A_0|\psi\rangle \approx -A_0A_1|\psi\rangle$ from step 1 (and $XZ = -ZX$), one checks

$$(V_AA_0\otimes I_B)|\psi\rangle \approx \big((Z\otimes I)V_A\otimes I_B\big)|\psi\rangle,$$

while $V_AA_1 = (X\otimes I)V_A$ holds exactly. The isometry V_B is constructed analogously from $(B_0 \pm B_1)/\sqrt{2}$.

(iii) *The state.* Let $|\Omega\rangle := (V_A\otimes V_B)|\psi\rangle$. By steps 1 and 2, $\big((Z\otimes I)\otimes(Z\otimes I)\big)|\Omega\rangle \approx (V_A\otimes V_B)(A_0\otimes\frac{B_0+B_1}{\sqrt{2}})|\psi\rangle \approx |\Omega\rangle$, and similarly with X. Since $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{4}(I\otimes I + X\otimes X + Z\otimes Z + XZ\otimes XZ)$, it follows that $\langle\Omega|(|\Phi^+\rangle\langle\Phi^+|\otimes I)|\Omega\rangle \geq 1 - O(\sqrt{\varepsilon})$, i.e., $|\Omega\rangle \approx |\Phi^+\rangle\otimes|\tilde{\psi}\rangle$. □

4.6 Commuting operator strategies

What is the "right" notion of a quantum strategy—should we allow infinite-dimensional Hilbert spaces? And how can we go beyond XOR games?

Definition 4.6.1: A *(quantum) commuting operator strategy* consists of

- a state ρ on a possibly infinite-dimensional (WLOG separable) Hilbert space $\mathcal{H}$, which need *not* be of the form $\mathcal{H}_A\otimes\mathcal{H}_B$,
- POVMs $\alpha(\cdot|x)$ on $\mathcal{H}$ for $x\in\mathcal{X}$ (Alice) and $\beta(\cdot|y)$ on $\mathcal{H}$ for $y\in\mathcal{Y}$ (Bob),

such that $[\alpha(a|x),\beta(b|y)] = 0$ for all a,b,x,y. WLOG the state is pure and the POVMs are projective. The corresponding *commuting operator correlation* is

$$p(a,b|x,y) = \operatorname{tr}\big[\rho\,\alpha(a|x)\beta(b|y)\big],$$

and the *commuting operator value* $\omega_{qc}(G)$ is the supremum (in fact maximum) of p_{win} over all such strategies.

Remark 4.6.2: (i) p is a correlation since $\alpha(a|x)\beta(b|y) = \sqrt{\alpha}\,\beta\sqrt{\alpha} \geq 0$ (commuting POVMs can be jointly measured). The set of commuting operator correlations is closed.

(ii) Every quantum correlation is a commuting operator correlation, even if $\mathcal{H}_A, \mathcal{H}_B$ are infinite-dimensional; hence $\omega_{qc}(G) \geq \omega_q(G)$. If $\mathcal{H}$ is finite-dimensional, the converse also holds ($\mathcal{H} \cong \bigoplus_i \mathcal{H}_{A_i} \otimes \mathcal{H}_{B_i}$).

(iii) *Nontrivial theorem:* there exists a game G with $\omega_{qc}(G) > \omega_q(G)$.

(iv) For XOR games, $\omega_q = \omega_{qc}$ (since $\omega_q \leq \omega_{qc} \leq \omega_{\text{vec}} = \omega_q$, with the vector bound proven as above).

(v) For general games, studying ω_{qc} often gives the best known upper bounds on ω_q.

4.7 The NPA hierarchy

Goal: a technique for proving (arbitrarily good) upper bounds on ω_{qc}, generalizing the Gram matrix idea from XOR games. Consider a commuting operator strategy with pure state $|\psi\rangle$ and projective POVMs, so

$$\alpha(a|x)\alpha(\tilde{a}|x) = \delta_{a\tilde{a}}\alpha(a|x), \qquad \beta(b|y)\beta(\tilde{b}|y) = \delta_{b\tilde{b}}\beta(b|y), \qquad [\alpha(a|x), \beta(b|y)] = 0.$$

First attempt. Consider the vectors $|v_{a|x}\rangle := \alpha(a|x)|\psi\rangle$ and $|w_{b|y}\rangle := \beta(b|y)|\psi\rangle$ and their Gram matrix M, with rows and columns labeled by $\{(a|x)\} \sqcup \{(b|y)\}$; e.g., $M_{(a|x),(\tilde{a}|\tilde{x})} = \langle\psi|\alpha(a|x)\alpha(\tilde{a}|\tilde{x})|\psi\rangle$. It satisfies:

(1) M is PSD,

(2) $M_{(a|x),(b|y)} = M_{(b|y),(a|x)}$ (commutativity),

(3) $\sum_a M_{*,(a|x)} = \sum_b M_{*,(b|y)}$ for all x, y (both equal the overlap with $|\psi\rangle$)),

(4) $\sum_a M_{(a|x),(a|x)} = 1$ and $\sum_b M_{(b|y),(b|y)} = 1$,

(5) $M_{(a|x),(\tilde{a}|x)} = 0$ if $a \neq \tilde{a}$, and $M_{(b|y),(\tilde{b}|y)} = 0$ if $b \neq \tilde{b}$.

Hence

$$\omega_q \leq \omega_{qc} \leq \max_{M \text{ sat. (1)–(5)}} \sum_{x,y,a,b} p(x,y)\, M_{(a|x),(b|y)}\, W(a,b|x,y),$$

which is a *semidefinite program* (SDP). This is a valid upper bound, tight for XOR games (Tsirelson), but in general not very good. Let us systematically improve it by looking at larger products.

Second attempt. Let $\Sigma := \{(a|x)\}_{a,x} \sqcup \{(b|y)\}_{b,y}$. For every symbol $\sigma \in \Sigma$ we have an operator $\hat{\sigma}$, namely $\alpha(a|x)$ if $\sigma = (a|x)$ and $\beta(b|y)$ if $\sigma = (b|y)$. Let $\Sigma^{\leq L}$ denote the set of strings of length at most L over Σ; for $s = \sigma_1 \cdots \sigma_\ell$ we set $\hat{s} := \hat{\sigma}_1 \cdots \hat{\sigma}_\ell$ (and $\hat{\epsilon} := I$ for the empty string ϵ). Note $\hat{s}^\dagger = \widehat{s^{\text{rev}}}$. The *moment matrix* of level L is the Gram matrix of the vectors $|s\rangle := \hat{s}|\psi\rangle$,

$$M_{s,\tilde{s}} := \langle s|\tilde{s}\rangle = \langle\psi|\widehat{s^{\text{rev}}\tilde{s}}|\psi\rangle \qquad \text{for } s, \tilde{s} \in \Sigma^{\leq L}.$$

It satisfies the following constraints:

(0) $M_{s,\tilde{s}} = M_{t,\tilde{t}}$ if $s^{\text{rev}}\tilde{s} = t^{\text{rev}}\tilde{t}$, i.e., $M_{s,\tilde{s}} = m(s^{\text{rev}}\tilde{s})$ for a function $m : \Sigma^{\leq 2L} \to \mathbb{C}$ with $m(\epsilon) = 1$,

(1) M is PSD,

(2) $m(s\,(a|x)\,(b|y)\,s') = m(s\,(b|y)\,(a|x)\,s')$,

(3) $\sum_a m(s\,(a|x)\,s') = m(ss') = \sum_b m(s\,(b|y)\,s')$,

(4) $m(s\,(a|x)(a|x)\,s') = m(s\,(a|x)\,s')$ and $m(s\,(b|y)(b|y)\,s') = m(s\,(b|y)\,s')$,

(5) $m(s\,(a|x)(\tilde{a}|x)\,s') = 0$ if $a \neq \tilde{a}$, and $m(s\,(b|y)(\tilde{b}|y)\,s') = 0$ if $b \neq \tilde{b}$,

for all strings s, s' (of appropriate length) and all $a, \tilde{a}, b, \tilde{b}, x, y$.

Definition 4.7.1: The *NPA (Navascués–Pironio–Acín) hierarchy* is the sequence of SDPs

$$\omega_{\mathrm{NPA}}^{(L)} := \max_{M \text{ sat. (0)–(5)}} \sum_{x,y,a,b} p(x,y)\, M_{(a|x),(b|y)}\, W(a,b|x,y), \qquad L = 1, 2, \ldots$$

Remark 4.7.2: Since every commuting operator correlation yields a feasible matrix at every level, and since a submatrix of a matrix satisfying the level-L constraints satisfies the level-$(L-1)$ constraints,

$$\omega_q \leq \omega_{qc} \leq \cdots \leq \omega_{\mathrm{NPA}}^{(L)} \leq \omega_{\mathrm{NPA}}^{(L-1)} \leq \cdots \leq \omega_{\mathrm{NPA}}^{(1)}.$$

Each level is an SDP and can be solved efficiently for fixed L (but its size grows exponentially in L).

Theorem 4.7.1: Convergence of the NPA hierarchy

$\lim_{L\to\infty} \omega_{\mathrm{NPA}}^{(L)} = \omega_{qc}$.

Proof sketch. For each level L choose a matrix $M^{(L)}$ achieving $\omega_{\mathrm{NPA}}^{(L)}$. By (0)–(3), $|M^{(L)}_{s,\tilde{s}}|^2 \leq M^{(L)}_{s,s} M^{(L)}_{\tilde{s},\tilde{s}} \leq 1$. Passing to a convergent subsequence (diagonal argument), there is an infinite matrix $M = (M_{s,\tilde{s}})_{s,\tilde{s}\in\Sigma^*}$ indexed by strings of arbitrary finite length with $M_{s,\tilde{s}} = \lim_L M^{(L)}_{s,\tilde{s}} = m(s^{\mathrm{rev}}\tilde{s})$, which satisfies (0)–(5) and all of whose finite submatrices are PSD. Now construct a Hilbert space: on $\mathcal{H}_0 := \mathbb{C}^{(\Sigma^*)} = \{\sum_s c_s|s\rangle : \text{finitely many } c_s \neq 0\}$ define the positive semidefinite form

$$\Big\langle \sum_s c_s|s\rangle, \sum_{\tilde{s}} c'_{\tilde{s}}|\tilde{s}\rangle \Big\rangle := \sum_{s,\tilde{s}} \bar{c}_s c'_{\tilde{s}}\, m(s^{\mathrm{rev}}\tilde{s}),$$

and let $\mathcal{H}$ be the completion of $\mathcal{H}_0/\{|\phi\rangle : \langle\phi,\phi\rangle = 0\}$. Define the state $|\psi\rangle := |\epsilon\rangle$ and operators $\alpha(a|x)|s\rangle := |(a|x)\,s\rangle$, $\beta(b|y)|s\rangle := |(b|y)\,s\rangle$. By (2)–(5) these are commuting projective measurements, and $\langle\psi|\alpha(a|x)\beta(b|y)|\psi\rangle = M_{(a|x),(b|y)}$. Hence $\omega_{qc} \geq \lim_L \omega_{\mathrm{NPA}}^{(L)}$. □

Remark 4.7.3: (i) We can approximate ω_{qc} from above (NPA) and ω_q from below (by enumerating all finite-dimensional strategies). If $\omega_q = \omega_{qc}$, this yields an approximation algorithm.

(ii) But in general $\omega_q < \omega_{qc}$: infinite-dimensional correlations are more powerful. In fact, there *cannot* be any algorithm that computes ω_q or ω_{qc}, even approximately: deciding whether $\omega_q \leq \frac{1}{2}$ or $\omega_q = 1$ is undecidable (and similarly for ω_{qc}).

Theorem 4.7.2: Flatness condition

If for a game there is a level L such that $\omega_{\mathrm{NPA}}^{(L)}$ is achieved by a matrix $M^{(L)}$ satisfying

$$\operatorname{rk} M^{(L)} = \operatorname{rk} M^{(L)}\big|_{\Sigma\leq L-1}, \tag{2}$$

then $\omega_q = \omega_{qc} = \omega_{\mathrm{NPA}}^{(L)} = \sum p(x,y) M^{(L)}_{(a|x),(b|y)} W(a,b|x,y)$ (and $= \omega_{\mathrm{NPA}}^{(L')}$ for all $L' \geq L$).

Proof idea. We only need to construct a *finite-dimensional* commuting operator strategy achieving the right-hand side. Write $M^{(L)}_{s,\tilde{s}} = \langle s|\tilde{s}\rangle$ as a Gram matrix of vectors $\{|s\rangle\}_{s\in\Sigma^{\le L}}$ and let $\mathcal{H} := \mathrm{span}\{|s\rangle\}_{s\in\Sigma^{\le L}}$; by (2), $\mathcal{H} = \mathrm{span}\{|s\rangle\}_{s\in\Sigma^{\le L-1}}$. Set $|\psi\rangle := |\epsilon\rangle$ and define $\alpha(a|x)|s\rangle := |(a|x)\, s\rangle$, $\beta(b|y)|s\rangle := |(b|y)\, s\rangle$ for $s \in \Sigma^{\le L-1}$; this uniquely defines the operators. They are well-defined even though the $|s\rangle$ may be linearly dependent: if $\sum_s c_s|s\rangle = 0$, then $\sum_s c_s|(a|x)s\rangle = 0$, using $M_{s',(a|x)s} = M_{(a|x)s',s}$. □

Remark 4.7.4: Conversely, if there exists an optimal finite-dimensional strategy, then at some point $\mathrm{span}\{|s\rangle : s \in \Sigma^{\le L}\} \subseteq \mathcal{H}$ must stabilize, so there exist L and $M^{(L)}$ as above—but they might not achieve $\omega^{(L)}_{\mathrm{NPA}}$.

5 Semidefinite programming

5.1 Motivating examples

We write $\mathrm{Herm}(n)$ for the Hermitian $n \times n$ matrices, $\mathrm{PSD}(n)$ for the PSD ones, and $\mathcal{S}(n)$ for the density matrices.

Example 5.1.1: P1. The maximal eigenvalue admits the variational characterization $\lambda_{\max}(\rho) = \max_{X\in\mathcal{S}(n)} \mathrm{tr}[X\rho]$.

P2. *Optimal measurement.* Alice prepares $\rho_i \in \mathcal{S}(n)$ with probability p_i $(i = 1, \ldots, m)$ and sends it to Bob, whose goal is to identify i. If Bob applies a measurement $\{\mu_i\}$, his success probability is $\sum_i \Pr[\text{outcome } i \,|\, \rho_i] = \sum_i p_i \,\mathrm{tr}[\rho_i\mu_i]$. The optimal success probability is

$$\max \sum_{i=1}^{m} p_i \,\mathrm{tr}[\rho_i\mu_i] \quad \text{s.t.} \quad \sum_{i=1}^{m} \mu_i = I,\ \mu_1, \ldots, \mu_m \in \mathrm{PSD}(n).$$

P3. By Tsirelson's theorem, the optimal bias of an XOR game with game matrix G is

$$\beta_q = \max \sum_{x,y} G_{xy}P_{xy} \quad \text{s.t.} \quad P_{xx} = P_{yy} = 1\ \forall x \in \mathcal{X}, y \in \mathcal{Y},\ P \in \mathrm{PSD}(|\mathcal{X}| + |\mathcal{Y}|),$$

where we index the PSD (Gram) matrix P by $\mathcal{X} \sqcup \mathcal{Y}$.

All three problems maximize a *linear* function over *PSD matrices* subject to *linear constraints*: they are semidefinite programs.

5.2 Primal and dual SDPs

Definition 5.2.1: The standard form of a *primal SDP* is

$$\text{(P)} \qquad \max\ \mathrm{tr}(CX) \quad \text{s.t.} \quad \mathrm{tr}(A_iX) = b_i\ (i = 1, \ldots, m), \quad X \in \mathrm{PSD}(n),$$

where $C, A_i \in \mathrm{Herm}(n)$ and $b \in \mathbb{R}^m$. The corresponding *dual SDP* is

$$\text{(D)} \qquad \min \sum_i y_i b_i \quad \text{s.t.} \quad \sum_i y_i A_i \ge C, \quad y \in \mathbb{R}^m.$$

We denote the optimal values by P_{opt} and D_{opt}.

Remark 5.2.2: An affine slice of the convex cone $\mathrm{PSD}(n)$ is called a *spectrahedron*; it is again convex. SDPs are optimizations of linear functions over spectrahedra.

Lemma 5.2.3 (Weak duality): $P_{\mathrm{opt}} \le D_{\mathrm{opt}}$. In particular, if X and y are feasible and $\mathrm{tr}[CX] = \sum_i y_i b_i$, then both are optimal.

Proof. For feasible X, y: $\operatorname{tr}(CX) \le \operatorname{tr}\big((\sum_i y_i A_i)X\big) = \sum_i y_i \operatorname{tr}(A_i X) = \sum_i y_i b_i$, since $\operatorname{tr}[(\sum_i y_i A_i - C)X] \ge 0$ for PSD matrices. □

Example 5.2.4: For $C \in \mathrm{Herm}(n)$ consider

$$\text{(P)}\ \max \operatorname{tr}[CX] \text{ s.t. } \operatorname{tr} X = 1,\ X \in \mathrm{PSD}(n), \qquad \text{(D)}\ \min y \text{ s.t. } yI \ge C.$$

The feasible values of (D) are $[\lambda_{\max}(C), \infty)$. If $C = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ with $p_1 \ge \cdots \ge p_n$, then $X = |\psi_1\rangle\langle\psi_1|$ is feasible with $\operatorname{tr}(CX) = p_1 = \lambda_{\max}(C)$. By weak duality the optimal value is $\lambda_{\max}(C)$. In fact, the feasible values of (P) are $[\lambda_{\min}(C), \lambda_{\max}(C)]$.

Definition 5.2.5: We say that *strong duality* holds if $P_{\text{opt}} = D_{\text{opt}}$ (the typical case).

Theorem 5.2.1: Slater's conditions

Strong duality holds in either of the following cases:

(i) (P) is feasible and there exists y with $\sum_i y_i A_i > C$ (strict dual feasibility),

(ii) (D) is feasible and there exists $X > 0$ with $\operatorname{tr}(A_i X) = b_i$ for all i (strict primal feasibility).

5.3 XOR games as SDPs

Using that any PSD matrix is a Gram matrix, Tsirelson's theorem gives (with real G)

$$\beta_q = \max\ \operatorname{tr}\left(\begin{pmatrix} 0 & \frac{1}{2}G \\ \frac{1}{2}G^T & 0 \end{pmatrix} Z\right) \quad \text{s.t.} \quad Z = \begin{pmatrix} P & X \\ X^T & Q \end{pmatrix} \ge 0, \quad Z_{xx} = 1,\ Z_{yy} = 1\ \ (x \in \mathcal{X}, y \in \mathcal{Y}),$$

with dual SDP

$$\min \sum_{x \in \mathcal{X}} a_x + \sum_{y \in \mathcal{Y}} b_y \quad \text{s.t.} \quad \begin{pmatrix} \operatorname{diag}(a) & -\frac{1}{2}G \\ -\frac{1}{2}G^T & \operatorname{diag}(b) \end{pmatrix} \ge 0.$$

Lemma 5.3.1: Let $\lambda \ge 0$ and let $X \in \mathbb{R}^{n \times n}$ satisfy $XX^T = \mu I_n$. Then $\begin{pmatrix} \lambda I_n & X \\ X^T & \lambda I_n \end{pmatrix} \ge 0$ if and only if $\mu \le \lambda^2$.

Proof. The condition is equivalent to $\lambda I_{2n} \ge \begin{pmatrix} 0 & -X \\ -X^T & 0 \end{pmatrix} =: R$. Since $R^2 = \mu I_{2n}$ and $\operatorname{tr} R = 0$, R has eigenvalues $\pm\sqrt{\mu}$ (each with multiplicity n). Hence the condition holds iff $\sqrt{\mu} \le \lambda$. □

Example 5.3.2 (CHSH): For $G = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, the primal choice $P = Q = I_2$, $X = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is feasible (here $XX^T = I$) with value $1/\sqrt{2}$. The dual choice $a_x = b_y = \frac{1}{4\sqrt{2}}$ is feasible (here $\frac{1}{4}GG^T = \frac{1}{32}I$ and $\lambda^2 = \frac{1}{32}$) with value $4 \cdot \frac{1}{4\sqrt{2}} = \frac{1}{\sqrt{2}}$. Hence $\beta_q = P_{\text{opt}} = D_{\text{opt}} = 1/\sqrt{2}$.

5.4 Fidelity as an SDP

Recall $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1 = \max_U |\operatorname{tr}[\sqrt{\rho}\sqrt{\sigma}U]|$.

Theorem 5.4.1: Fidelity SDP

For states ρ, σ,

$$F(\rho,\sigma) = \max\ \tfrac{1}{2}\operatorname{tr}(X) + \tfrac{1}{2}\operatorname{tr}(X^\dagger) \quad \text{s.t.} \quad W := \begin{pmatrix} \rho & X \\ X^\dagger & \sigma \end{pmatrix} \geq 0.$$

This is an SDP: the objective is $\operatorname{tr}\left[\begin{pmatrix} 0 & \frac{1}{2}I \\ \frac{1}{2}I & 0 \end{pmatrix} W\right]$ and the constraints fix the diagonal blocks of $W \geq 0$. Its dual is (after a change of variables)

$$\min\ \tfrac{1}{2}\operatorname{tr}(Y\rho) + \tfrac{1}{2}\operatorname{tr}(Z\sigma) \quad \text{s.t.} \quad \begin{pmatrix} Y & -I \\ -I & Z \end{pmatrix} \geq 0.$$

Proof. It is enough to prove this for $\rho, \sigma > 0$ (by continuity). *Primal:* Choose a unitary U with $F(\rho,\sigma) = \operatorname{tr}[\sqrt{\rho}\sqrt{\sigma}U]$ (multiply U by a phase if necessary) and set $X := \sqrt{\rho}\,U\sqrt{\sigma}$. Then

$$\begin{pmatrix} \sqrt{\rho} & U\sqrt{\sigma} \end{pmatrix}^\dagger \begin{pmatrix} \sqrt{\rho} & U\sqrt{\sigma} \end{pmatrix} = \begin{pmatrix} \rho & \sqrt{\rho}U\sqrt{\sigma} \\ \sqrt{\sigma}U^\dagger\sqrt{\rho} & \sigma \end{pmatrix} \geq 0$$

is feasible, with value $\frac{1}{2}\operatorname{tr} X + \frac{1}{2}\overline{\operatorname{tr} X} = \operatorname{tr}[\sqrt{\rho}U\sqrt{\sigma}] = F(\rho,\sigma)$. Hence $F \leq P_{\text{opt}}$.

Dual: If $Y > 0$, then $\begin{pmatrix} Y & -I \\ -I & Y^{-1} \end{pmatrix} \geq 0$, since

$$\begin{pmatrix} I & 0 \\ -Y^{-1} & I \end{pmatrix} \begin{pmatrix} Y & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} I & -Y^{-1} \\ 0 & I \end{pmatrix} = \begin{pmatrix} Y & -I \\ -I & Y^{-1} \end{pmatrix}.$$

Set $Y := \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}$ and $Z := Y^{-1} = \rho^{1/2}(\rho^{1/2}\sigma\rho^{1/2})^{-1/2}\rho^{1/2}$. Then

$D_{\text{opt}} \leq \frac{1}{2}\operatorname{tr}[Y\rho] + \frac{1}{2}\operatorname{tr}[Z\sigma] = \frac{1}{2}\operatorname{tr}\left[(\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}})^{\frac{1}{2}}\right] + \frac{1}{2}\operatorname{tr}\left[\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}}(\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}})^{-\frac{1}{2}}\right] = \operatorname{tr}\left[(\rho^{\frac{1}{2}}\sigma\rho^{\frac{1}{2}})^{\frac{1}{2}}\right] = F(\rho,\sigma)$,

using $\|X\|_1 = \operatorname{tr}\sqrt{XX^\dagger}$. Together with weak duality, $D_{\text{opt}} \leq F \leq P_{\text{opt}} \leq D_{\text{opt}}$, so all are equal. □

6 Quantum channels

6.1 Channels in a classical world

A classical (memoryless) channel with input $x \in \Sigma_X$ and output $y \in \Sigma_Y$ is described by a *conditional probability distribution* (or *transition matrix*)

$$p(y|x) = \Pr(\text{output } y \,|\, \text{input } x), \qquad p(y|x) \geq 0, \quad \sum_y p(y|x) = 1.$$

If the input has distribution $p(x)$, the output distribution is $p(y) = \sum_x p(y|x)p(x)$, i.e., $p_Y = p_{Y|X} \cdot p_X$ (matrix-vector multiplication); in particular $p_X \mapsto p_Y$ is linear. The joint distribution is $p(x,y) = p(x)p(y|x)$.

Example 6.1.1: (i) *Binary symmetric channel:* flips the bit with probability ε, i.e., $p(0|0) = p(1|1) = 1-\varepsilon$, $p(1|0) = p(0|1) = \varepsilon$.

(ii) *Binary erasure channel:* loses the bit with probability ε, i.e., $p(0|0) = p(1|1) = 1-\varepsilon$, $p(\perp|0) = p(\perp|1) = \varepsilon$.

Just like probability distributions describe the most general states, channels describe the most general operations. Historically, they were introduced to study reliable communication in the presence of uncertainty and noise. Two channels used in parallel are described by $p(y,w|x,z) = p_1(y|x)p_2(w|z)$, which is again a channel.

6.2 Superoperators

Since quantum states are operators, a quantum channel should be a map $\Phi : \mathrm{L}(\mathcal{H}_A) \to \mathrm{L}(\mathcal{H}_B)$. We abbreviate $\mathrm{L}(A) := \mathrm{L}(\mathcal{H}_A)$.

Definition 6.2.1: A *superoperator* is a linear map $\Phi \in \mathrm{L}(\mathrm{L}(A), \mathrm{L}(B))$, written $\Phi_{A\to B}$. Linearity ensures that $\Phi[\sum_i p_i \rho_i] = \sum_i p_i \Phi[\rho_i]$. The *identity* is $\mathcal{I}_A[M_A] = M_A$. The *tensor product* of superoperators is defined by

$$(\Phi_{A\to B} \otimes \Psi_{C\to D})[M_A \otimes N_C] := \Phi_{A\to B}[M_A] \otimes \Psi_{C\to D}[N_C],$$

extended by linearity.

A first attempt at the conditions a channel should satisfy: (1) *positivity*, $M \geq 0 \Rightarrow \Phi[M] \geq 0$; (2) *trace preservation*, $\operatorname{tr} \Phi[M] = \operatorname{tr} M$. Together, these imply that states are mapped to states. However, this is not good enough:

Example 6.2.2 (Transpose map): The transpose map $\mathcal{T}[M] := M^T$ (with respect to a fixed basis) satisfies (1) and (2). But applied to half of the EPR pair,

$$(\mathcal{T} \otimes \mathcal{I}_R)\big[|\Phi^+\rangle\langle\Phi^+|\big] = \tfrac{1}{2}\big(|00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|\big) = \tfrac{1}{2}\,\mathrm{SWAP},$$

which has trace one but the eigenvalue $-\frac{1}{2}$ (eigenvector $|01\rangle - |10\rangle$). So it does not map states to states!

Definition 6.2.3: A *quantum channel* is a superoperator $\Phi_{A\to B}$ such that

(i) Φ is *completely positive*: for all $\mathcal{H}_R$ and $M_{AR} \geq 0$, $(\Phi_{A\to B} \otimes \mathcal{I}_R)[M_{AR}] \geq 0$,

(ii) Φ is *trace-preserving*: $\operatorname{tr} \Phi_{A\to B}[M_A] = \operatorname{tr} M_A$ for all M_A.

We write $\mathrm{C}(A, B) := \mathrm{CP}(A, B) \cap \mathrm{TP}(A, B)$ and $\mathrm{C}(A) := \mathrm{C}(A, A)$.

Example 6.2.4: The following are quantum channels (exercise/homework):

(i) *Base change:* $\Phi[\rho] = U\rho U^\dagger$ for a unitary or isometry U.

(ii) *Adding a state:* $\Phi[\rho_A] = \rho_A \otimes \sigma_B$ for a state σ_B.

(iii) *Partial trace:* $\Phi = \operatorname{tr}_A$.

(iv) *Measurement channel:* $\Phi[\rho] = \sum_x \langle x|\rho|x\rangle\, |x\rangle\langle x|$, or more generally $\Phi[\rho] = \sum_x \operatorname{tr}[\rho\, \mu(x)]|x\rangle\langle x|$ for a POVM μ; the output is classical.

(v) *Measure-and-prepare:* $\Phi[\rho] = \sum_\omega \operatorname{tr}[\rho\, \mu(\omega)]\, \sigma_\omega$.

(vi) *Qubit flip channel* $\Phi_f[\rho] = (1-f)\rho + fX\rho X$, *dephasing channel* $\Phi_f[\rho] = (1-f)\rho + fZ\rho Z$, and *depolarizing channel* $\mathcal{D}_f[\rho] = (1-f)\rho + f\ \operatorname{tr}[\rho]\, I/d$ on $\mathbb{C}^d$.

Moreover, if $\Phi_{A\to B}$, $\Psi_{C\to D}$, $\Phi'_{B\to C}$ are channels, so are $\Phi \otimes \Psi$ and $\Phi' \circ \Phi$. We will see that every channel is a composition of the examples above (isometries and partial traces suffice).

6.3 The Choi operator

Definition 6.3.1: The *Choi operator* of a superoperator $\Phi_{A\to B}$ (with respect to a fixed basis $\{|x\rangle\}$ of $\mathcal{H}_A$) is

$$J^\Phi_{AB} := \sum_{x,y} |x\rangle\langle y| \otimes \Phi[|x\rangle\langle y|] = (\mathcal{I}_A \otimes \Phi_{A\to B})\Big[\sum_{x,y} |xx\rangle\langle yy|\Big] \in \mathrm{L}(\mathcal{H}_A \otimes \mathcal{H}_B),$$

i.e., Φ applied to half of the unnormalized maximally entangled state. The normalized version $\omega^\Phi_{AB} := J^\Phi_{AB}/d_A = (\mathcal{I}_A \otimes \Phi)[|\Phi^+\rangle\langle\Phi^+|]$ is called the *Choi state*.

Example 6.3.2: For the *completely dephasing channel* (basis measurement) $\Delta[\rho] = \sum_z \langle z|\rho|z\rangle |z\rangle\langle z|$, we get

$$J^\Delta = \sum_{x,y} |x\rangle\langle y| \otimes \Delta[|x\rangle\langle y|] = \sum_x |x\rangle\langle x| \otimes |x\rangle\langle x|.$$

Lemma 6.3.3 (Choi–Jamiołkowski isomorphism): The map $\Phi \mapsto J^\Phi$ is an isomorphism $\mathrm{L}(\mathrm{L}(A), \mathrm{L}(B)) \to \mathrm{L}(\mathcal{H}_A \otimes \mathcal{H}_B)$, with inverse

$$\Phi_{A\to B}[M_A] = \mathrm{tr}_A\big[(M_A^T \otimes I_B) J^\Phi_{AB}\big].$$

Proof.

$$\mathrm{tr}_A[(M^T \otimes I) J^\Phi] = \sum_{x,y} \langle x|M|y\rangle\, \Phi[|x\rangle\langle y|] = \Phi\Big[\sum_{x,y} |x\rangle\langle x|M|y\rangle\langle y|\Big] = \Phi[M]. \qquad \square$$

6.4 Characterization of complete positivity

Theorem 6.4.1: Complete positivity

For a superoperator $\Phi_{A\to B}$, the following are equivalent:

(i) $\Phi_{A\to B}$ is completely positive,

(ii) $\Phi_{A\to B} \otimes \mathcal{I}_A$ is positive (it suffices to check R = a copy of A),

(iii) $J^\Phi_{AB} \geq 0$,

(iv) *Kraus representation:* there exist $K_1, \ldots, K_r \in \mathrm{L}(\mathcal{H}_A, \mathcal{H}_B)$ (*Kraus operators*) such that $\Phi[M] = \sum_i K_i M K_i^\dagger$ for all M,

(v) *Stinespring representation:* there exist $\mathcal{H}_E$ and $V \in \mathrm{L}(\mathcal{H}_A, \mathcal{H}_B \otimes \mathcal{H}_E)$ such that $\Phi[M] = \mathrm{tr}_E[VMV^\dagger]$ for all M (*Stinespring extension*).

Moreover, in (iv) and (v), r and $\dim \mathcal{H}_E$ can be chosen equal to $\mathrm{rk}\, J^\Phi_{AB} \leq d_A d_B$.

Proof. (i)⇒(ii)⇒(iii): Clear, since $J^\Phi = (\mathcal{I}_A \otimes \Phi)[\sum_{x,y} |xx\rangle\langle yy|]$ and $\sum_{x,y} |xx\rangle\langle yy| \geq 0$.

(iii)⇒(iv): Write $J^\Phi = \sum_i |v_i\rangle\langle v_i|$ with $|v_i\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ (spectral decomposition; $\|v_i\|^2$ are the eigenvalues). Turn the vectors into operators, $K_i := \sum_{a,b} \langle ab|v_i\rangle |b\rangle\langle a| \in \mathrm{L}(\mathcal{H}_A, \mathcal{H}_B)$. Then

$$\begin{aligned}\Phi[M] &= \sum_i \mathrm{tr}_A\big[(M^T \otimes I)|v_i\rangle\langle v_i|\big] = \sum_i \sum_{a,b,\tilde a,\tilde b} \langle ab|v_i\rangle\langle v_i|\tilde a\tilde b\rangle\, \mathrm{tr}_A\big[(M^T \otimes I)|ab\rangle\langle \tilde a\tilde b|\big] \\ &= \sum_i \sum_{a,b,\tilde a,\tilde b} \langle ab|v_i\rangle\langle v_i|\tilde a\tilde b\rangle\, \langle a|M|\tilde a\rangle\, |b\rangle\langle\tilde b| = \sum_i K_i M K_i^\dagger.\end{aligned}$$

(iv)⇒(v): Let $\mathcal{H}_E := \mathbb{C}^r$ and $V := \sum_{i=1}^r K_i \otimes |i\rangle$. Then $\mathrm{tr}_E[VMV^\dagger] = \sum_{i,j} K_i M K_j^\dagger \,\mathrm{tr}[|i\rangle\langle j|] = \sum_i K_i M K_i^\dagger$. (Conversely, (v)⇒(iv) with $K_i := (I_B \otimes \langle i|)V$.)

(v)⇒(i): Both $M \mapsto VMV^\dagger$ and tr_E are completely positive. □

Example 6.4.1: The completely dephasing channel Δ has Kraus operators $K_z = |z\rangle\langle z|$ and Stinespring isometry $V = \sum_z |zz\rangle\langle z| = \sum_z |z\rangle\langle z| \otimes |z\rangle$ with $\mathcal{H}_E = \mathcal{H}_A$. In particular, Δ is completely positive.

6.5 Trace preservation and Stinespring dilations

Lemma 6.5.1: A completely positive superoperator $\Phi_{A\to B}$ is trace-preserving if and only if

(i) (Choi) $\mathrm{tr}_B[J^\Phi_{AB}] = I_A$,

(ii) (Kraus) $\sum_i K_i^\dagger K_i = I_A$, for one/any Kraus representation,

(iii) (Stinespring) $V^\dagger V = I_A$, i.e., V is an isometry, for one/any Stinespring extension.

Proof. Kraus: $\mathrm{tr}[M_A] = \mathrm{tr}\,\Phi[M_A] = \mathrm{tr}[\sum_i K_i M_A K_i^\dagger] = \mathrm{tr}[M_A \sum_i K_i^\dagger K_i]$ for all M_A iff $\sum_i K_i^\dagger K_i = I_A$. Stinespring: $\mathrm{tr}[VM_AV^\dagger] = \mathrm{tr}[M_AV^\dagger V]$. Choi: similar. □

Remark 6.5.2 (Stinespring dilation): Thus every quantum channel can be written as (1) an isometry $V_{A\to BE}$ followed by (2) forgetting (tracing out) the subsystem E. Equivalently (homework), it can be implemented by (1) adding a pure state $|0\rangle_C$, (2) applying a unitary U on $AC \cong BF$, and (3) forgetting F. (Idea: if $\mathcal{K} \subseteq \mathcal{H}$, any isometry $\mathcal{K} \to \mathcal{H}$ can be extended to a unitary $\mathcal{H} \to \mathcal{H}$.) This is a good justification of our definition of channels.

6.6 Adjoints and the Heisenberg picture

Recall that $\mathrm{L}(\mathcal{H})$ is itself a Hilbert space with the Hilbert–Schmidt inner product $(X, Y) = \mathrm{tr}[X^\dagger Y]$. Hence every superoperator $\Phi_{A\to B}$ has an *adjoint* $\Phi^\dagger_{B\to A} := (\Phi_{A\to B})^\dagger \in \mathrm{L}(\mathrm{L}(B), \mathrm{L}(A))$, given by

$$\mathrm{tr}\big[(\Phi^\dagger[M_B])^\dagger N_A\big] = \mathrm{tr}\big[M_B^\dagger\, \Phi[N_A]\big] \qquad \text{for all } M_B, N_A.$$

If Φ is Hermiticity-preserving, $\Phi[M^\dagger] = \Phi[M]^\dagger$ for all M (e.g., if Φ is CP or a difference of CP maps), then the same holds for $\Phi^\dagger$, and $\Phi^\dagger$ is characterized by

$$\mathrm{tr}\big[\Phi_{A\to B}[M_A]\, N_B\big] = \mathrm{tr}\big[M_A\, \Phi^\dagger_{B\to A}[N_B]\big] \qquad \text{for all } M_A, N_B. \tag{3}$$

Lemma 6.6.1: (i) If Φ has Kraus operators $\{K_j\}$, then $\Phi^\dagger$ has Kraus operators $\{K_j^\dagger\}$. Hence Φ is CP if and only if $\Phi^\dagger$ is CP. Also $(\Phi^\dagger)^\dagger = \Phi$.

(ii) Φ is trace-preserving if and only if $\Phi^\dagger$ is *unital*, i.e., $\Phi^\dagger[I] = I$.

Proof. (i): $\mathrm{tr}[\sum_j K_j M K_j^\dagger N] = \mathrm{tr}[M \sum_j K_j^\dagger N K_j]$. (ii): $\mathrm{tr}\,\Phi[M_A] = \mathrm{tr}[\Phi^\dagger[I]M_A]$ equals $\mathrm{tr}[M_A] = \mathrm{tr}[I\, M_A]$ for all M_A iff $\Phi^\dagger[I] = I$. □

Remark 6.6.2 (Interpretation): If Φ is a quantum channel, it describes what happens to states (*Schrödinger picture*), while $\Phi^\dagger$ describes what happens to observables (*Heisenberg picture*). Indeed, (3) reads

$$\mathrm{tr}\big[\Phi[\rho_A]\, X_B\big] = \mathrm{tr}\big[\rho_A\, \Phi^\dagger[X_B]\big] :$$

we may either evolve the state "forward" or the observable (e.g., a POVM element) "backwards".

7 No-cloning and quantum money

7.1 No-cloning for grown-ups

Suppose someone hands us a box that they claim can perfectly clone qubit states, i.e., output $|\psi\rangle \otimes |\psi\rangle$ when given an arbitrary $|\psi\rangle \in \mathbb{C}^2$. We can test their claim with the following *cloning test*:

(i) Initialize a qubit in a state $|\psi\rangle$ chosen uniformly at random from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$.

(ii) Input it into the box.

(iii) Measure each of the two output qubits in the corresponding basis (standard basis for $|0\rangle, |1\rangle$, Hadamard basis for $|\pm\rangle$).

(iv) Accept if both outcomes are as they should be.

A perfect cloner would pass this test with probability one. How well can one do? We need to consider all possible quantum channels $\Phi_{A\to BC}$, where A, B, C are qubits.

Theorem 7.1.1: Quantitative no-cloning

For every channel $\Phi_{A\to BC}$, $\Pr(\text{passing the cloning test}) \le \frac{3}{4}$. This bound is tight.

Proof. Using the inverse of the Choi–Jamiołkowski isomorphism,

$$\begin{aligned}\Pr(\text{pass} \,|\, \text{input } \psi) &= \operatorname{tr}\Big[(|\psi\rangle\langle\psi|_B \otimes |\psi\rangle\langle\psi|_C)\ \operatorname{tr}_A\big[(|\psi\rangle\langle\psi|_A^T \otimes I_{BC}) J_{ABC}^{\Phi}\big]\Big] \\ &= \operatorname{tr}\big[(|\psi\rangle\langle\psi|^T \otimes |\psi\rangle\langle\psi| \otimes |\psi\rangle\langle\psi|) J_{ABC}^{\Phi}\big].\end{aligned}$$

All four states are real, so $|\psi\rangle\langle\psi|^T = |\psi\rangle\langle\psi|$. Averaging,

$$\begin{aligned}\Pr(\text{pass}) &= \operatorname{tr}\Big[\underbrace{\tfrac{1}{2}\big(|000\rangle\langle 000| + |111\rangle\langle 111| + |{+}{+}{+}\rangle\langle {+}{+}{+}| + |{-}{-}{-}\rangle\langle {-}{-}{-}|\big)}_{=:Q_{ABC}\ \ge 0}\ \underbrace{\tfrac{1}{2} J_{ABC}^{\Phi}}_{=\omega_{ABC}^{\Phi}}\Big] \\ &\le \|Q\|_\infty \|\omega^{\Phi}\|_1 = \lambda_{\max}(Q) = \tfrac{3}{4},\end{aligned}$$

where ω^{Φ} is the Choi state and the last step is a computation with an 8×8 matrix. Tightness is a homework problem. □

Remark 7.1.1: This generalizes to n-qubit states $|\Psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$ with each $|\psi_j\rangle \in \{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ chosen uniformly at random: if we input $|\Psi\rangle$ into any channel Φ from n to $2n$ qubits and measure each qubit of both outputs in the appropriate basis,

$$\Pr(\text{pass}) \le \left(\frac{3}{4}\right)^n, \tag{4}$$

since the relevant operator is $Q^{\otimes n}$ and $\|Q^{\otimes n}\|_\infty = \|Q\|_\infty^n$.

7.2 Wiesner's quantum money

Wiesner's idea (1970s): the no-cloning principle should allow one to create and verify banknotes that are impossible to forge. Nowadays such schemes are referred to as *quantum money*. A bill consists of a classical serial number s and an n-qubit quantum state $|\Psi\rangle$, where n plays the role of a *security parameter*.

Issuing bills. The central bank chooses a serial number s (e.g., a random bit string of length n) as well as a random state $|\Psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$, where each $|\psi_j\rangle$ is chosen independently and uniformly at random from $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. The bank issues the bill $(s, |\Psi\rangle)$ and stores s together with a classical description of Ψ (a string in $\{0, 1, +, -\}^n$) privately in its vault.

Verifying bills. Given a purported bill $(s, |\Psi\rangle)$, the bank looks up the description of the state corresponding to s, measures each qubit of $|\Psi\rangle$ in the corresponding basis, and checks that each outcome is as expected. This is exactly the test in (4), applied to the n qubits of the bill.

7.3 Security

Is the scheme secure? First we need to define which notion of security we care about. Typically this is phrased in terms of *security games* between honest parties (e.g., the bank) and adversaries (e.g., a counterfeiter).

Definition 7.3.1 (Simple counterfeiting attack): The bank issues a bill $(s, |\Psi\rangle)$ to the counterfeiter, who outputs two n-qubit registers (s, B), (s, C). The counterfeiter wins if both are accepted by the bank.

Proposition 7.3.2: For every counterfeiting strategy,

$$\Pr(\text{both bills accepted}) = \Pr(\text{passing the cloning test}) \leq (3/4)^n.$$

Hence the scheme is secure against such attacks.

Remark 7.3.3: Is this the "right" definition of security? Let us try to poke holes:

(i) Why not allow the counterfeiter to output (possibly distinct) serial numbers? Good point, the game should be modified to allow this (to make it nontrivial, the bank needs to issue more bills at the beginning). One can show that the probability of guessing a valid distinct serial number correctly is exponentially small.

(ii) The game assumes there is only one money bill in the world (in the counterfeiter's hands). One could instead give the counterfeiter $m = \text{poly}(n)$ bills and ask them to output $m + 1$ valid bills. This is still secure.

(iii) The counterfeiter interacts with the bank only once. What if the bank can be queried whether a purported bill is valid, and returns the (post-measurement) bill even if it is illegitimate? Such a *verification oracle* is not realistic in the bank model, but it is motivated by decentralized settings where bills can be verified without going to the bank ("public-key quantum money").

7.4 An attack using a verification oracle

Key idea: since $|\Psi\rangle$ is a product state, correct qubits are left unchanged by the verification even if other qubits are wrong. So we can repeatedly modify individual qubits and verify, in order to learn $|\Psi\rangle$.

Proposition 7.4.1: Given a valid bill $(s, |\Psi\rangle)$ and access to a verification oracle, there is an attack which outputs the classical description of Ψ with failure probability $\tilde{O}(2^{-n})$ using $O(n^2)$ oracle queries.

Proof. For $j = 1, \ldots, n$ and for each $\phi \in \{0, 1, +, -\}$, repeat R times: replace the j-th qubit by a fresh qubit in state $|\phi\rangle$ and query the verification oracle. If the oracle accepted all R times,

conclude $|\psi_j\rangle = |\phi\rangle$ and continue with the next j. (Finally, restore the bill using the learned description.)

If we replace the j-th qubit of a valid bill by $|\phi\rangle$ and verify, then $\Pr(\text{accept}) = 1$ if $|\phi\rangle = |\psi_j\rangle$, while $\Pr(\text{accept}) \leq \frac{1}{2}$ if $|\phi\rangle \neq |\psi_j\rangle$, and in any case only the j-th qubit changes. Since orthogonal states are always rejected, only two wrong candidates can be accepted, so

$$\Pr(\text{incorrectly determine } \psi_j) \leq 2{\cdot}(1/2)^R, \qquad \Pr(\text{incorrectly determine } \Psi) \leq O(n2^{-R}) = \tilde{O}(2^{-n})$$

for $R = n$, with $4nR = O(n^2)$ queries. □

Remark 7.4.2 (Lessons learned): Finding the right security definition is subtle (and there are many options). Public-key quantum money needs entanglement and computational assumptions (i.e., security against efficient attackers).

8 Decoupling and quantum error correction

8.1 Recoverability and the decoupling principle

When can we "undo" a quantum channel? Motivation: Alice wants to send a qubit to Bob but the channel is noisy; a quantum computer sometimes introduces errors; in teleportation, Bob can perfectly recover Alice's qubit given his half of the EPR pair and two classical bits—why?

Definition 8.1.1: A channel $\mathcal{T}_{A\to B}$ can be *recovered* (or *undone*) if there exists a channel $\mathcal{R}_{B\to A}$ (the *recovery channel*) such that $\mathcal{R}_{B\to A} \circ \mathcal{T}_{A\to B} = \mathcal{I}_A$.

Definition 8.1.2: Let $V_{A\to BE}$ be a Stinespring isometry of $\mathcal{T}_{A\to B}$, i.e., $\mathcal{T}[\rho_A] = \operatorname{tr}_E[V\rho_A V^\dagger]$. The *complementary channel* is

$$\mathcal{T}^c_{A\to E}[\rho_A] := \operatorname{tr}_B\big[V\rho_A V^\dagger\big].$$

It describes everything an eavesdropper or the environment can possibly learn. It depends on the choice of Stinespring extension, but only up to an isometry on E, which is not a problem.

Theorem 8.1.1: Decoupling principle

The following are equivalent:

(i) $\mathcal{T}_{A\to B}$ can be recovered,

(ii) the Choi state of $\mathcal{T}^c$ is a product: $\omega^{\mathcal{T}^c}_{AE} = \omega_A \otimes \omega_E$,

(iii) $\mathcal{T}^c_{A\to E}$ is a constant channel, i.e., $\mathcal{T}^c[\rho_A]$ is the same for all states ρ_A.

Proof. (ii)⇔(iii) follows from the Choi–Jamiołkowski isomorphism.

Key observation. Let $\mathcal{R}_{B\to A}$ be a channel with Stinespring isometry $W_{B\to A'F}$. Then WV is a Stinespring isometry of $\mathcal{R}\circ\mathcal{T}$ (with environment FE), and since extensions of pure states are product states,

$$\mathcal{R}\circ\mathcal{T} = \mathcal{I}_A \iff (I_A \otimes W_{B\to A'F}V_{A\to BE})|\Phi^+_{AA}\rangle = |\Phi^+_{AA'}\rangle \otimes |\psi_{EF}\rangle \text{ for some } |\psi_{EF}\rangle. \quad (5)$$

(i)⇒(ii): If $\mathcal{R}$ is a recovery channel, then by (5), $\omega^{\mathcal{T}^c}_{AE} = \operatorname{tr}_{A'F}\big[|\Phi^+_{AA'}\rangle\langle\Phi^+_{AA'}| \otimes |\psi_{EF}\rangle\langle\psi_{EF}|\big] = \frac{I_A}{|A|} \otimes \operatorname{tr}_F|\psi_{EF}\rangle\langle\psi_{EF}|$ is a product.

(ii)⇒(i): Consider the pure Choi state $|\Omega_{ABE}\rangle := (I_A \otimes V)|\Phi^+_{AA}\rangle$ of the isometry V. It is a purification of $\omega_{AE} = \omega^{\mathcal{T}^c}_{AE}$, and $\omega_A = I_A/|A|$ is maximally mixed. Since $\omega_{AE} = \omega_A \otimes \omega_E$, another purification is $|\tilde{\Omega}_{AA'EF}\rangle := |\Phi^+_{AA'}\rangle \otimes |\psi_{EF}\rangle$, where $|\psi_{EF}\rangle$ is any purification of ω_E. WLOG $|B| \leq |A'F|$; by uniqueness of purifications there is an isometry $W_{B\to A'F}$ such that $|\tilde{\Omega}_{AA'EF}\rangle = (I_{AE} \otimes W_{B\to A'F})|\Omega_{ABE}\rangle$. Thus (5) holds, and W is the Stinespring isometry of a recovery map. □

8.2 Application: quantum erasure codes

Suppose we want to encode one "logical" qudit L into 3 "physical" qudits $P_1P_2P_3$ such that we can correct for the loss of any single qudit (*erasure model*: we know which one got lost). The encoder is an isometry $V_{L\to P_1P_2P_3}$, and the loss of the j-th qudit is the partial trace tr_{P_j}. By the decoupling principle, we can correct for the loss of the j-th qudit if and only if the Choi state of the complementary channel is a product,

$$\omega_{LP_j} = \omega_L \otimes \omega_{P_j}.$$

(Why three? With two, we could clone...)

Example 8.2.1 (Three-qutrit code): Let $d = 3$ and arithmetic mod 3. The isometry

$$V = \frac{1}{\sqrt{3}} \sum_{x,y} |y,\, y+x,\, y-x\rangle\langle x|, \qquad |\Omega_{LP_1P_2P_3}\rangle = \frac{1}{3} \sum_{x,y} |x\rangle_L |y,\, y+x,\, y-x\rangle_{P_1P_2P_3},$$

defines a code whose Choi state is maximally entangled across any bipartition into two pairs (an "absolutely maximally entangled state" or "perfect tensor"): $\omega_{LP_j} = \frac{I}{9} = \frac{I_L}{3} \otimes \frac{I_{P_j}}{3}$ for all j. Hence the code corrects the loss of any single qutrit. (This is not possible for qubits.)

8.3 Application: quantum error correction conditions

Now we solve the general case. Encode with an arbitrary isometry $V_{L\to P}$, with projection $\Pi := VV^\dagger$ onto the *code subspace*. The noise is a channel $\mathcal{N}_{P\to P'}[\rho] = \sum_j K_j \rho K_j^\dagger$. When can we recover $\mathcal{T} := \mathcal{N} \circ V(\cdot)V^\dagger$? A Stinespring isometry of $\mathcal{T}$ is $\sum_j (K_j V) \otimes |j\rangle_E$, so

$$\mathcal{T}^c_{L\to E}[\rho] = \sum_{i,j} \mathrm{tr}\left[K_i V \rho V^\dagger K_j^\dagger\right] |i\rangle\langle j|_E = \sum_{i,j} \mathrm{tr}\left[V^\dagger K_j^\dagger K_i V \rho\right] |i\rangle\langle j|_E.$$

By the decoupling principle, recovery is possible iff this does not depend on ρ:

Theorem 8.3.1: Quantum error correction conditions

The encoding V allows to recover from the noise $\mathcal{N}$ if and only if, for all i, j,

$$V^\dagger K_j^\dagger K_i V \propto I_L, \qquad \text{equivalently} \qquad \Pi K_j^\dagger K_i \Pi \propto \Pi.$$

Remark 8.3.1: These famous conditions only depend on the code subspace and on the (arbitrary choice of) Kraus operators of the noise.

8.4 Application: why teleportation must be possible

How to convince your friends that teleportation must be possible if you only remember half of the protocol: in teleportation, Alice performs a basis measurement (the Bell measurement) on her input S and her half A of an EPR pair, which yields uniformly random outcomes x_1, x_2 *regardless of the input state*. When can the channel $\mathcal{T}_{S\to X_1X_2B}$ be recovered? By the decoupling principle, iff the complementary channel is constant. To compute it:

- For the measurement channel $\mathcal{M}_{C\to X}[\rho] = \sum_x |x\rangle\langle x| \langle e_x|\rho|e_x\rangle$ in an arbitrary basis $\{|e_x\rangle\}$, a Stinespring isometry is $V_{C\to XX'} = \sum_x |xx\rangle\langle e_x|$, and the complementary channel is the "same" basis measurement channel.
- Thus the complementary channel of the first part of teleportation is just the Bell measurement on S and A (with $\rho_A = I_A/|A|$), which outputs uniformly random x_1, x_2 for every input. It is a constant channel, so Bob *must* be able to recover the input state!

9 Data compression

9.1 Classical one-shot compression

What does it mean to compress, and what does entropy have to do with it? We consider a "data source" described by a probability distribution p_X: an encoder maps X into r bits, from which a decoder produces $\hat{X}$, and we want $\hat{X} = X$ with probability $\geq 1 - \varepsilon$. (It is easy to convert this into a compression scheme that always works and uses an expected number of bits $\approx r$.)

Definition 9.1.1: A *classical* (ε, r)*-compression code* for $p_X \in \mathcal{P}(\Sigma_X)$ consists of functions $E : \Sigma_X \to \Sigma_C$ and $D : \Sigma_C \to \Sigma_X$ such that

$$\log|\Sigma_C| \leq r \qquad \text{and} \qquad \Pr\big(D(E(X)) = X\big) \geq 1 - \varepsilon. \tag{$*$}$$

The minimal number of bits needed to compress with error ε is

$$C^\varepsilon(p_X) := \min\{r \mid \exists\, (\varepsilon, r)\text{-code for } p_X\}.$$

Logarithms are always base 2.

Definition 9.1.2: The *smooth Rényi-0 (max) entropy* of p_X is

$$H_0^\varepsilon(X) = H_0^\varepsilon(p_X) := \min\big\{\log|S| \;\big|\; S \subseteq \Sigma_X,\ \Pr(X \in S) \geq 1 - \varepsilon\big\}.$$

Theorem 9.1.1: Classical one-shot compression

$C^\varepsilon(p_X) = H_0^\varepsilon(p_X)$.

Proof. ($\geq$) Given a code, $S := \{x : D(E(x)) = x\}$ has $\Pr(X \in S) \geq 1 - \varepsilon$ and $|S| \leq |\Sigma_C|$. ($\leq$) Given S, let $\Sigma_C := S$, $E(x) := x$ for $x \in S$ and arbitrary otherwise, and $D := \mathrm{id}$. □

Boring! How to do better? Next we compress many symbols at a time (IID source); the optimal rate will be $\frac{1}{n} H_0^\varepsilon(X_1, \ldots, X_n) \to H(p) = \sum_x p_x \log \frac{1}{p_x}$. But first, let us generalize to the quantum setting:

- the data source is described by a quantum state ρ_A describing its average output,
- encoder and decoder are quantum channels $\mathcal{E}_{A\to C}$, $\mathcal{D}_{C\to A}$ with $\log|C| \leq r$ ("r qubits"),
- but what replaces $(*)$?

9.2 Quantum one-shot compression

A first guess, $\mathcal{D}[\mathcal{E}[\rho_A]] \approx_\varepsilon \rho_A$, is trivial, even for $\varepsilon = 0$ and in the classical case: take $r = 0$, $\mathcal{E}[\sigma] = \mathrm{tr}[\sigma]$ and $\mathcal{D}[\sigma] = \mathrm{tr}[\sigma]\, \rho_A$. The key issue is that we need to preserve *correlations* between the source and the decoded message. Classically, random variables take care of this, since X and $D(E(X))$ have a joint distribution. We can rephrase this in the language of states:

Lemma 9.2.1: $(*)$ holds if and only if for every joint distribution p_{XZ} with marginal p_X, the distribution q_{XZ} of $(D(E(X)), Z)$ satisfies $\mathrm{TV}(p_{XZ}, q_{XZ}) \leq \varepsilon$.

Proof. Use the coupling inequality: for any joint distribution of random variables A, B, $\mathrm{TV}(p_A, p_B) \leq \Pr(A \neq B)$ (exercise). ($\Rightarrow$) Apply it with $A = (X, Z)$, $B = (D(E(X)), Z)$: $\mathrm{TV}(p_{XZ}, q_{XZ}) \leq \Pr(X \neq D(E(X))) \leq \varepsilon$. ($\Leftarrow$) Take $Z := X$, i.e., $p(x, z) = p(x)\delta_{xz}$. Then $\mathrm{TV}(p_{XZ}, q_{XZ}) = \Pr(X \neq D(E(X)))$. □

This perspective generalizes to the quantum case: we require that $(\mathcal{D}\circ\mathcal{E}\otimes\mathcal{I}_R)[\rho_{AR}]\approx\rho_{AR}$ for all extensions ρ_{AR} of ρ_A.

Definition 9.2.2: The *entanglement fidelity* of a channel $\mathcal{N}_{A\to A}$ and a state ρ_A is

$$F_E(\mathcal{N},\rho_A) := \min_{\mathcal{H}_R,\ \rho_{AR}} F\big((\mathcal{N}\otimes\mathcal{I}_R)[\rho_{AR}],\rho_{AR}\big),$$

where the minimum is over all extensions ρ_{AR} of ρ_A. A *quantum (ε,r)-compression code* for $\rho_A\in\mathcal{S}(\mathcal{H}_A)$ consists of channels $\mathcal{E}\in\mathrm{C}(\mathcal{H}_A,\mathcal{H}_C)$ and $\mathcal{D}\in\mathrm{C}(\mathcal{H}_C,\mathcal{H}_A)$ such that

$$\log\dim\mathcal{H}_C\le r \qquad\text{and}\qquad F_E(\mathcal{D}\circ\mathcal{E},\rho_A)\ge 1-\varepsilon.$$

We let $C^\varepsilon(\rho_A) := \min\{r \mid \exists\,(\varepsilon,r)\text{-code for }\rho_A\}$.

Remark 9.2.3: (i) We use the fidelity instead of the trace distance just for convenience.

(ii) The entanglement fidelity satisfies (exercise, good practice!)

$$F_E(\mathcal{N},\rho_A) = F\big((\mathcal{N}\otimes\mathcal{I}_R)[\rho_{AR}],\rho_{AR}\big) = \sqrt{\sum_j |\mathrm{tr}[K_j\rho_A]|^2}$$

for any fixed purification ρ_{AR} of ρ_A, where $\{K_j\}$ are Kraus operators of $\mathcal{N}$.

Definition 9.2.4: The *smooth Rényi-0 (max) entropy* of $\rho_A\in\mathcal{S}(\mathcal{H}_A)$ is

$$H_0^\varepsilon(A)=H_0^\varepsilon(\rho_A) := \min\big\{\log\mathrm{rk}\,P_A \;\big|\; P_A \text{ projection on } \mathcal{H}_A,\ \mathrm{tr}[P_A\rho_A]\ge 1-\varepsilon\big\}.$$

Lemma 9.2.5: $H_0^\varepsilon(\rho)=H_0^\varepsilon(p)$, where $p=(p_1,\dots,p_n)$ are the eigenvalues of ρ.

Proof. Homework. □

Theorem 9.2.1: Quantum one-shot compression

$H_0^{2\varepsilon}(\rho_A)\le C^\varepsilon(\rho_A)\le H_0^\varepsilon(\rho_A)$.

Proof. Achievability ("≤"). Let P_A be an orthogonal projection with $\mathrm{tr}[P_A\rho_A]\ge 1-\varepsilon$. Write $P_A=VV^\dagger$ with an isometry $V:\mathcal{H}_C\to\mathcal{H}_A$, $\dim\mathcal{H}_C=\mathrm{rk}\,P_A$, and define

$$\mathcal{E}[M_A] := V^\dagger M_A V + \mathrm{tr}\big[(I-P_A)M_A\big]\,\tau_C, \qquad \mathcal{D}[N_C] := VN_CV^\dagger,$$

where the second term in $\mathcal{E}$ ensures trace preservation. Since $P_A=VV^\dagger$ is one of the Kraus operators of $\mathcal{D}\circ\mathcal{E}$, we get $F_E(\mathcal{D}\circ\mathcal{E},\rho_A)\ge\mathrm{tr}[P_A\rho_A]\ge 1-\varepsilon$. This is an (ε,r)-code with $r=\log\mathrm{rk}\,P_A$.

Converse ("≥"). Let $(\mathcal{E},\mathcal{D})$ be an (ε,r)-code with $\mathcal{E}[M]=\sum_j X_jMX_j^\dagger$ and $\mathcal{D}[N]=\sum_k Y_kNY_k^\dagger$. Then $Z_\ell=Y_kX_j$ are Kraus operators for $\mathcal{D}\circ\mathcal{E}$ with $\mathrm{rk}\,Z_\ell\le|C|\le 2^r$. Let P_ℓ be the projection onto the image of Z_ℓ. Then, using Cauchy–Schwarz $|\mathrm{tr}[A^\dagger B]|^2\le\mathrm{tr}[A^\dagger A]\,\mathrm{tr}[B^\dagger B]$,

$$\begin{aligned}1-2\varepsilon\le(1-\varepsilon)^2\le F_E(\mathcal{D}\circ\mathcal{E},\rho_A)^2 &= \sum_\ell|\mathrm{tr}[Z_\ell\rho_A]|^2\\ &= \sum_\ell|\mathrm{tr}\big[\sqrt{\rho_A}P_\ell\cdot Z_\ell\sqrt{\rho_A}\big]|^2 \le \sum_\ell \mathrm{tr}[Z_\ell^\dagger Z_\ell\rho_A]\;\mathrm{tr}[P_\ell\rho_A].\end{aligned}$$

Since $\mathcal{D}\circ\mathcal{E}$ is trace-preserving, $\{\mathrm{tr}[Z_\ell^\dagger Z_\ell\rho_A]\}_\ell$ is a probability distribution. Hence there is ℓ with $\mathrm{tr}[P_\ell\rho_A]\ge 1-2\varepsilon$ and $\log\mathrm{rk}\,P_\ell\le r$. □

9.3 Asymptotic compression

One-shot compression has two problems: $H_0^\varepsilon(X)$ is in general quite messy, and a single (qu)bit cannot be compressed in any nontrivial way. For example, if $p(0) = 0.9$, $p(1) = 0.1$ and $\varepsilon = 0.05$, we need one full bit. The solution to both problems is to compress *blocks* (strings) of symbols $x^n = x_1 \cdots x_n$, with $p(x^n) = p(x_1) \cdots p(x_n)$. In the example, for $n = 3$ the strings 000 (probability 0.729) and $100, 010, 001$ (probability 0.081 each) together have probability 0.972, so 2 bits suffice to achieve error $0.028 < 0.05$: a rate of $2/3$ bits per symbol.

Definition 9.3.1: The *optimal compression rate* of a source at vanishing error is

$$r(p_X) := \lim_{\varepsilon\to 0} \lim_{n\to\infty} \frac{1}{n} C^\varepsilon(p_{X^n}) = \lim_{\varepsilon\to 0} \lim_{n\to\infty} \frac{1}{n} H_0^\varepsilon(p_{X^n}), \qquad r(\rho_A) := \lim_{\varepsilon\to 0} \lim_{n\to\infty} \frac{1}{n} C^\varepsilon(\rho_A^{\otimes n}).$$

In the quantum case, $r(\rho_A) = \lim_{\varepsilon\to 0} \lim_{n\to\infty} \frac{1}{n} H_0^\varepsilon(\rho_A^{\otimes n})$, since $H_0^{2\varepsilon} \leq C^\varepsilon \leq H_0^\varepsilon$ is good enough. Moreover, since $H_0^\varepsilon(\rho) = H_0^\varepsilon(p)$ for the eigenvalues p, it suffices to study probability distributions.

9.4 Shannon and von Neumann entropy

Definition 9.4.1: The *Shannon entropy* of a probability distribution p is

$$H(p) := \sum_{x:\, p(x)>0} p(x) \log \frac{1}{p(x)} = \mathbb{E}_{X\sim p}\Big[\log \frac{1}{p(X)}\Big],$$

with the convention $0 \log \frac{1}{0} = 0$. Its unit is the *bit*. We write $H(X) := H(p_X)$.

Example 9.4.2: If X is uniform on $\mathcal{A}$, then $H(X) = \log|\mathcal{A}|$; e.g., one uniformly random bit has entropy 1. If $X \sim \text{Bernoulli}(p)$, then $H(X) = h(p) := p \log \frac{1}{p} + (1-p) \log \frac{1}{1-p} \in [0, 1]$, the *binary entropy*.

Lemma 9.4.3: $H(X)$ only depends on the probabilities (not on the labels), $0 \leq H(X) \leq \log|\{x : p(x) > 0\}| \leq \log|\Sigma_X|$, with $H(X) = 0$ iff X is deterministic and $H(X) = \log|\Sigma_X|$ iff X is uniform. H is continuous and concave.

Definition 9.4.4: The *von Neumann entropy* of $\rho \in \mathcal{S}(\mathcal{H})$ is

$$H(\rho) := H(p) = -\sum_i p_i \log p_i = -\operatorname{tr}[\rho \log \rho],$$

where p is the vector of eigenvalues of ρ (repeated according to multiplicity). We write $H(A) := H(\rho_A)$.

Lemma 9.4.5: (i) $H(V\rho V^\dagger) = H(\rho)$ for any isometry V,

(ii) $0 \leq H(\rho) \leq \log \operatorname{rk} \rho \leq \log d$, where $d = \dim \mathcal{H}$; $H(\rho) = 0$ iff ρ is pure, and $H(\rho) = \log d$ iff $\rho = I/d$,

(iii) H is continuous (if $\rho \approx \sigma$, then the sorted eigenvalues are close) and concave: $H(\sum_x p_x \rho_x) \geq \sum_x p_x H(\rho_x)$.

9.5 Asymptotic equipartition and the theorems of Shannon and Schumacher

Theorem 9.5.1: Asymptotic equipartition

For every probability distribution p_X and every $0 < \varepsilon < 1$,

$$\lim_{n\to\infty} \frac{1}{n} H_0^\varepsilon(p_{X^n}) = H(p_X).$$

It follows at once that $\lim_{n\to\infty} \frac{1}{n} H_0^\varepsilon(\rho_A^{\otimes n}) = H(\rho_A)$.

As a direct consequence:

Corollary 9.5.1 (Shannon's and Schumacher's theorems): $r(p_X) = H(X)$ and $r(\rho_A) = H(A)$.

Our main tool is the typical set.

Definition 9.5.2: The *typical set* is

$$T_{n,\delta}(p) := \Big\{ x^n \in \Sigma^n \;\Big|\; \Big| \frac{1}{n} \log \frac{1}{p(x^n)} - H(p) \Big| \le \delta \Big\} = \Big\{ x^n \in \Sigma^n \;\Big|\; \Big| \frac{1}{n} \sum_{k=1}^n \log \frac{1}{p(x_k)} - H(p) \Big| \le \delta \Big\}.$$

Lemma 9.5.3 (Asymptotic equipartition property): (0) $2^{-n(H(p)+\delta)} \le p(x^n) \le 2^{-n(H(p)-\delta)}$ for all $x^n \in T_{n,\delta}$,

(1) $|T_{n,\delta}| \le 2^{n(H(p)+\delta)}$,

(2) $\Pr(X^n \notin T_{n,\delta}) \le \frac{\sigma^2}{n\delta^2} \to 0$, where $\sigma^2 = \mathrm{Var}(\log \frac{1}{p(X)})$ for $X \sim p$.

In words: for large n, typical probabilities are $2^{-n(H(p)\pm\delta)}$.

Proof. (0) holds by definition. (1): $1 \ge \Pr(X^n \in T_{n,\delta}) \ge |T_{n,\delta}|\, 2^{-n(H(p)+\delta)}$. (2): Let $R_k := \log \frac{1}{p(X_k)}$. Then $R_1, \dots, R_n$ are IID with mean $\mathbb{E}[R_k] = H(p)$, and by Chebyshev's inequality

$$\Pr(X^n \notin T_{n,\delta}) = \Pr\Big(\Big| \frac{1}{n} \sum_{k=1}^n R_k - H(p) \Big| > \delta \Big) \le \frac{\mathrm{Var}(\frac{1}{n}\sum_k R_k)}{\delta^2} = \frac{\sigma^2}{n\delta^2}.$$

(This is the weak law of large numbers.) □

Proof of Theorem 9.5.1. We need to show: (A) for all $R > H(p)$ there is n_0 such that $\frac{1}{n} H_0^\varepsilon(p_{X^n}) \le R$ for all $n \ge n_0$, and (B) for all $R < H(p)$ there is n_0 such that $\frac{1}{n} H_0^\varepsilon(p_{X^n}) \ge R$ for all $n \ge n_0$.

(A) Choose $\delta = R - H(p) > 0$. By (1), $|T_{n,\delta}| \le 2^{nR}$ for all n, and by (2) there is n_0 such that $\Pr(X^n \notin T_{n,\delta}) \le \varepsilon$ for all $n \ge n_0$ (as $\varepsilon > 0$). Thus $\frac{1}{n} H_0^\varepsilon(p_{X^n}) \le \frac{1}{n} \log |T_{n,\delta}| \le R$.

(B) If this did not hold, there would be $R < H(p)$ and sets S_n for infinitely many n with $|S_n| < 2^{nR}$ and $\Pr(X^n \in S_n) \ge 1 - \varepsilon$. Choosing $\delta := \frac{H(p)-R}{2} > 0$,

$$\begin{aligned} 1 - \varepsilon &\le \Pr(X^n \in S_n \cap T_{n,\delta}) + \Pr(X^n \notin T_{n,\delta}) \\ &\le |S_n| 2^{-n(H(p)-\delta)} + o(1) \le 2^{-n(H(p)-R-\delta)} + o(1) = 2^{-n\delta} + o(1) \to 0, \end{aligned}$$

a contradiction since $\varepsilon < 1$. □

Remark 9.5.4: In the quantum case, if $\rho_A = \sum_x p_x |e_x\rangle\langle e_x|$ is an eigendecomposition, the typical set gives rise to the projection onto the *typical subspace* $S_{n,\delta} = \mathrm{span}\{|e_{x_1}\rangle \otimes \cdots \otimes |e_{x_n}\rangle : x^n \in T_{n,\delta}(p)\} \subseteq \mathcal{H}_A^{\otimes n}$.

10 Entropy inequalities

10.1 Entropies of multiple parties

If $\rho_{ABC\dots}$ is a quantum state of multiple systems, the entropies $H(A) := H(\rho_A)$, $H(B)$, $H(AB)$, $H(ABC)$, ... are *not* independent. We discuss their relations.

Definition 10.1.1: A state ρ_{XA} is called *classical on* X (or a *cq-state*) if

$$\rho_{XA} = \sum_x p(x)|x\rangle\langle x| \otimes \rho_{A,x}$$

for the computational basis of $\mathcal{H}_X = \mathbb{C}^{\Sigma_X}$. This is the same data as an ensemble of quantum states $\{p_x, \rho_{A,x}\}$. (Note that ρ_{XA} being cq is stronger than ρ_X being classical.)

Lemma 10.1.2 (Basic properties): (i) *Subadditivity:* $H(AB) \leq H(A) + H(B)$, with equality iff $\rho_{AB} = \rho_A \otimes \rho_B$.

(ii) If ρ_{AB} is pure, then $H(AB) = 0$ and $H(A) = H(B)$, which is > 0 iff ρ_{AB} is entangled. Hence, in general, $H(AB) \ngeq H(A), H(B)$.

(iii) *Araki–Lieb:* $H(AB) \geq |H(A) - H(B)|$.

(iv) If ρ_{AX} is a cq-state, then $H(AX) = \sum_x p(x)H(\rho_{A,x}) + H(p)$ and $H(X) = H(p)$. Hence $H(AX) \geq H(X)$, and by subadditivity $H(A) \geq H(AX) - H(X)$, i.e.,

$$H\Big(\sum_x p_x \rho_x\Big) \geq \sum_x p_x H(\rho_x) \qquad \text{(concavity of the entropy)}.$$

In fact, $H(AX) \geq H(A)$ also holds.

(v) More generally, if $\rho_{AB} = \sum_x p(x)\rho_{A,x} \otimes \rho_{B,x}$ is separable, then the entropy is *monotone*: $H(AB) \geq H(A), H(B)$ (proof below). Thus $H(AB) < H(A)$ or $H(AB) < H(B)$ is a signature (sufficient condition) of entanglement.

Proof. (i) One way to compress $\rho_{AB}^{\otimes n}$ is to compress the A systems and then the B systems, but there might be better ways; hence $r(AB) \leq r(A)+r(B)$, and the claim follows from Schumacher's theorem. (ii) follows from the Schmidt decomposition. (iii) Purify to ρ_{ABC}. Then by (ii) and subadditivity, $H(A) = H(BC) \leq H(B) + H(C) = H(B) + H(AB)$. (iv) Direct computation, since ρ_{AX} is block diagonal. □

Theorem 10.1.1: Strong subadditivity and weak monotonicity

For all states ρ_{ABC}:

$$\begin{aligned} &\text{(SSA)} \qquad H(AC) + H(BC) \geq H(ABC) + H(C), \\ &\text{(WM)} \qquad H(AB) + H(BC) \geq H(A) + H(C). \end{aligned}$$

SSA reduces to subadditivity if C is trivial, and WM to Araki–Lieb. The two inequalities are equivalent via purification (exercise).

These are the most powerful entropy inequalities known and are widely used. To systematize, we introduce some derived quantities.

10.2 Conditional entropy and mutual information

Definition 10.2.1: The *conditional entropy* is $H(A|B) := H(AB) - H(B)$ ("remaining entropy if you have access to B"), and the *mutual information* is

$$I(A:B) := H(A) + H(B) - H(AB) = H(A) - H(A|B) = H(B) - H(B|A)$$

("correlations between A and B").

Lemma 10.2.2: (i) $H(A|B) \leq H(A) \leq \log|A|$, with equality in the first inequality iff $\rho_{AB} = \rho_A \otimes \rho_B$ (subadditivity).

(ii) $H(A|B) \geq -\min(H(A), H(B))$ (Araki–Lieb), and $H(A|B) \geq 0$ if ρ_{AB} is separable; i.e., $H(A|B) < 0$ implies that ρ_{AB} is entangled.

(iii) If ρ_{XA} is a cq-state, $H(A|X) = \sum_x p_x H(\rho_{A,x}) \geq 0$ (the average entropy of the ensemble).

(iv) SSA is equivalent to $H(A|C) \geq H(A|BC)$: "if you lose information, entropy can only increase".

(v) $I(A:B) \geq 0$, with equality iff $\rho_{AB} = \rho_A \otimes \rho_B$; $I(A:B) \leq 2\min(H(A), H(B))$ in general, and $I(A:B) \leq \min(H(A), H(B))$ if ρ_{AB} is separable.

(vi) SSA is equivalent to $I(A:C) \leq I(A:BC)$.

Example 10.2.3: For the maximally entangled state Φ_{AB} on $\mathbb{C}^d \otimes \mathbb{C}^d$: $H(A|B) = -\log d$ and $I(A:B) = 2\log d$. For the maximally classically correlated state $\rho_{XY} = \frac{1}{d}\sum_x |xx\rangle\langle xx|$: $H(X|Y) = 0$ and $I(X:Y) = \log d$. Note that we can get from Φ to ρ by a local measurement.

Remark 10.2.4: The common Venn diagram picture, with $H(AB)$ as the union of two circles $H(A)$ and $H(B)$ whose intersection is $I(A:B)$, is misleading in the quantum case (e.g., $H(A|B)$ can be negative).

Theorem 10.2.1: Data processing inequality (DPI)

For every state ρ_{AB} and quantum channel $\Phi_{B\to C}$,

$$H(A|C)_{(\mathcal{I}\otimes\Phi)[\rho]} \geq H(A|B)_\rho \qquad \text{and} \qquad I(A:C)_{(\mathcal{I}\otimes\Phi)[\rho]} \leq I(A:B)_\rho.$$

Both statements are equivalent to SSA.

Proof. Let $\Phi_{B\to C}[\sigma] = \mathrm{tr}_E[V_{B\to CE}\,\sigma\,V^\dagger]$ be a Stinespring representation. Since entropies are invariant under local isometries,

$$H(A|B)_\rho = H(A|CE)_{(I\otimes V)\rho(I\otimes V^\dagger)} \leq H(A|C)_{(\mathcal{I}\otimes\Phi)[\rho]}$$

by SSA. The second statement follows similarly. □

Corollary 10.2.5: If ρ_{AB} is separable, then $H(A|B) \geq 0$, i.e., $H(AB) \geq H(B)$.

Proof. Write $\rho_{AB} = \sum_x p_x \rho_{A,x} \otimes \rho_{B,x} = (\mathcal{I}_A \otimes \Phi_{X\to B})[\rho_{AX}]$, where $\rho_{AX} = \sum_x p_x \rho_{A,x} \otimes |x\rangle\langle x|$ is a cq-state and $\Phi[\sigma] = \sum_x \langle x|\sigma|x\rangle \rho_{B,x}$ is a channel. By the DPI, $H(A|B) \geq H(A|X) \geq 0$. □

Small mutual information implies a bound on correlations:

Lemma 10.2.6 (Pinsker's inequality): $I(A:B) \geq \frac{2}{\ln 2} T(\rho_{AB}, \rho_A \otimes \rho_B)^2 = \frac{1}{2\ln 2}\|\rho_{AB} - \rho_A \otimes \rho_B\|_1^2$. Consequently, for all observables O_A, O_B,

$$\frac{|\langle O_A \otimes O_B\rangle - \langle O_A\rangle\langle O_B\rangle|}{\|O_A\|\|O_B\|} \leq \sqrt{2\ln(2)\, I(A:B)}.$$

Proof. The first statement is quoted. For the second, $|\langle O_A O_B\rangle - \langle O_A\rangle\langle O_B\rangle|^2 = |\mathrm{tr}[(\rho_{AB} - \rho_A \otimes \rho_B)(O_A \otimes O_B)]|^2 \leq \|\rho_{AB} - \rho_A \otimes \rho_B\|_1^2 \|O_A \otimes O_B\|_\infty^2 \leq 2\ln(2)\, I(A:B)\|O_A\|^2\|O_B\|^2$. □

10.3 Proof of weak monotonicity

Proof of (WM). WLOG ρ_{ABC} is invertible (by continuity).

(i) *Idea.* Write WM as $H(AB) - H(C) \geq H(A) - H(BC)$. Since $\log(M \otimes N) = \log M \otimes I + I \otimes \log N$, this is equivalent to

$$\operatorname{tr}\big[\rho_{ABC} \log(\rho_{AB}^{-1} \otimes \rho_C)\big] \geq \operatorname{tr}\big[\rho_{ABC} \log(\rho_A^{-1} \otimes \rho_{BC})\big].$$

(ii) Because the logarithm is *operator monotone* ($M \leq N \Rightarrow \log M \leq \log N$; not so obvious! Exercise: it is easy for $M \mapsto -M^{-1}$, then use $\log x = \int_0^\infty \big(\frac{1}{1+t} - \frac{1}{x+t}\big)dt$), we only need to show

$$\rho_{AB}^{-1} \otimes \rho_C \geq \rho_A^{-1} \otimes \rho_{BC} \iff \big\|(\rho_{AB}^{1/2} \otimes \rho_C^{-1/2})(\rho_A^{-1/2} \otimes \rho_{BC}^{1/2})\big\|_\infty \leq 1,$$

using that for $P > 0$, $P \geq Q \iff P^{-1/2} Q P^{-1/2} \leq I \iff \|P^{-1/2} Q^{1/2}\|_\infty \leq 1$.

(iii) Let $|\Gamma\rangle_{BB'} := \sum_b |bb\rangle$ be the unnormalized maximally entangled state on B and a copy B', and recall the "teleportation" identity $(\langle\Gamma|_{B'B} \otimes I_{B''})(I_{B'} \otimes |\Gamma\rangle_{BB''}) = I$. Inserting it on the B wire between $\rho_{BC}^{1/2}$ and $\rho_{AB}^{1/2}$ shows

$$(\rho_{AB}^{1/2} \otimes \rho_C^{-1/2})(\rho_A^{-1/2} \otimes \rho_{BC}^{1/2}) = (I_{AB} \otimes W^\dagger)(V \otimes I_{BC}),$$

where

$$V := \big(\rho_{AB}^{1/2}(\rho_A^{-1/2} \otimes I_B) \otimes I_{B'}\big)\big(I_A \otimes |\Gamma\rangle_{BB'}\big) : \mathcal{H}_A \to \mathcal{H}_{ABB'},$$
$$W := \big(I_{B'} \otimes \rho_{BC}^{1/2}(I_B \otimes \rho_C^{-1/2})\big)\big(|\Gamma\rangle_{B'B} \otimes I_C\big) : \mathcal{H}_C \to \mathcal{H}_{B'BC}.$$

(Exercise: translate this diagrammatic argument into ordinary equations.)

(iv) V and W are isometries, hence the operator norm is at most $\|W\|\|V\| \leq 1$. Indeed, using $\langle\Gamma|(M \otimes I)|\Gamma\rangle = \operatorname{tr} M$,

$$V^\dagger V = \operatorname{tr}_B\big[(\rho_A^{-1/2} \otimes I_B)\rho_{AB}(\rho_A^{-1/2} \otimes I_B)\big] = \rho_A^{-1/2}\rho_A\rho_A^{-1/2} = I_A,$$

and similarly $W^\dagger W = I_C$. □

11 The Holevo bound and communication

11.1 The Holevo bound

Definition 11.1.1: For an ensemble $\{p_x, \rho_x\}$ with corresponding cq-state $\rho_{XB} = \sum_x p_x |x\rangle\langle x| \otimes \rho_{B,x}$, the *Holevo χ-quantity* is

$$\chi(\{p_x, \rho_x\}) := I(X:B) = H(B) - H(B|X) = H\Big(\sum_x p_x \rho_x\Big) - \sum_x p_x H(\rho_x) \leq \log|B|.$$

Relevance: Imagine Alice wants to communicate a classical message $x \in \Sigma_X$ to Bob and she can send qubits. She encodes x into $\rho_{B,x}$, and Bob measures to obtain an estimate $\hat{x}$. How many qubits are needed?

Theorem 11.1.1: Holevo bound

For every measurement Bob might perform, $I(X:\hat{X}) \leq \chi(\{p_x, \rho_x\}) = I(X:B)$.

Proof. Bob's measurement is a channel $\mathcal{M}[\sigma] = \sum_{\hat{x}} \operatorname{tr}[\mu(\hat{x})\sigma]|\hat{x}\rangle\langle\hat{x}|$; apply the DPI. □

Interpretation. Assume we can decode x perfectly from ρ_x. If X is uniformly random, then $\hat{X} = X$ and $I(X : \hat{X}) = H(X) + H(\hat{X}) - H(X\hat{X}) = \log|\Sigma_X|$. By Holevo, $\log|\Sigma_X| \leq I(X : B) \leq \log|B|$. Thus, *to communicate n bits perfectly, one needs to send n qubits.* This is also obvious because only orthogonal states can be perfectly distinguished, but Holevo's bound makes it quantitative:

Lemma 11.1.2 (Fano's inequality): If $\varepsilon := \Pr(X \neq \hat{X})$, then $H(X|\hat{X}) \leq h(\varepsilon) + \varepsilon \log(|\Sigma_X| - 1)$.

By Fano and the DPI, $H(X|B) \leq h(\varepsilon) + \varepsilon \log(|\Sigma_X| - 1)$. If X is uniform, this gives

$$\varepsilon \geq \frac{H(X|B) - h(\varepsilon)}{\log(|\Sigma_X| - 1)} \geq \frac{\log|\Sigma_X| - I(X : B) - h(\varepsilon)}{\log|\Sigma_X|}, \qquad \text{i.e.,} \qquad I(X : B) \geq (1-\varepsilon)\log|\Sigma_X| - h(\varepsilon).$$

Corollary 11.1.3: To communicate n bits with average error ε, one needs to send at least $I(X : B) \geq (1 - \varepsilon)n - h(\varepsilon)$ qubits.

11.2 Further applications

Optimality of superdense coding. Suppose now that Alice and Bob share an entangled state $\rho_{E_A E_B}$ (of arbitrary dimension), Alice encodes x by applying a channel $\mathcal{E}_x$ to E_A with output B, and sends B to Bob. The state is $\rho_{XBE_B} = \sum_x p(x)|x\rangle\langle x| \otimes (\mathcal{E}_x \otimes \mathcal{I})[\rho_{E_A E_B}]$. Since $\rho_{XE_B} = \rho_X \otimes \rho_{E_B}$, $I(X : E_B) = 0$ and by the chain rule

$$I(X : \hat{X}) \leq I(X : BE_B) = I(XE_B : B) + I(X : E_B) - I(B : E_B) \leq I(XE_B : B) \leq 2\log|B|.$$

Thus, in the presence of entanglement (no matter how much), Alice can communicate at most $2n$ bits reliably by sending n qubits. This is achieved by superdense coding, which is therefore optimal.

Classical communication only. If Alice can only send classical information Y (e.g., the result of a standard basis measurement of her encoded system), then the state is classical on Y and

$$I(X : \hat{X}) \leq I(XE_B : Y) \leq \log|\Sigma_Y|.$$

Entanglement does *not* help here: no matter how much entanglement Alice and Bob share, they can communicate at most n bits if they send n bits.

Optimality of teleportation. By superdense coding, any protocol that communicates m qubits can be converted into one that communicates $2m$ bits. Hence, in the presence of entanglement (no matter how much), Alice can communicate at most m qubits reliably if she sends $2m$ bits. This is achieved by teleportation, which is therefore optimal.

11.3 Channel capacities

So far we analyzed situations where Alice can send qubits or bits perfectly. Similar bounds can be derived when they are forced to use a noisy quantum channel $\mathcal{N}_{A\to B}$:

- *Classical communication* (without shared entanglement): $I(X : \hat{X}) \leq \chi(\mathcal{N})$ (exercise), where the *Holevo information* of the channel is

$$\chi(\mathcal{N}) := \max_{\{p_x, \rho_{A,x}\}} \chi\big(\{p_x, \mathcal{N}[\rho_{A,x}]\}\big) = \max_{\rho_{XA}\ \text{cq}} I(X : B)_{(\mathcal{I}_X \otimes \mathcal{N})[\rho_{XA}]}.$$

- *Entanglement-assisted classical communication*: $I(X : \hat{X}) \leq I(\mathcal{N})$ (exercise), where the *mutual information* of the channel is

$$I(\mathcal{N}) := \max_{\rho_{RA}} I(R:B)_{(\mathcal{I}_R \otimes \mathcal{N})[\rho_{RA}]}.$$

Theorem 11.3.1: Capacity theorems

(i) (Holevo–Schumacher–Westmoreland) The *classical capacity*, i.e., the optimal asymptotic rate in bits per channel use, is

$$C(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n}\chi(\mathcal{N}^{\otimes n}).$$

(ii) The *entanglement-assisted classical capacity* is $C_E(\mathcal{N}) = I(\mathcal{N})$. Since entanglement is free, by superdense coding and teleportation the *entanglement-assisted quantum capacity* (in qubits per channel use) is $Q_E(\mathcal{N}) = \frac{1}{2}I(\mathcal{N})$.

(iii) (Lloyd–Shor–Devetak) The *quantum capacity* is $Q(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} I_c(\mathcal{N}^{\otimes n})$, where the *coherent information* is $I_c(\mathcal{N}) := \max_{\rho_{RA}} \big(-H(R|B)\big)$.

12 Learning pure states and representation theory

12.1 Pure state learning

Goal: design a POVM which, given n copies $|\psi\rangle^{\otimes n}$ of an arbitrary unknown pure state $|\psi\rangle \in \mathbb{C}^d$, outputs an estimate $\hat{\psi}$ such that $|\langle\psi|\hat{\psi}\rangle|^2 \geq 1-\varepsilon^2$ (in expectation, or even with high probability). Equivalently, the trace distance satisfies $T(\hat{\psi}, \psi) \leq \varepsilon$. How many copies (samples) are needed? Intuition:

- $\dim_{\mathbb{C}}\{|\psi\rangle\langle\psi|\} = d-1$ parameters need to be estimated,
- if $\psi \approx \psi'$, it is impossible to distinguish them well by any POVM (so $n=1$ is not possible),
- $(\langle\psi|^{\otimes n})(|\psi'\rangle^{\otimes n}) = \langle\psi|\psi'\rangle^n \to 0$ if $\psi \neq \psi'$: many copies become nearly orthogonal. $(*)$

We thus need a POVM with outcomes in the (uncountably infinite!) set $\Omega = \{|\psi\rangle\langle\psi| : |\psi\rangle \in \mathbb{C}^d,\ \|\psi\| = 1\}$.

Definition 12.1.1: POVMs can be generalized to outcomes in a measure space (Ω, dx): a POVM is a map $\mu : \Omega \to \mathrm{PSD}(\mathcal{H})$ with $\int_\Omega \mu(x)\,dx = I$. The *Born rule* says that $\mathrm{tr}[\mu(x)\rho]$ is the probability density of the outcome with respect to dx, i.e.,

$$\mathbb{E}_x f(x) = \int f(x)\ \mathrm{tr}[\mu(x)\rho]\,dx, \qquad \Pr(x \in S) = \int_S \mathrm{tr}[\mu(x)\rho]\,dx.$$

Remark 12.1.2: These are physical: if $\mathcal{H}$ is finite-dimensional, they can always be implemented by a finite POVM chosen at random. Examples of measures: counting/uniform measure on a finite set (permutation symmetry), Lebesgue measure on $\mathbb{R}$ (translation symmetry), rotation-invariant measure on the sphere S^n (rotation symmetry); in each case it is unique up to normalization.

Lemma 12.1.3 (Haar measure on pure states): There exists a unique probability measure $d\psi$ on the set of pure states on $\mathbb{C}^d$ such that

$$\int d\psi\, f(\psi) = \int d\psi\, f(U\psi U^\dagger) \qquad \text{for all integrable } f \text{ and unitaries } U$$

("uniform", "unitarily invariant", or "Haar" probability measure). Concretely: pick $z_j = x_j + iy_j$ with x_j, y_j independent standard Gaussians, define $|\tilde{\psi}\rangle = \sum_j z_j|j\rangle$ and $|\psi\rangle := |\tilde{\psi}\rangle/\|\tilde{\psi}\|$.

Hayashi's idea: $(*)$ suggests that $\mu(\psi) \propto |\psi\rangle\langle\psi|^{\otimes n}$ might be a good idea. But what is $\int d\psi\, |\psi\rangle\langle\psi|^{\otimes n}$? Let us study the symmetries.

12.2 The symmetric subspace

The state $|\psi\rangle^{\otimes n} \in (\mathbb{C}^d)^{\otimes n}$ is invariant under permuting the subsystems.

Definition 12.2.1: The *symmetric group* S_n consists of all permutations (bijections) $\pi : \{1,\dots,n\} \to \{1,\dots,n\}$, with composition as the group operation. For $\pi \in S_n$ define the unitary R_π on $(\mathbb{C}^d)^{\otimes n}$ by

$$R_\pi\big(|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle\big) := |\psi_{\pi^{-1}(1)}\rangle \otimes \cdots \otimes |\psi_{\pi^{-1}(n)}\rangle.$$

Then $R_{\mathrm{id}} = I$ and $R_\pi R_\tau = R_{\pi\tau}$ (a unitary representation of S_n, see below); in particular $R_{\pi^{-1}} = R_\pi^\dagger$.

Definition 12.2.2: The *symmetric subspace* is

$$\mathrm{Sym}^n(\mathbb{C}^d) := \big\{|\Phi\rangle \in (\mathbb{C}^d)^{\otimes n} \;\big|\; R_\pi|\Phi\rangle = |\Phi\rangle \;\forall \pi \in S_n\big\}.$$

It contains $|\psi\rangle^{\otimes n}$ for all $|\psi\rangle \in \mathbb{C}^d$, but not all its elements are of this form.

Example 12.2.3: $\mathrm{Sym}^2(\mathbb{C}^2)$ has basis $|00\rangle$, $|11\rangle$, $\frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$; missing is $\frac{1}{\sqrt{2}}(|10\rangle - |01\rangle)$.

Lemma 12.2.4: The operator $\Pi_{n,d} := \frac{1}{n!}\sum_{\pi\in S_n} R_\pi$ ("symmetrizer") is the orthogonal projection onto $\mathrm{Sym}^n(\mathbb{C}^d)$: $\Pi_{n,d}|\Phi\rangle \in \mathrm{Sym}^n(\mathbb{C}^d)$ for all $|\Phi\rangle$, $\Pi_{n,d}|\Phi\rangle = |\Phi\rangle$ if $|\Phi\rangle \in \mathrm{Sym}^n(\mathbb{C}^d)$, and $\Pi_{n,d}^\dagger = \Pi_{n,d}$. A basis of $\mathrm{Sym}^n(\mathbb{C}^d)$ is given by

$$\Pi_{n,d}|\underbrace{1\cdots1}_{w_1}\underbrace{2\cdots2}_{w_2}\cdots\underbrace{d\cdots d}_{w_d}\rangle \;\propto\; |1\rangle^{\otimes w_1} \otimes \cdots \otimes |d\rangle^{\otimes w_d} + \text{permutations},$$

labeled by *types* (occupation numbers) $w_j \geq 0$ with $\sum_{j=1}^d w_j = n$. Hence

$$\operatorname{tr}\Pi_{n,d} = \dim \mathrm{Sym}^n(\mathbb{C}^d) = \binom{n+d-1}{n} = \frac{(n+d-1)!}{n!\,(d-1)!}.$$

Theorem 12.2.1: Integral formula for the symmetric subspace

$$\Pi_{n,d} = \binom{n+d-1}{n}\int d\psi\, |\psi\rangle\langle\psi|^{\otimes n}.$$

The proof is given in Section 12.5, using representation theory. Consequently, $\{|\psi\rangle^{\otimes n}\}$ is an "overcomplete basis" of $\mathrm{Sym}^n(\mathbb{C}^d)$: every $|\Phi\rangle \in \mathrm{Sym}^n(\mathbb{C}^d)$ can be written as $|\Phi\rangle = \Pi_{n,d}|\Phi\rangle = \int d\psi\, |\psi\rangle^{\otimes n}\binom{n+d-1}{n}\big(\langle\psi|^{\otimes n}|\Phi\rangle\big)$.

12.3 The estimator

By Theorem 12.2.1,

$$\mu(\hat\psi) := \binom{n+d-1}{n}|\hat\psi\rangle\langle\hat\psi|^{\otimes n}$$

is a POVM on $\mathrm{Sym}^n(\mathbb{C}^d)$ (with respect to $d\hat\psi$). It can be completed to a POVM on $(\mathbb{C}^d)^{\otimes n}$ by adding the outcome $\mu(\perp) := I - \Pi_{n,d}$, but we do not care about the outside of $\mathrm{Sym}^n(\mathbb{C}^d)$.

Theorem 12.3.1: Pure state learning

Measuring μ on $|\psi\rangle^{\otimes n}$ yields

$$\mathbb{E}_{\hat{\psi}}|\langle\psi|\hat{\psi}\rangle|^2 = \frac{n+1}{n+d} = 1 - \frac{d-1}{n+d} \geq 1 - \frac{d}{n}, \qquad \Pr_{\hat{\psi}}\big(T(\hat{\psi},\psi) > \varepsilon\big) \leq \frac{d}{n\varepsilon^2}.$$

Hence $n = O(d/\varepsilon^2)$ copies suffice to learn an unknown pure state $|\psi\rangle\langle\psi|$ on $\mathbb{C}^d$ up to trace distance ε, with success probability 99% (or any other constant).

Proof. By the Born rule and Theorem 12.2.1 applied with $n+1$,

$$\begin{aligned}\mathbb{E}_{\hat{\psi}}|\langle\psi|\hat{\psi}\rangle|^2 &= \int d\hat{\psi}\ \mathrm{tr}\big[\mu(\hat{\psi})|\psi\rangle\langle\psi|^{\otimes n}\big]|\langle\psi|\hat{\psi}\rangle|^2 = \binom{n+d-1}{n}\int d\hat{\psi}\,|\langle\hat{\psi}|\psi\rangle|^{2n+2} \\ &= \binom{n+d-1}{n}\langle\psi|^{\otimes(n+1)}\Big(\int d\hat{\psi}\,|\hat{\psi}\rangle\langle\hat{\psi}|^{\otimes(n+1)}\Big)|\psi\rangle^{\otimes(n+1)} \\ &= \binom{n+d-1}{n}\binom{n+d}{n+1}^{-1} = \frac{n+1}{n+d}.\end{aligned}$$

For the second claim, using $T(\hat{\psi},\psi) = \sqrt{1-|\langle\psi|\hat{\psi}\rangle|^2}$ and Markov's inequality,

$$\Pr\big(T(\hat{\psi},\psi) > \varepsilon\big) = \Pr\big(1-|\langle\psi|\hat{\psi}\rangle|^2 > \varepsilon^2\big) \leq \frac{\mathbb{E}\big[1-|\langle\psi|\hat{\psi}\rangle|^2\big]}{\varepsilon^2} \leq \frac{d}{n\varepsilon^2}. \qquad \square$$

Remark 12.3.1: The estimator $|\hat{\psi}\rangle\langle\hat{\psi}|$ is biased (exercise: make it unbiased). Open questions at this point: why does the integral formula hold, is our measurement optimal, and how to go beyond pure states?

12.4 Groups and representations

Definition 12.4.1: A *group* G is a set with a multiplication $g \cdot h \in G$ (often omitted), a neutral element 1 with $g \cdot 1 = 1 \cdot g = g$, and inverses g^{-1} with $gg^{-1} = g^{-1}g = 1$, such that the multiplication is associative.

Example 12.4.2: The symmetric group S_n (with $(\pi \cdot \tau)(x) = \pi(\tau(x))$, $1 = \mathrm{id}$); the unitary group $U(d)$ of unitary $d \times d$ matrices (with matrix multiplication, $U^{-1} = U^\dagger$); its subgroup $SU(d) = \{U \in U(d) : \det U = 1\}$; the special and general linear groups $SL(d) = \{g \in \mathbb{C}^{d\times d} : \det g = 1\} \subseteq GL(d) = \{g \in \mathbb{C}^{d\times d} \text{ invertible}\}$. Other groups: D_8, $\mathbb{Z}/n\mathbb{Z}$, ...

Definition 12.4.3: A *(unitary) representation* of a group G consists of a Hilbert space $\mathcal{H}$ (for us always finite-dimensional) and unitary operators R_g on $\mathcal{H}$ for $g \in G$ such that

$$R_g R_h = R_{gh} \quad \forall g,h \in G, \qquad R_1 = I_{\mathcal{H}}.$$

We often simply say "the representation $\mathcal{H}$".

Example 12.4.4: (i) $(\mathbb{C}^d)^{\otimes n}$ is a representation of S_n (via R_π) and of $U(d)$ (via $T_U = U^{\otimes n}$). Since $[R_\pi, T_U] = 0$ for all π, U, it is even a representation of $S_n \times U(d)$.

(ii) Representations of S_3: the *trivial representation* $\mathcal{H} = \mathbb{C}|0\rangle$, $R_\pi|0\rangle = |0\rangle$ (exists for any group); the *sign representation* $R_\pi|0\rangle = \mathrm{sign}(\pi)|0\rangle$; and $\mathcal{H} = \mathbb{C}^3$ with R_π permuting the coordinates, e.g., $R_{1\leftrightarrow 2}(\alpha,\beta,\gamma)^T = (\beta,\alpha,\gamma)^T$. (†)

To understand a representation, we want to decompose it into smaller building blocks.

Definition 12.4.5: An *invariant subspace* (or *subrepresentation*) is a subspace $\tilde{\mathcal{H}} \subseteq \mathcal{H}$ such that $R_g|\psi\rangle \in \tilde{\mathcal{H}}$ for all $|\psi\rangle \in \tilde{\mathcal{H}}$ and $g \in G$. A representation is called *irreducible* (an *irrep*) if $\{0\}$ and $\mathcal{H}$ are its only invariant subspaces.

Remark 12.4.6: If $\tilde{\mathcal{H}}$ is an invariant subspace, then so is $\tilde{\mathcal{H}}^\perp$ (by unitarity). Thus $\mathcal{H} = \tilde{\mathcal{H}} \oplus \tilde{\mathcal{H}}^\perp$ and R_g is block diagonal with two "smaller" representations. Iterating, we obtain a decomposition

$$\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \cdots \oplus \mathcal{H}_m$$

into irreducible representations (the elementary building blocks). Representation theory tells us the irreps and how to decompose.

Example 12.4.7: (i) Any one-dimensional representation is irreducible.

(ii) (†) is not an irrep of S_3: it has the nontrivial invariant subspaces $\tilde{\mathcal{H}} = \{(\alpha, \beta, \gamma)^T : \alpha + \beta + \gamma = 0\}$ (exercise: this is an irrep) and $\tilde{\mathcal{H}}^\perp = \mathbb{C}(1,1,1)^T$ (also an irrep). This is its decomposition into irreps.

(iii) $\mathrm{Sym}^n(\mathbb{C}^d) \subseteq (\mathbb{C}^d)^{\otimes n}$ is invariant for S_n but not irreducible (indeed $\mathbb{C}|\psi\rangle^{\otimes n}$ is S_n-invariant for all $|\psi\rangle$). It is also invariant for $U(d)$, and in fact *irreducible* (see below).

12.5 Schur's lemma and the proof of the integral formula

How can we relate and compare representations?

Definition 12.5.1: Given two representations $\{R_g\}$ on $\mathcal{H}$ and $\{R'_g\}$ on $\mathcal{H}'$, an *intertwiner* is a linear map $J : \mathcal{H} \to \mathcal{H}'$ such that $JR_g = R'_g J$ for all $g \in G$. If J is unitary, we say that the two representations are *(unitarily) equivalent*, written $\mathcal{H} \cong \mathcal{H}'$; then $JR_gJ^\dagger = R'_g$ for all g ("a basis change that works for all g").

Theorem 12.5.1: Schur's lemma

Let J be an intertwiner between two irreps.

(i) If they are the same irrep ($\mathcal{H} = \mathcal{H}'$, $R_g = R'_g$), then $J \propto I_{\mathcal{H}}$: symmetry determines the operator up to a scalar!

(ii) In general, $J = 0$ or $J \propto$ unitary; in the latter case the irreps are equivalent.

Proof. (i) Let λ be an eigenvalue of J (which exists since we work over $\mathbb{C}$). Then $\ker(J - \lambda I) \neq \{0\}$ is an invariant subspace, hence $\ker(J - \lambda I) = \mathcal{H}$ and $J = \lambda I$. (ii) If J is an intertwiner, so is $J^\dagger$ (taking adjoints of $JR_g = R'_g J$ and using unitarity), and hence so are $J^\dagger J$ and $JJ^\dagger$. By (i), $J^\dagger J = \lambda I_{\mathcal{H}}$ and $JJ^\dagger = \lambda' I_{\mathcal{H}'}$ with $\lambda, \lambda' \geq 0$; since $J^\dagger J$ and $JJ^\dagger$ have the same nonzero eigenvalues, $\lambda = \lambda'$ unless both vanish. If $\lambda = 0$, then $J = 0$. If $\lambda > 0$, then $J/\sqrt{\lambda}$ is unitary. □

Proof of Theorem 12.2.1. Let $M := \int d\psi\, |\psi\rangle\langle\psi|^{\otimes n}$. By unitary invariance of $d\psi$,

$$M = \int d\psi\, (U|\psi\rangle\langle\psi|U^\dagger)^{\otimes n}, \qquad \text{hence} \qquad U^{\otimes n} M = M U^{\otimes n} \quad \text{for all } U \in U(d).$$

Moreover, M is supported on $\mathrm{Sym}^n(\mathbb{C}^d)$: with respect to $(\mathbb{C}^d)^{\otimes n} = \mathrm{Sym}^n(\mathbb{C}^d) \oplus \mathrm{Sym}^n(\mathbb{C}^d)^\perp$, both M and $\Pi_{n,d}$ are block diagonal with a single nonzero block. Since $\mathrm{Sym}^n(\mathbb{C}^d)$ is an irrep of $U(d)$, Schur's lemma gives $M = \lambda \Pi_{n,d}$ for some $\lambda \in \mathbb{C}$. To determine λ, compare traces: $\mathrm{tr}\, M = \int d\psi\ \mathrm{tr}[|\psi\rangle\langle\psi|^{\otimes n}] = 1$ and $\mathrm{tr}\, \Pi_{n,d} = \binom{n+d-1}{n}$. □

12.6 Irreducibility of the symmetric subspace

For $M \in \mathbb{C}^{d\times d}$ define the "Lie algebra representation"

$$\tilde{M} := M \otimes I \otimes \cdots \otimes I + \cdots + I \otimes \cdots \otimes I \otimes M \quad \text{on } (\mathbb{C}^d)^{\otimes n}.$$

Lemma 12.6.1 (Key fact): If $\tilde{\mathcal{H}} \subseteq (\mathbb{C}^d)^{\otimes n}$ is an invariant subspace for $U^{\otimes n}$ (i.e., $U^{\otimes n}\tilde{\mathcal{H}} \subseteq \tilde{\mathcal{H}}$ for all $U \in U(d)$), then also $\tilde{M}\tilde{\mathcal{H}} \subseteq \tilde{\mathcal{H}}$ for all $M \in \mathbb{C}^{d\times d}$.

Proof. By linearity in M, WLOG $M = M^\dagger$. Then $\tilde{M} = -i\frac{d}{dt}\big|_{t=0} e^{i\tilde{M}t} = -i\frac{d}{dt}\big|_{t=0}\big(e^{iMt}\big)^{\otimes n}$, which lies in the (closed) span of $\{U^{\otimes n}\}$. □

Proposition 12.6.2: $\mathrm{Sym}^n(\mathbb{C}^d)$ is an irreducible representation of $U(d)$.

Proof. We give the proof for $d = 2$ (exercise: generalize to any d). Recall that $\mathrm{Sym}^n(\mathbb{C}^2)$ has the $n+1$ basis vectors

$$|\tilde{k}\rangle \propto \text{sum of all strings with } k \text{ ones and } n-k \text{ zeros}, \qquad k = 0, \ldots, n,$$

e.g., $|\tilde{0}\rangle = |0\cdots 0\rangle$, $|\tilde{1}\rangle \propto |10\cdots 0\rangle + |010\cdots 0\rangle + \cdots + |0\cdots 01\rangle$, $|\tilde{n}\rangle = |1\cdots 1\rangle$.

(i) For $Z = \mathrm{diag}(1,-1)$: $\tilde{Z}|\tilde{k}\rangle = (\#0\text{'s} - \#1\text{'s})|\tilde{k}\rangle = (n-2k)|\tilde{k}\rangle$, so the $|\tilde{k}\rangle$ are eigenvectors of $\tilde{Z}$ with distinct eigenvalues.

(ii) For $M_+ = |1\rangle\langle 0|$ and $M_- = |0\rangle\langle 1|$: $\tilde{M}_+|\tilde{k}\rangle \propto \widetilde{|k+1\rangle}$ (nonzero unless $k = n$) and $\tilde{M}_-|\tilde{k}\rangle \propto \widetilde{|k-1\rangle}$ (nonzero unless $k = 0$).

Now let $\tilde{\mathcal{H}} \subseteq \mathrm{Sym}^n(\mathbb{C}^2)$ be a nonzero invariant subspace. By the key fact, $\tilde{Z}$ and $\tilde{M}_\pm$ preserve $\tilde{\mathcal{H}}$. Since $\tilde{Z}$ is Hermitian, by (i) $\tilde{\mathcal{H}}$ contains at least one eigenvector, i.e., some $|\tilde{k}\rangle$; by (ii) it then contains all $|\tilde{k}\rangle$. Hence $\tilde{\mathcal{H}} = \mathrm{Sym}^n(\mathbb{C}^2)$. □

Remark 12.6.3: The proof shows that $\mathrm{Sym}^n(\mathbb{C}^d)$ is even an irrep of $SU(d)$: it suffices to consider traceless M, since $\det e^{iM} = e^{i\,\mathrm{tr}\,M}$, and $\tilde{I} = nI$ preserves every subspace.

13 Learning mixed states and a matching lower bound

Recall that we learned an unknown pure state from $n = O(d/\varepsilon^2)$ copies using Hayashi's POVM, whose analysis relied on the integral formula (Theorem 12.2.1), which in turn followed from two ingredients: Schur's lemma and the irreducibility of $\mathrm{Sym}^n(\mathbb{C}^d)$ as a representation of $U(d)$.

13.1 A lower bound for pure state estimation

Theorem 13.1.1: Optimality of Hayashi's measurement

Let μ be any POVM on $\mathrm{Sym}^n(\mathbb{C}^d)$ (with outcomes $\hat{\psi}$) such that for every pure state $\psi = |\psi\rangle\langle\psi|$, measuring $\psi^{\otimes n}$ yields, with probability at least 0.99, an outcome $\hat{\psi}$ with $T(\hat{\psi}, \psi) \le \varepsilon$. Then $n = \Omega(d/\varepsilon^2)$.

Remark 13.1.1: Restricting to POVMs on $\mathrm{Sym}^n(\mathbb{C}^d)$ is no loss of generality, since $\psi^{\otimes n}$ is supported on the symmetric subspace (replace μ by $\Pi_{n,d}\,\mu\,\Pi_{n,d}$).

Proof. Write $\mathbb{E}_{\hat{\psi}|\psi}$ for the expectation over the outcome $\hat{\psi}$ if we measure $\psi^{\otimes n}$, and let $k \in \mathbb{N}$ be a parameter to be chosen. Since $T(\hat{\psi}, \psi) \le \varepsilon$ iff $|\langle\hat{\psi}|\psi\rangle|^2 \ge 1 - \varepsilon^2$,

$$\mathbb{E}_{\hat{\psi}|\psi}\big[|\langle\hat{\psi}|\psi\rangle|^{2k}\big] \ge 0.99\,(1-\varepsilon^2)^k \ge 0.99\,(1 - k\varepsilon^2) \qquad \text{for all } \psi.$$

Now average over a uniformly random ψ and write $\mu(\hat{\psi})\,d\hat{\psi}$ for the POVM. Using the integral formula for $n+k$ copies,

$$\begin{aligned}\mathbb{E}_{\psi}\mathbb{E}_{\hat{\psi}|\psi}\big[|\langle\hat{\psi}|\psi\rangle|^{2k}\big] &= \iint \operatorname{tr}\Big[\big(|\hat{\psi}\rangle\langle\hat{\psi}|^{\otimes k}\otimes\mu(\hat{\psi})\big)|\psi\rangle\langle\psi|^{\otimes(k+n)}\Big]\,d\psi\,d\hat{\psi}\\ &= \binom{n+k+d-1}{n+k}^{-1}\int \operatorname{tr}\Big[\big(|\hat{\psi}\rangle\langle\hat{\psi}|^{\otimes k}\otimes\mu(\hat{\psi})\big)\Pi_{n+k,d}\Big]\,d\hat{\psi}\\ &\le \binom{n+k+d-1}{n+k}^{-1}\operatorname{tr}\Big[\int\mu(\hat{\psi})\,d\hat{\psi}\Big] = \binom{n+k+d-1}{n+k}^{-1}\binom{n+d-1}{n},\end{aligned}$$

where we used $\Pi_{n+k,d}\le I$, $\operatorname{tr}|\hat{\psi}\rangle\langle\hat{\psi}|^{\otimes k}=1$, and $\int\mu(\hat{\psi})\,d\hat{\psi}=\Pi_{n,d}$ since μ is a POVM on $\mathrm{Sym}^n(\mathbb{C}^d)$. The right-hand side equals

$$\prod_{j=1}^{k}\frac{n+j}{n+j+d-1}\le\Big(\frac{n+k}{n+k+d-1}\Big)^{k}=\Big(1-\frac{d-1}{n+k+d-1}\Big)^{k}\le\exp\Big(-\frac{k(d-1)}{n+k+d-1}\Big),$$

using $1+x\le e^{x}$. Thus

$$0.99\,(1-k\varepsilon^2)\le\exp\Big(-\frac{k(d-1)}{n+k+d-1}\Big).$$

Choose $k=\Theta(1/\varepsilon^2)$ such that the left-hand side is at least $e^{-1/2}$ (e.g., $k=\lfloor 1/(3\varepsilon^2)\rfloor$ for small ε). Then $\frac{1}{2}\ge\frac{k(d-1)}{n+k+d-1}$, i.e.,

$$n\ge 2k(d-1)-(k+d-1)=(2k-1)(d-1)-k\ge(k-1)(d-1)=\Omega\Big(\frac{d}{\varepsilon^2}\Big).\qquad\square$$

13.2 Random purifications

Idea (2025): can we construct a quantum channel $\mathcal{P}$ with $\rho^{\otimes n}\mapsto|\psi_\rho\rangle\langle\psi_\rho|^{\otimes n}$, where $|\psi_\rho\rangle$ is a purification of ρ? Then we could reduce learning mixed states to learning pure states.

Lemma 13.2.1: This is *not* possible as stated.

Proof. Let $\tau=I/d$. For every unit vector $|\Phi\rangle\in\mathcal{H}^{\otimes n}$ we can decompose $\tau^{\otimes n}=I/d^n=\frac{1}{d^n}|\Phi\rangle\langle\Phi|+\sigma$ with $\sigma\ge 0$. Hence $\mathcal{P}(\tau^{\otimes n})=\frac{1}{d^n}\mathcal{P}(|\Phi\rangle\langle\Phi|)+\mathcal{P}(\sigma)$ with both terms PSD. Since the left-hand side is pure, $\mathcal{P}(|\Phi\rangle\langle\Phi|)=\mathcal{P}(\tau^{\otimes n})$ for all $|\Phi\rangle$, i.e., $\mathcal{P}$ is constant on states. But then it cannot output purifications of different ρ. $\square$

It *is* possible if we map $\rho^{\otimes n}$ to n copies of a *random* purification. Recall that every ρ_A has a purification $|\psi_{AR}\rangle$ with $\mathcal{H}_R=\mathcal{H}_A$, and any other purification on the same space is of the form $(I_A\otimes U_R)|\psi_{AR}\rangle$ for a unitary U_R. So we can define a random purification by picking U_R at random.

Lemma 13.2.2 (Haar measure): There is a unique probability measure dU on $U(d)$, called the *Haar measure*, such that U and VUW have the same distribution for all fixed $V,W\in U(d)$. If U is Haar random, then $U|0\rangle$ is a random pure state distributed according to $d\psi$, and $(I_A\otimes U_R)|\psi_{AR}\rangle$ is a *random purification* of ρ_A.

Theorem 13.2.1: Random purification channel

For all $\mathcal{H}_A$ and n there exists a quantum channel $\mathcal{P}_{A^n\to A^nR^n}$ (the *random purification channel*) such that for all ρ_A,

$$\mathcal{P}\big[\rho_A^{\otimes n}\big]=\mathbb{E}_{U_R}\Big[\big((I\otimes U_R)|\psi_\rho^{\mathrm{fix}}\rangle\langle\psi_\rho^{\mathrm{fix}}|_{AR}(I\otimes U_R^\dagger)\big)^{\otimes n}\Big],$$

where $|\psi_\rho^{\mathrm{fix}}\rangle$ is any fixed purification of ρ_A on $\mathcal{H}_A \otimes \mathcal{H}_R$, $\mathcal{H}_R = \mathcal{H}_A$.

Corollary 13.2.3 (Learning mixed states): $n = O(d^2/\varepsilon^2)$ copies suffice to learn an unknown (mixed) state ρ on $\mathbb{C}^d$ up to trace distance ε, with any constant probability of success (e.g., 99%).

Proof. Apply $\mathcal{P}$ to $\rho_A^{\otimes n}$, then measure Hayashi's POVM on $(AR)^{\otimes n}$ to obtain $\hat{\psi}_{AR}$, and output $\hat{\rho}_A := \mathrm{tr}_R\, \hat{\psi}_{AR}$. For any *fixed* purification ψ_{AR} of ρ_A, by monotonicity of the trace distance and Theorem 12.3.1 applied in dimension $\dim \mathcal{H}_{AR} = d^2$,

$$\Pr\big(T(\rho_A, \hat{\rho}_A) > \varepsilon\big) \le \Pr\big(T(\psi_{AR}, \hat{\psi}_{AR}) > \varepsilon\big) \le \frac{d^2}{n\varepsilon^2}.$$

The same holds if ψ_{AR} is a random purification, since the bound holds for every purification and we may average over U_R (try spelling out the argument in detail!). □

Proof sketch of Theorem 13.2.1. Take $\mathcal{H}_R = \mathcal{H}_A = \mathbb{C}^d$. By Schur–Weyl duality, decompose

$$\mathcal{H}_A^{\otimes n} \cong \bigoplus_\lambda \mathcal{H}_{A,\lambda} \otimes \mathbb{C}^{m_\lambda} \cong \mathcal{H}_R^{\otimes n},$$

where the $\mathcal{H}_{A,\lambda}$ are irreps of $U(d)$, inequivalent for different λ, appearing with multiplicity m_λ. Since $\rho^{\otimes n}$ lies in the span of $\{U^{\otimes n}\}$ (for invertible $\rho = e^M$ we have $\rho^{\otimes n} = e^{\tilde{M}}$), it decomposes as $\rho_A^{\otimes n} \cong \bigoplus_\lambda \rho_{A,\lambda} \otimes I_{m_\lambda}$. Using the standard purification $|\psi_\rho^{\mathrm{fix}}\rangle = (\sqrt{\rho} \otimes I) \sum_x |xx\rangle$,

$$|\psi_\rho^{\mathrm{fix}}\rangle^{\otimes n} = \bigoplus_\lambda \Big(\sqrt{\rho_{A,\lambda}} \otimes I_{R,\lambda}\Big) \sum_x |xx\rangle_{A_\lambda R_\lambda} \otimes \sum_y |yy\rangle_{m_\lambda m_\lambda} \in \bigoplus_{\lambda,\lambda'} \mathcal{H}_{A,\lambda} \otimes \mathcal{H}_{R,\lambda'} \otimes \mathbb{C}^{m_\lambda} \otimes \mathbb{C}^{m_{\lambda'}}.$$

Averaging over $U_R^{\otimes n}$ and applying Schur's lemma on the R side kills all off-diagonal blocks and twirls each $\mathcal{H}_{R,\lambda}$ block, giving

$$\mathbb{E}_{U_R}[\cdots] = \bigoplus_\lambda \rho_{A,\lambda} \otimes \frac{I_{R,\lambda}}{d_{R,\lambda}} \otimes m_\lambda |\Phi^+\rangle\langle\Phi^+|_{m_\lambda m_\lambda}.$$

This depends on ρ only through the blocks $m_\lambda \rho_{A,\lambda} = \mathrm{tr}_{\mathbb{C}^{m_\lambda}}[\Pi_\lambda \rho^{\otimes n} \Pi_\lambda]$ and is therefore the output of a quantum channel: project onto the blocks λ, trace out the multiplicity space, and append the states $I_{R,\lambda}/d_{R,\lambda}$ and $|\Phi^+\rangle\langle\Phi^+|_{m_\lambda m_\lambda}$. □

Remark 13.2.4 (Follow-ups): (i) To implement $\mathcal{P}$, one needs to implement the decomposition $\rho^{\otimes n} \cong \bigoplus_\lambda \rho_\lambda \otimes I_{m_\lambda}$, which is achieved by the *quantum Schur transform*.

(ii) If we do not want fully entangled measurements, we can group $\rho^{\otimes n} = \rho^{\otimes k} \otimes \cdots \otimes \rho^{\otimes k}$ into n/k blocks, apply an unbiased estimator to each block to obtain $\hat{\rho}_1, \ldots, \hat{\rho}_{n/k}$, and output $\hat{\rho} = \frac{k}{n} \sum_j \hat{\rho}_j$ (exercise).

14 Quantum query complexity

14.1 Computational models

Our focus now is on the *limitations* of quantum computers: proving that quantum algorithms for some problems cannot be too fast. We write $[n] := \{1, \ldots, n\}$.

An algorithm solves a problem: it receives an input and produces an output. For example, the input is a bit string $x \in \{0,1\}^n$ and the output is 1 iff there is $i \in [n]$ with $x_i = 1$ (the decision problem OR), or the output is such an i (the search version). The *complexity* of a problem is how

much resource an algorithm needs to solve it. *Randomized* algorithms can use random coins and only need to succeed with high probability (e.g., $\geq 2/3$, or more generally $\geq 1-\varepsilon$), as opposed to deterministic algorithms, which are always correct.

The most common model is *time complexity*: the input is given to the algorithm, and we count the number of basic steps needed to compute the output. In *query complexity* (also called decision tree complexity), by contrast,

- the input is not given explicitly, but only via an *oracle* (black-box access),
- only the number of oracle calls is counted (in between, arbitrarily powerful computation is allowed).

Why does this model make sense? It models some situations well (e.g., a very large input stored on a far-away server). It is the appropriate model for proving lower bounds: a query lower bound is also a lower bound on the time. And for many problems (search, cryptography, ...) the time complexity is Θ(query complexity); it is the "right" model.

Definition 14.1.1: *Oracle access* to an input $x \in \Sigma^I$ means access to the map $O_x : I \to \Sigma$. From now on we mostly consider $x \in \{0,1\}^n$ and $O_x(i) = x_i$ for $i \in [n]$. For a (total or partial) function f, the minimal number of oracle calls needed to compute f deterministically or with randomized algorithms (success probability $\geq 2/3$) is denoted $D(f)$ and $R(f)$, respectively.

Remark 14.1.2: n queries always suffice, so query complexity is interesting if it is $o(n)$, or for lower bounds. E.g., $f = \mathrm{OR}$ or $f = \mathrm{PARITY}$ ($= 1$ iff x has an odd number of ones). The $\Omega(n \log n)$ lower bound for sorting n numbers is a lower bound on comparison queries ("is $x_i < x_j$?", $\approx n^2$ options).

A quantum algorithm can make queries in superposition: the oracle is the unitary

$$O_x|i,b\rangle = |i, b \oplus x_i\rangle = (1-x_i)|i,b\rangle + x_i|i,1-b\rangle \qquad (i \in [n],\ b \in \{0,1\}).$$

Alternatively, we can use the *phase query* $O_x^{\pm}|i,b\rangle = (-1)^{b \cdot x_i}|i,b\rangle$. One can be simulated by the other:

Lemma 14.1.3: $O_x^{\pm} = (I \otimes H)\, O_x\, (I \otimes H)$, where H is the Hadamard gate.

Proof. $(I\otimes H)O_x(I\otimes H)|i,b\rangle = (I\otimes H)O_x\frac{|i,0\rangle+(-1)^b|i,1\rangle}{\sqrt{2}} = (I\otimes H)\frac{|i,x_i\rangle+(-1)^b|i,1-x_i\rangle}{\sqrt{2}}$, which equals

$$\frac{1+(-1)^b}{2}|i,0\rangle + \frac{(-1)^{x_i}+(-1)^b(-1)^{1-x_i}}{2}|i,1\rangle = \begin{cases} |i,0\rangle & b=0, \\ (-1)^{x_i}|i,1\rangle & b=1, \end{cases}$$

i.e., $O_x^{\pm}|i,b\rangle$. □

Definition 14.1.4: The *quantum query complexity* $Q(f)$ is the minimal number of queries of a quantum algorithm computing f with success probability $\geq 2/3$ on every input (equivalently, any $\geq 1-\varepsilon$ for a constant $\varepsilon < 1/2$, up to constant factors).

Remark 14.1.5: Clearly $Q(f) \leq R(f) \leq D(f) \leq n$. In fact, for total Boolean functions $f : \{0,1\}^n \to \{0,1\}$, the three measures are polynomially related: $D(f) = O(R(f)^3)$ and $D(f) = O(Q(f)^4)$. Large superpolynomial speedups thus require partial functions (promise problems).

By the principle of deferred measurement (same output distribution, some extra qubits used), all measurements can be moved to the end. Hence any T-query quantum algorithm has the form

$$U_T O_x U_{T-1} \cdots O_x U_1 O_x U_0 |\psi_0\rangle,$$

followed by a measurement, where $|\psi_0\rangle$ is an input-independent initial state and the U_t are input-independent unitaries acting on a query register X (the index i), an answer register Y (the bit b), and a workspace W (which will not matter to us).

Example 14.1.6 (Grover search): Given $x \in \{0,1\}^n$, find i with $x_i = 1$. Using the phase oracle $O_x^{\pm}|i\rangle = (-1)^{x_i}|i\rangle$:

(i) prepare the uniform superposition $|s\rangle = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} |i\rangle$,

(ii) repeat T times: apply $O_x^{\pm}$, then apply $G = 2|s\rangle\langle s| - I$,

(iii) measure in the standard basis.

Analysis: both $O_x^{\pm}$ and G act as reflections in the same two-dimensional space spanned by $|s\rangle$ and $|g\rangle$ (the normalized superposition of solutions). Together they implement a rotation by an angle $\approx 1/\sqrt{n}$ (for one solution), and the total rotation needed is a constant ($\approx \pi/2$). Hence $T = O(\sqrt{n})$ repetitions suffice. More generally, with k solutions one can find one using $O(\sqrt{n/k})$ queries.

There are two families of quantum query lower bound methods:

(i) *Polynomial methods:* lower bounds by proving that approximating polynomials must have high degree; symmetry is important; there is a dual version.

(ii) *Adversary methods:* hybrid, basic, positive-weight, negative-weight (and multiplicative) adversary; "recording" queries. Intuition: (a) initially, the state of the algorithm is the same for every input; (b) at the end, if $f(x) \neq f(y)$, the corresponding final states must be nearly orthogonal; (c) to distinguish x and y, the algorithm needs to query i such that $x_i \neq y_i$.

14.2 The adversary method

Let $|x|_H := |\{i : x_i = 1\}|$ denote the Hamming weight.

Definition 14.2.1: Let $P \subseteq f^{-1}(1)$, $N \subseteq f^{-1}(0)$, and let $R \subseteq P \times N$ be a relation ("hard inputs"). For $i \in [n]$ let $R_i := \{(x,y) \in R : x_i \neq y_i\}$. Define

$$m := \min_{x \in P} |\{y \in N : (x,y) \in R\}|, \qquad m' := \min_{y \in N} |\{x \in P : (x,y) \in R\}|,$$

$$\ell := \max_{x \in P,\, i \in [n]} |\{y \in N : (x,y) \in R,\ x_i \neq y_i\}|, \qquad \ell' := \max_{y \in N,\, i \in [n]} |\{x \in P : (x,y) \in R,\ x_i \neq y_i\}|.$$

That is, m, m' are the minimal degrees of a point on the left/right of the bipartite graph R, and ℓ, ℓ' the maximal degrees according to any R_i. Note $|R| \geq m|P|$, $|R| \geq m'|N|$, and $|R_i| \leq \ell|P|$, $|R_i| \leq \ell'|N|$.

Theorem 14.2.1: Deterministic adversary bound

$D(f) \geq \max\{\frac{m}{\ell}, \frac{m'}{\ell'}\}$.

Proof. Intuitively, the algorithm has to distinguish all pairs in R, and a query to i only helps to distinguish the pairs in R_i. Formally, fix $x \in P$ and consider the T indices $i_1, \dots, i_T$ queried on input x. Every y with $(x, y) \in R$ must lead to a different output, hence must differ from x in some queried index i_t (otherwise the algorithm behaves identically on x and y). For each t, at most ℓ such y differ from x at i_t. Hence $m \le T\ell$. The bound m'/ℓ' follows symmetrically. □

Example 14.2.2: For OR: $N = \{0^n\}$, $P = \{x : |x|_H = 1\}$, $R = P \times N$. Then $m = 1$, $m' = n$, $\ell = \ell' = 1$, so $D(\mathrm{OR}) \ge n$.

Theorem 14.2.2: Basic quantum adversary method

In the same setting, $Q(f) = \Omega\Big(\sqrt{\frac{mm'}{\ell\ell'}}\Big)$.

Example 14.2.3: For OR as above, $Q(\mathrm{OR}) = \Omega(\sqrt{n})$; hence Grover's algorithm is optimal.

Proof. Consider a T-query algorithm and let $|\psi_t^x\rangle$ be its state after t queries on input x. Define the *progress measure*

$$\Delta(t) := \sum_{(x,y)\in R} |\langle \psi_t^x | \psi_t^y \rangle|.$$

(a) Before the first query, $\langle \psi_0^x | \psi_0^y \rangle = 1$ since the state is input-independent; thus $\Delta(0) = |R|$.

(b) Before the final measurement, $|\langle \psi_T^x | \psi_T^y \rangle| \le \sqrt{8/9}$, because the algorithm must distinguish x and y with probability $2/3$. Indeed, measuring the output register Y in the standard basis, $\|(I \otimes |1\rangle\langle 1|_Y \otimes I)|\psi_T^x\rangle\|^2 = 1 - \varepsilon_1 \ge 2/3$ and $\|(I \otimes |1\rangle\langle 1|_Y \otimes I)|\psi_T^y\rangle\|^2 = \varepsilon_2 \le 1/3$, so by the triangle and Cauchy–Schwarz inequalities

$$|\langle \psi_T^x | \psi_T^y \rangle| \le \sqrt{\varepsilon_1(1-\varepsilon_2)} + \sqrt{(1-\varepsilon_1)\varepsilon_2} \le 2\sqrt{\tfrac{1}{3}\cdot\tfrac{2}{3}} = \sqrt{\tfrac{8}{9}}.$$

Hence $\Delta(T) \le \sqrt{8/9}\,|R|$.

(c) Each query changes the inner products only a little:

$$|\Delta(t+1) - \Delta(t)| \le 2|R|\sqrt{\frac{\ell\ell'}{mm'}}. \tag{6}$$

Combining (a)–(c), $(1 - \sqrt{8/9})|R| \le T \cdot 2|R|\sqrt{\ell\ell'/(mm')}$, which proves the theorem.

To show (6), write $|\psi_t^x\rangle = \sum_{i,b} \alpha_{i,b}|i,b\rangle \otimes |\phi_{i,b}\rangle$ and $|\psi_t^y\rangle = \sum_{i,b} \alpha'_{i,b}|i,b\rangle \otimes |\phi'_{i,b}\rangle$ with normalized workspace states, so that $\langle \psi_t^x | \psi_t^y \rangle = \sum_{i,b} \bar{\alpha}_{i,b}\alpha'_{i,b}\langle \phi_{i,b} | \phi'_{i,b} \rangle$. Input-independent unitaries do not change inner products, and after one more query

$$\langle \psi_{t+1}^x | \psi_{t+1}^y \rangle = \sum_{i,b,b'} \bar{\alpha}_{i,b}\alpha'_{i,b'}\langle b \oplus x_i | b' \oplus y_i \rangle \langle \phi_{i,b} | \phi'_{i,b'} \rangle.$$

If $x_i = y_i$, then $b' = b$ and the term is the same as before (no progress); if $x_i \ne y_i$, then $b' = 1 - b$. Hence, with the *query weights* $p_t^x(i) := |\alpha_{i,0}|^2 + |\alpha_{i,1}|^2$ (the probability of querying i next; $\sum_i p_t^x(i) = 1$),

$$|\langle \psi_{t+1}^x | \psi_{t+1}^y \rangle - \langle \psi_t^x | \psi_t^y \rangle| \le \sum_{i:\, x_i \ne y_i} \big(|\alpha_{i,0}| + |\alpha_{i,1}|\big)\big(|\alpha'_{i,0}| + |\alpha'_{i,1}|\big) \le 2 \sum_{i:\, x_i \ne y_i} \sqrt{p_t^x(i)\, p_t^y(i)}.$$

By the reverse triangle inequality and Cauchy–Schwarz,

$$|\Delta(t+1)-\Delta(t)| \le 2\sum_{(x,y)\in R}\sum_{i:\,x_i\neq y_i}\sqrt{p_t^x(i)p_t^y(i)} \le 2\sqrt{\sum_{(x,y)\in R}\sum_{i:\,x_i\neq y_i}p_t^x(i)}\sqrt{\sum_{(x,y)\in R}\sum_{i:\,x_i\neq y_i}p_t^y(i)}$$
$$\le 2\sqrt{\ell|P|\cdot\ell'|N|} \le 2|R|\sqrt{\frac{\ell\ell'}{mm'}},$$

where we reordered the sums, $\sum_{x\in P}\sum_i p_t^x(i)\,|\{y : (x,y)\in R, x_i\neq y_i\}| \le \ell|P|$, and used $|R|\ge m|P|$, $|R|\ge m'|N|$. □

Example 14.2.4: (i) $\mathrm{THR}_k(x)=1$ iff $|x|_H\ge k$. Take $P=\{x:|x|_H=k\}$, $N=\{x:|x|_H=k-1\}$, and $(x,y)\in R$ iff their Hamming distance is 1. Then $m=k$, $m'=n-k+1$, $\ell=\ell'=1$, so $Q(\mathrm{THR}_k)=\Omega(\sqrt{k(n-k+1)})$. In particular $Q(\mathrm{MAJORITY})=Q(\mathrm{THR}_{\lceil n/2\rceil})=\Omega(n)$.

(ii) PARITY: take P,N to be the strings of Hamming weight $\lfloor n/2\rfloor$ and $\lfloor n/2\rfloor+1$ (whichever is odd forms P), with R as for THR. Then $m\approx m'\approx n/2$ and $\ell=\ell'=1$, so $Q(\mathrm{PARITY})=\Omega(n)$.

Remark 14.2.5 (Generalized adversary): We can put weights on the pairs in R ("importance of the pair"): arrange them in a symmetric nonnegative matrix $\Gamma\in\mathbb{R}^{2^n\times 2^n}$ with $\Gamma(x,y)=0$ whenever $f(x)=f(y)$. Inspired by ℓ,ℓ', define $\Delta_i\in\{0,1\}^{2^n\times 2^n}$ by $\Delta_i(x,y)=1$ iff $x_i\neq y_i$, and

$$\mathrm{Adv}(f):=\max_{\Gamma}\frac{\|\Gamma\|}{\max_i\|\Gamma\circ\Delta_i\|},$$

where $\circ$ is the entrywise product. Then $Q(f)=\Omega(\mathrm{Adv}(f))$. If negative weights are allowed, it turns out that $Q(f)=\Theta(\mathrm{Adv}^{\pm}(f))$. This can be written as an SDP, whose dual gives an upper bound (an algorithm) for $Q(f)$.

15 The polynomial method

15.1 Polynomials and query algorithms

We consider real *multilinear* polynomials in the variables $x_1,\dots,x_n$: since the variables are Boolean, $x_i^2=x_i$, so every polynomial can be written as

$$p(x)=\sum_{S\subseteq[n]}\alpha_S\prod_{i\in S}x_i,\qquad \alpha_S\in\mathbb{R},\qquad \deg(p)=\max_{S:\,\alpha_S\neq 0}|S|.$$

The polynomial method relates $Q(f)$ to the existence of low-degree polynomials, so that quantum query lower bounds follow from (classical) Boolean function analysis, for which many tools are known.

Definition 15.1.1: A polynomial $p:\{0,1\}^n\to\mathbb{R}$ *ε-approximates* f if $|f(x)-p(x)|<\varepsilon$ for all $x\in\{0,1\}^n$. The *ε-approximate degree* $\widetilde{\deg}_\varepsilon(f)$ is the smallest d such that there exists a polynomial of degree d that ε-approximates f. The *exact degree* $\deg(f)$ is the case $\varepsilon=0$.

Remark 15.1.2: Every $f:\{0,1\}^n\to\mathbb{R}$ has a unique exact multilinear polynomial representation. For example, $\mathrm{OR}(x_1,\dots,x_n)=1-(1-x_1)\cdots(1-x_n)$, so $\deg(\mathrm{OR})=n$.

Lemma 15.1.3: $D(f)\ge\deg(f)$.

Proof. By induction on the depth of a decision tree computing f. Depth 0: f is constant. If the root queries x_i and the subtrees compute polynomials p_L (for $x_i=0$) and p_R (for $x_i=1$), then $p=(1-x_i)p_L+x_ip_R$ represents f, and the degree increases by at most 1. For example, $D(\mathrm{OR})=n$. □

Theorem 15.1.1: Polynomial method

Any quantum algorithm that computes f with bounded error $\varepsilon < 1/2$ has to make at least $\widetilde{\deg}_\varepsilon(f)/2$ queries.

Proof. Write the state after t queries as $|\psi_t^x\rangle = U_t O_x U_{t-1} \cdots O_x U_0 |0\rangle = \sum_{i,b} \alpha_{i,b}^t |i,b\rangle$, with $\alpha_{i,b}^t = \langle i,b|\psi_t^x\rangle$ (absorbing the workspace into the index).

Claim: $\alpha_{i,b}^t$ is a polynomial of degree $\le t$ in the variables $x_1, \dots, x_n$. For $t = 0$, $U_0|0\rangle$ is independent of x (degree 0). Assume the claim for t. A query maps $\sum_{i,b} \alpha_{i,b}^t |i,b\rangle$ to $\sum_{i,b} \alpha_{i,b}^t \big((1-x_i)|i,b\rangle + x_i|i,1-b\rangle\big)$, so the new amplitudes $\alpha_{i,b}^t(1-x_i) + \alpha_{i,1-b}^t x_i$ have degree at most $t+1$. The input-independent unitary U_{t+1} does not increase the degree: with $\beta_{jc} := \langle i,b|U_{t+1}|j,c\rangle$ independent of x,

$$\alpha_{i,b}^{t+1} = \langle i,b|U_{t+1}O_x|\psi_t^x\rangle = \sum_{j,c} \beta_{jc}\big(\alpha_{j,c}^t(1-x_j) + \alpha_{j,1-c}^t x_j\big).$$

Consequently, after T queries the amplitudes are polynomials of degree $\le T$, and the acceptance probability

$$\Pr[\mathcal{A}(x) = 1] = \big\|(I \otimes |1\rangle\langle 1|)|\psi_T^x\rangle\big\|^2 = \sum_i |\alpha_{i,1}^T|^2$$

is a polynomial in x of degree $\le 2T$ (real and imaginary parts are real polynomials). An ε-bounded-error algorithm satisfies $|\Pr[\mathcal{A}(x) = 1] - f(x)| < \varepsilon$ for all x, so the acceptance probability ε-approximates f. Thus $\widetilde{\deg}_\varepsilon(f) \le 2T$. □

15.2 Symmetrization

Arguing about every approximating multivariate polynomial can still be hard. A useful tool is *symmetrization*: if f is symmetric (i.e., only depends on $|x|_H$), then instead of a polynomial in the n Boolean variables $x_1, \dots, x_n$ we consider one in the single real variable $|x|_H$.

Theorem 15.2.1: Minsky–Papert symmetrization

Let $p(x_1, \dots, x_n)$ be a polynomial over $\{0,1\}^n$. Then there exists a univariate polynomial $q : \mathbb{R} \to \mathbb{R}$ with $\deg(q) \le \deg(p)$ such that

$$q(k) = \mathbb{E}_{x \in \{0,1\}^n : |x|_H = k}[p(x)] \qquad \text{for all } k \in \{0, 1, \dots, n\}.$$

Proof. By linearity it suffices to consider monomials $p(x) = \prod_{i \in S} x_i$ with $d := |S|$. Let

$$q(k) := \frac{k(k-1)\cdots(k-d+1)}{n(n-1)\cdots(n-d+1)}, \qquad \deg(q) = d.$$

If $k \in \{0, \dots, d-1\}$, then $q(k) = 0$ (there is a zero in the numerator) and $\mathbb{E}_{|x|_H = k}[\prod_{i \in S} x_i] = 0$ since $k < d$. If $k \in \{d, \dots, n\}$, then $q(k) = \binom{n-d}{k-d} \big/ \binom{n}{k}$, which is the probability that a uniformly random x with $|x|_H = k$ has $x_i = 1$ for all $i \in S$ (number of ways to choose the rest of x after fixing S to be 1, divided by the total number of choices). □

Example 15.2.1 (PARITY): Let p be any ε-approximating polynomial for PARITY (with $\varepsilon = 1/3$), and let q be its symmetrization. Since $\mathrm{PARITY}(x) = 1$ iff $|x|_H$ is odd, $q(k) \le 1/3$ for even k and $q(k) \ge 2/3$ for odd k. Hence $q - 1/2$ changes sign between consecutive integers in $\{0, \dots, n\}$ and has at least n roots, so $n \le \deg(q) \le \deg(p)$. Thus $\widetilde{\deg}_\varepsilon(\mathrm{PARITY}) \ge n$ and $Q(\mathrm{PARITY}) \ge n/2$.

For OR (also symmetric), the symmetrized polynomial must "jump" from $q(0) \le \varepsilon$ to $q(1) \ge 1-\varepsilon$ while staying bounded on $\{1,\dots,n\}$. Such a jump forces large degree, by a result from approximation theory:

Theorem 15.2.2: Paturi

Let $q : \mathbb{R} \to \mathbb{R}$ be a polynomial, $a < b$ integers, and $\xi \in [a,b]$ a real number such that, for some constants $c_1, c_2 > 0$,

- $|q(i)| \le c_1$ for all integers $i \in [a,b]$,
- $|q(\lfloor \xi \rfloor) - q(\xi)| \ge c_2$.

Then $\deg(q) = \Omega\big(\sqrt{(\xi - a + 1)(b - \xi + 1)}\big)$.

Example 15.2.2 (OR): Let q be the symmetrization of an ε-approximating polynomial for OR. Then $|q(i)| \le 1 + \varepsilon =: c_1$ for $i \in \{0,\dots,n\}$, $q(0) \le \varepsilon$ and $q(1) \ge 1-\varepsilon$. By continuity, there is $\xi = 1 - \delta \in (0,1)$ with $\lfloor \xi \rfloor = 0$ and $|q(\xi) - q(0)| \ge c_2 := \frac{1-2\varepsilon}{2}$. Paturi's theorem with $a = 0$, $b = n$ gives $\deg(q) = \Omega(\sqrt{(2-\delta)(n+\delta)}) = \Omega(\sqrt{n})$. Hence $Q(\mathrm{OR}) = \Omega(\sqrt{n})$, again showing that Grover is optimal.

15.3 The dual polynomial method

Here we use $f : \{\pm 1\}^n \to \{\pm 1\}$, replacing $b \in \{0,1\}$ by $1 - 2b$; this preserves the approximate degree. We can write $\widetilde{\deg}_\varepsilon$ as a linear program: the minimal ε achievable at degree $< d$ is

$$\min_{p,\varepsilon} \ \varepsilon \quad \text{s.t.} \quad |p(x) - f(x)| \le \varepsilon \ \ \forall x \in \{\pm 1\}^n, \quad \deg(p) < d, \quad \varepsilon \ge 0,$$

and after rewriting, its dual is

$$\max_{\varphi} \ \sum_x \varphi(x) f(x) \quad \text{s.t.} \quad \sum_x |\varphi(x)| = 1, \quad \sum_x \varphi(x) p(x) = 0 \ \ \forall p \text{ with } \deg(p) < d.$$

Theorem 15.3.1: Dual polynomials

$\widetilde{\deg}_\varepsilon(f) \ge d$ if there exists a *dual polynomial* $\varphi : \{\pm 1\}^n \to \mathbb{R}$ satisfying

(i) normalization: $\sum_x |\varphi(x)| = 1$,

(ii) high correlation with f: $\sum_x \varphi(x) f(x) > \varepsilon$,

(iii) pure high degree d: $\sum_x \varphi(x) p(x) = 0$ for all polynomials p of degree $< d$.

Proof. If p had degree $< d$ and $|p(x) - f(x)| \le \varepsilon$ for all x, then by (iii) and (i), $\sum_x \varphi(x) f(x) = \sum_x \varphi(x)\big(f(x) - p(x)\big) \le \varepsilon \sum_x |\varphi(x)| = \varepsilon$, contradicting (ii). □

Remark 15.3.1: Condition (iii) holds iff φ, as a multilinear polynomial, has no monomial of degree $< d$ with nonzero coefficient. Indeed, if φ had a monomial $\prod_{i \in S} x_i$ with $|S| < d$, take $p(x) = \prod_{i \in S} x_i$: all other monomials of φ cancel in $\sum_x \varphi(x) p(x)$ (some x_i appears in one but not the other, so the terms with $x_i = \pm 1$ cancel), while this monomial contributes 2^n times its coefficient. Conversely, if φ has no such monomial, every product of a monomial of φ with one of p contains some variable, so the sum vanishes.

The dual method replaces arguing about every approximating polynomial by exhibiting a single "witness" of high degree.

Example 15.3.2 (PARITY): Let $f = \mathrm{PARITY}$, i.e., $f(x) = x_1 \cdots x_n$ on $\{\pm 1\}^n$, and take $\varphi(x) = f(x)/2^n$. Then (i) holds, (ii) $\sum_x \varphi(x) f(x) = 1 > \varepsilon$, and (iii) holds since φ has no monomial of degree $< n$. This gives the lower bound $\Omega(n)$.

15.4 Importance of the polynomial method

The basic and positive-weight adversary methods cannot give good (better than constant) lower bounds for some problems (certificate complexity barrier, property testing barrier). The general (dual) polynomial method can, but it is not always intuitive and can be hard to use.

Example 15.4.1 (Collision problem): Given $x \in [M]^N$, find $i \neq j$ with $x_i = x_j$, promised that the input is 2-to-1 (similar to Simon's problem; important in cryptography). The decision version asks whether x is 1-to-1 or 2-to-1. This is a partial function with non-Boolean input, but it is symmetric: only the frequencies of the characters matter. Classically, the birthday paradox gives complexity $\Theta(\sqrt{N})$. A quantum algorithm with $O(N^{1/3})$ queries was known, but no nontrivial lower bound. Using the polynomial method (Boolean encoding, a 3-variate polynomial defined via symmetry, and repeated use of Paturi's theorem) it was proved in 2002 that the problem has quantum query complexity $\Omega(N^{1/3})$.

Remark 15.4.2: Even nowadays, new lower bounds are proven using the (dual) polynomial method, e.g., for k-collision: decide whether there are 0 or at least εn k-collisions.

16 Monogamy of entanglement and de Finetti theorems

16.1 Monogamy

Recall that if ρ_{AB} is pure, then $\rho_{ABC} = \rho_{AB} \otimes \rho_C$, so in particular $\rho_{AC} = \rho_A \otimes \rho_C$ and $\rho_{BC} = \rho_B \otimes \rho_C$: a system cannot share a *pure* entangled state with more than one system ("monogamy"). In contrast, classical correlations can be shared: for $\rho_{ABC} = \frac{1}{2}(|000\rangle\langle 000| + |111\rangle\langle 111|)$ we have $\rho_{AB} = \rho_{AC} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$. Note that the GHZ state $|\psi_{ABC}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ does *not* have the EPR pair $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ as two-party marginal; its marginals are the classically correlated states above.

What happens beyond pure states, and can we make this quantitative?

(i) *Symmetric pure states.* If $|\Phi_{A_1 \cdots A_N}\rangle \in \mathrm{Sym}^N(\mathbb{C}^d)$, we would expect the two-party marginals $\Phi_{A_i A_j}$ (which are all the same!) not to be very entangled. Indeed, by the quantum de Finetti theorem below, if $k \ll N$, then

$$\Phi_{A_1 \cdots A_k} \approx \int d\psi \, p(\psi) \, |\psi\rangle\langle\psi|^{\otimes k}$$

for some probability density $p(\psi)$, i.e., the marginal is close to a mixture of product states $|\psi\rangle\langle\psi|^{\otimes k}$ (a *de Finetti state*), which is separable. The GHZ state is an extreme case: $N = 3$, $k = 2$, and the marginal is exactly separable. Physics applications: n bosons; mean-field theory (if $H = \sum_{i<j} h_{ij}$ with the same interaction h_{ij} between all pairs and the ground state is non-degenerate, then it lies in the symmetric subspace).

(ii) *Mixed states.* More generally, if $\rho_{A_1 \cdots A_N}$ is a state such that all marginals $\rho_{A_{i_1} \cdots A_{i_k}}$ are the same, and $k \ll N$, then

$$\rho_{A_1 \cdots A_k} \approx \int d\mu(\sigma) \, \sigma^{\otimes k}$$

for a probability measure μ on mixed states (a de Finetti state).

(iii) *Extendibility.* If a state ρ_{AB_1} can be extended to $\rho_{AB_1 \cdots B_N}$ with $\rho_{AB_1} = \cdots = \rho_{AB_N}$ (N-*extendible*), then ρ_{AB_1} is $\varepsilon = O_d(1/N)$-close to a separable state $\sum_j p_j \rho_A^{(j)} \otimes \rho_B^{(j)}$. Every separable state is N-extendible for all N, and the sets of N-extendible states shrink towards the separable states as N grows. This gives a *separability hierarchy*; for each fixed N, testing N-extendibility is an SDP.

Remark 16.1.1: Statement (ii) can be reduced to (i) (a nice exercise): the symmetrized state $\tilde{\rho} := \frac{1}{N!}\sum_\pi R_\pi \rho R_\pi^\dagger$ has the same k-party marginals as ρ and commutes with all R_π. Hence its standard purification $|\tilde{\Phi}\rangle = (\sqrt{\tilde{\rho}} \otimes I)|\Phi^+\rangle$ lies in $\mathrm{Sym}^N\big((\mathbb{C}^d)^{\otimes 2}\big)$, using the general fact that $(U \otimes \bar{U})|\Phi^+\rangle = |\Phi^+\rangle$ for unitaries U (here $U = R_\pi$). Now apply (i) with local dimension d^2 and trace out the purifying systems.

16.2 The quantum de Finetti theorem

Theorem 16.2.1: Quantum de Finetti theorem for the symmetric subspace

Let $k \leq N$. For every $|\Phi\rangle \in \mathrm{Sym}^N(\mathbb{C}^d)$ there exists a probability density $p(\psi)$ such that

$$\frac{1}{2}\Big\|\Phi_{A_1\cdots A_k} - \int d\psi\, p(\psi)\, |\psi\rangle\langle\psi|^{\otimes k}\Big\|_1 \leq \sqrt{\frac{dk}{N}}.$$

Idea: write $N = k + n$. If we measure Hayashi's POVM $\{\binom{n+d-1}{n}|\psi\rangle\langle\psi|^{\otimes n}\}$ on the last n systems and get outcome ψ, then the first k systems should be $\approx |\psi\rangle\langle\psi|^{\otimes k}$.

Proof. Let $\rho := \mathrm{tr}_{k+1,\ldots,N}[|\Phi\rangle\langle\Phi|]$ (by symmetry, it does not matter which $N - k = n$ systems we trace out). Since $(I \otimes \Pi_{n,d})|\Phi\rangle = |\Phi\rangle$, the integral formula gives

$$\begin{aligned}\rho &= \mathrm{tr}_{k+1,\ldots,N}\big[(I \otimes \Pi_{n,d})|\Phi\rangle\langle\Phi|\big] = \binom{n+d-1}{n}\int d\psi\ \mathrm{tr}_{k+1,\ldots,N}\big[(I \otimes |\psi\rangle\langle\psi|^{\otimes n})|\Phi\rangle\langle\Phi|\big] \\ &= \binom{n+d-1}{n}\int d\psi\, (I_k \otimes \langle\psi|^{\otimes n})|\Phi\rangle\langle\Phi|(I_k \otimes |\psi\rangle^{\otimes n}) = \int d\psi\, p(\psi)\, |v_\psi\rangle\langle v_\psi|,\end{aligned}$$

where we defined the probability density $p(\psi)$ and unit vectors $|v_\psi\rangle \in \mathrm{Sym}^k(\mathbb{C}^d)$ by

$$\sqrt{p(\psi)}\,|v_\psi\rangle := \binom{n+d-1}{n}^{1/2}(I_k \otimes \langle\psi|^{\otimes n})|\Phi\rangle.$$

Claim: $\rho \approx \tilde{\rho} := \int d\psi\, p(\psi)|\psi\rangle\langle\psi|^{\otimes k}$. To see this, first note that

$$\begin{aligned}\int d\psi\, p(\psi)\, |\langle\psi|^{\otimes k}|v_\psi\rangle|^2 &= \binom{n+d-1}{n}\int d\psi\, |\langle\psi|^{\otimes(k+n)}|\Phi\rangle|^2 \\ &= \binom{n+d-1}{n}\langle\Phi|\Big(\int d\psi\, |\psi\rangle\langle\psi|^{\otimes N}\Big)|\Phi\rangle \\ &= \binom{N+d-1}{N}^{-1}\binom{n+d-1}{n}\langle\Phi|\Pi_{N,d}|\Phi\rangle = \frac{(n+d-1)\cdots(n+1)}{(N+d-1)\cdots(N+1)} \\ &\geq \Big(\frac{n+1}{N+1}\Big)^{d-1} \geq 1 - \frac{dk}{N}.\end{aligned}$$

Hence, using the formula for the trace distance of pure states and Jensen's inequality for the concave square root,

$$\begin{aligned}\frac{1}{2}\|\tilde{\rho} - \rho\|_1 &\leq \int d\psi\, p(\psi)\, \frac{1}{2}\big\||\psi\rangle\langle\psi|^{\otimes k} - |v_\psi\rangle\langle v_\psi|\big\|_1 = \int d\psi\, p(\psi)\sqrt{1 - |\langle\psi|^{\otimes k}|v_\psi\rangle|^2} \\ &\leq \sqrt{\int d\psi\, p(\psi)\big(1 - |\langle\psi|^{\otimes k}|v_\psi\rangle|^2\big)} = \sqrt{1 - \int d\psi\, p(\psi)|\langle\psi|^{\otimes k}|v_\psi\rangle|^2} \leq \sqrt{\frac{dk}{N}}.\end{aligned}$$

□